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1 – OVERVIEW

This procedure will attempt to assist the FE with isolating failures within the System Support Module (SSM). This procedure is not intended as a guide to troubleshooting the various problem scenarios associated with all the components driven or sourced by the SSM. The primary goal of this procedure is to assist the FE with determining if a system problem can be isolated to the SSM. If the problem can be isolated to the SSM, then the secondary goal is to help the FE isolate the problem to an SSM sub-component that can be replaced. A flowchart is provided that the FE will follow during the course of the troubleshooting process. The flowchart will, at certain points, refer the FE to various troubleshooting sections in this document. The FE will perform the instructions in the sections to determine if the problem is internal to the SSM. If the problem is determined to be internal to the SSM then the instructions will provide the FE with a troubleshooting process for isolating the problem down to a replaceable part.

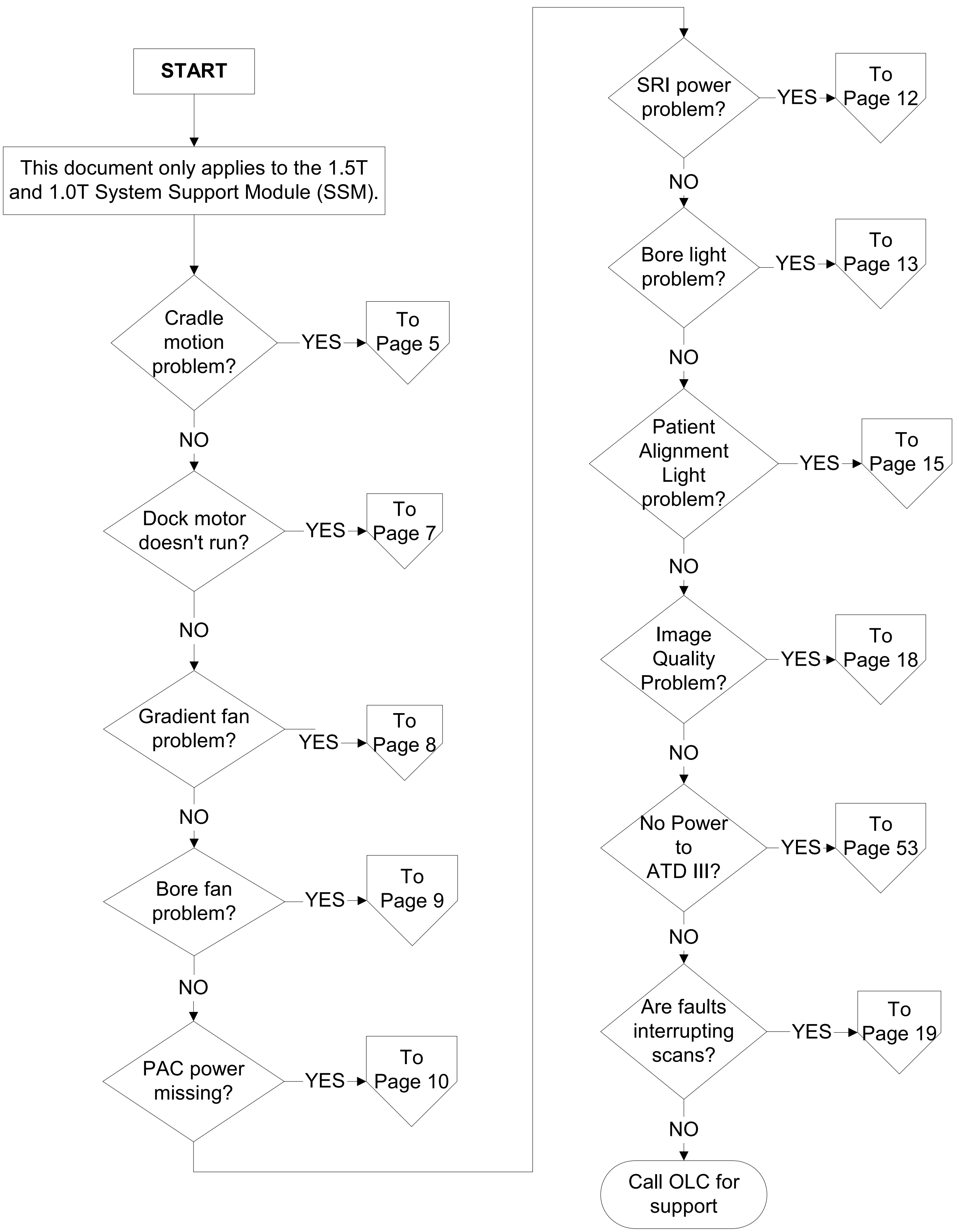
1-1 Required Tools

TABLE 1-1
EQUIPMENT REQUIRED

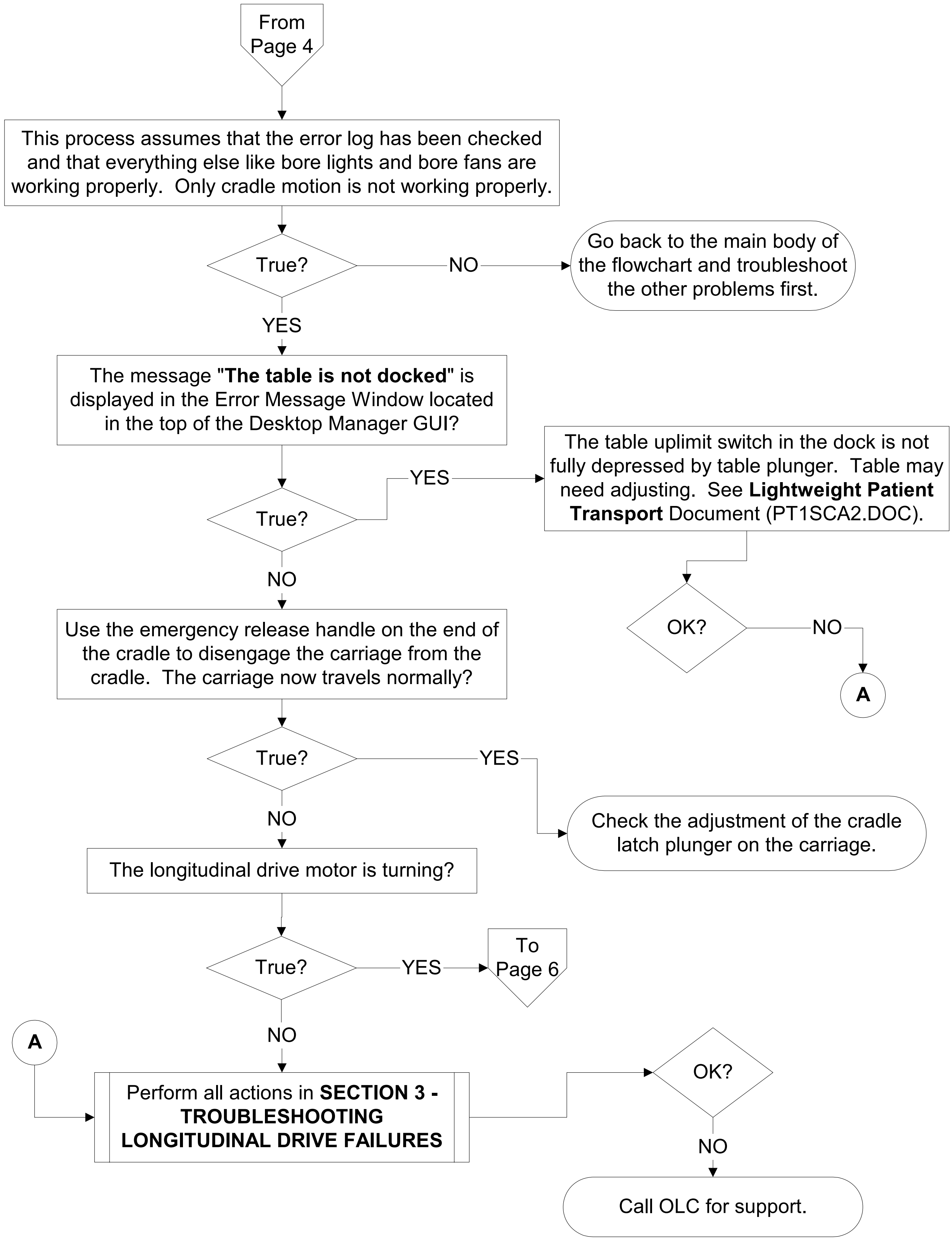
Item	Description	Part Number
1.	100 MHz Scope (equivalent or greater)	46-183029P61
2.	RF Power Measurement Kit	46-317724G1 or G2
3.	50 ohm, 200 Watt, 30dB Attenuator (dummy load) **NOTE:** Only required with above 46-317724G1 kit.	46-317724P14
4.	RF Test Cables Kit	46-255816G1
5.	Digital Multimeter (DMM)	46-194427P49
6.	TPS RF Service Interface Kit	46-301927G1
7.	Copper Tape (2 inch width)	46-258218P5

2 – 1.5T AND 1.0T SSM TROUBLESHOOTING FLOWCHART

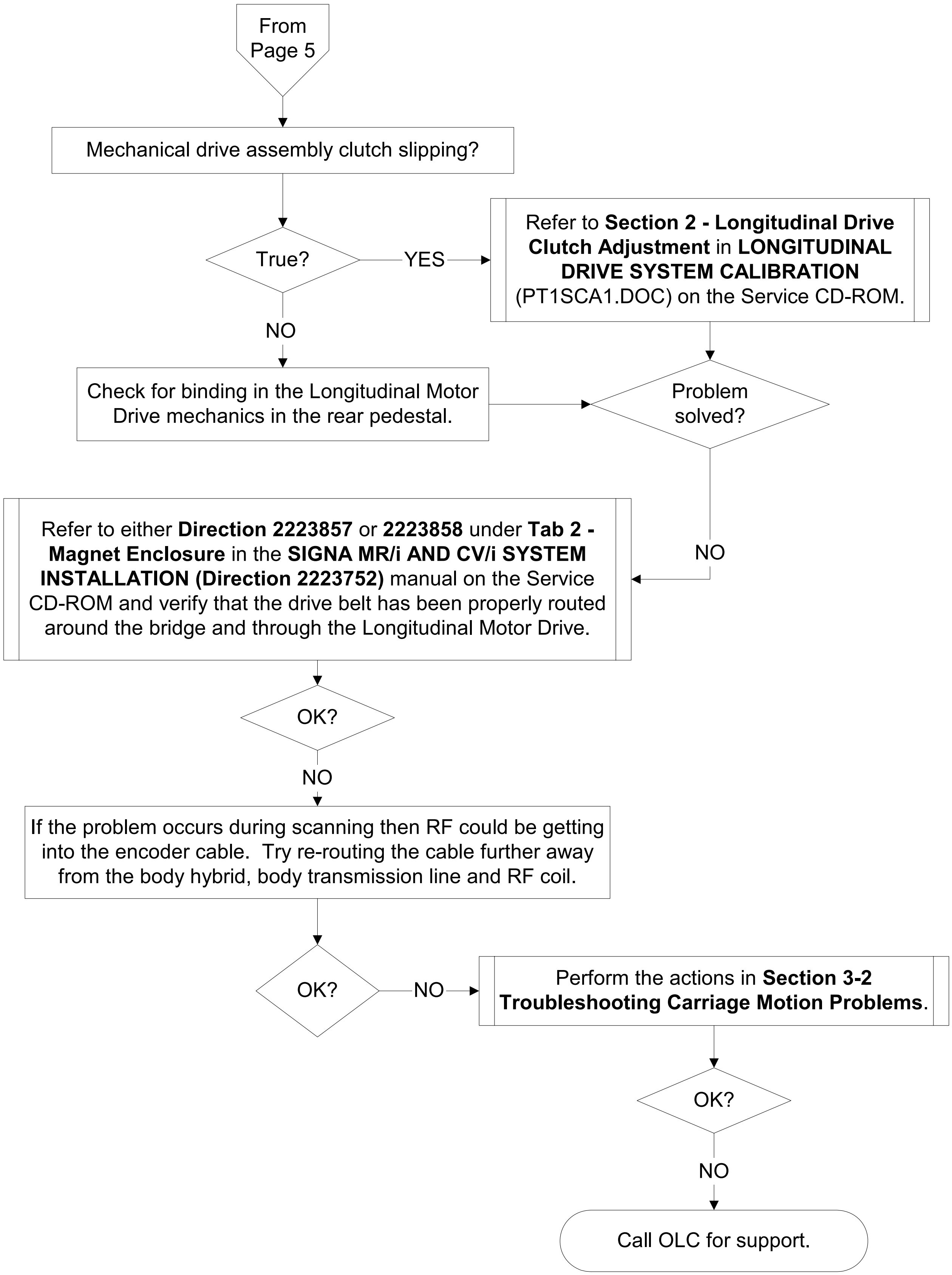
2-1 Main Body



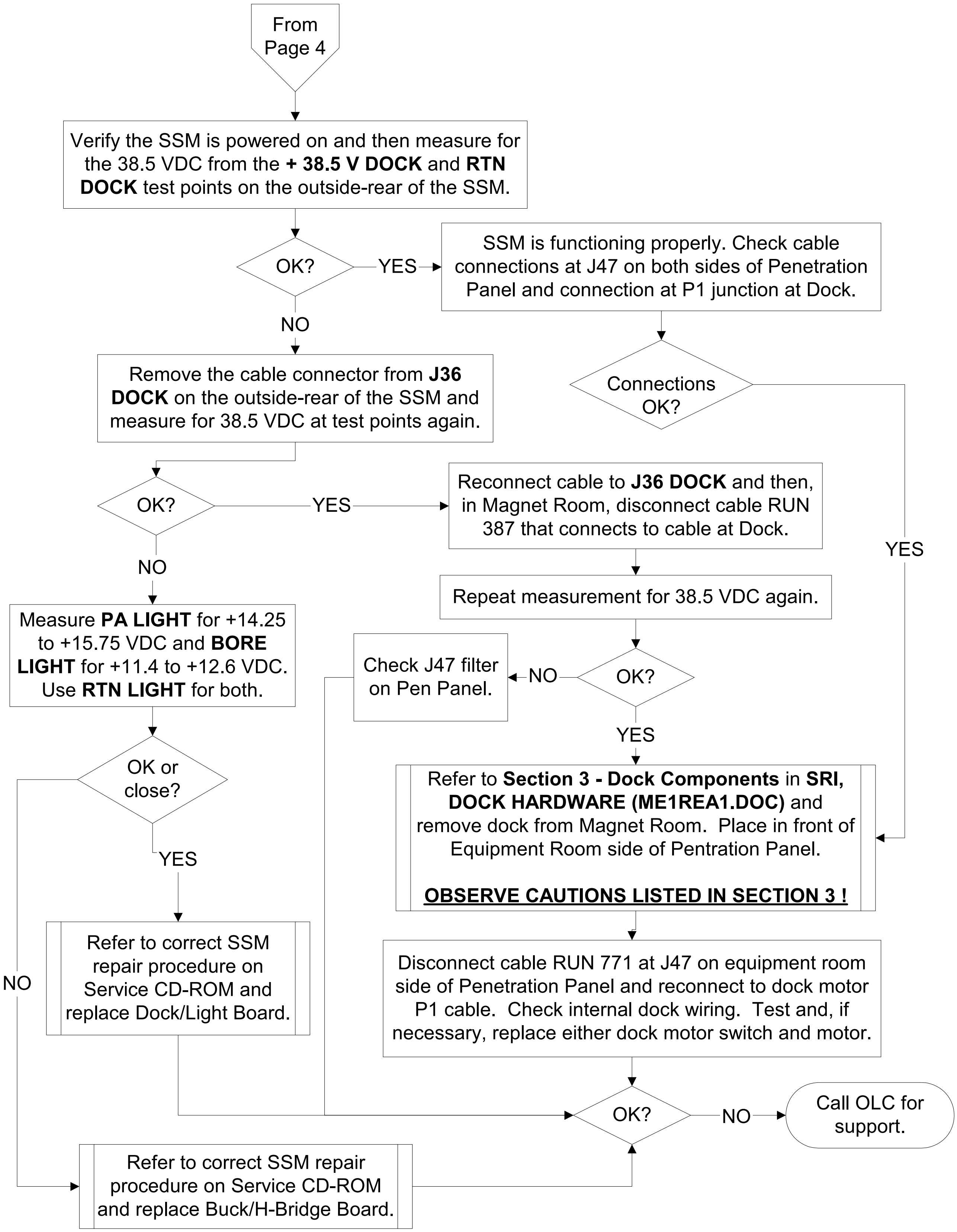
2-2 Cradle Motion Troubleshooting



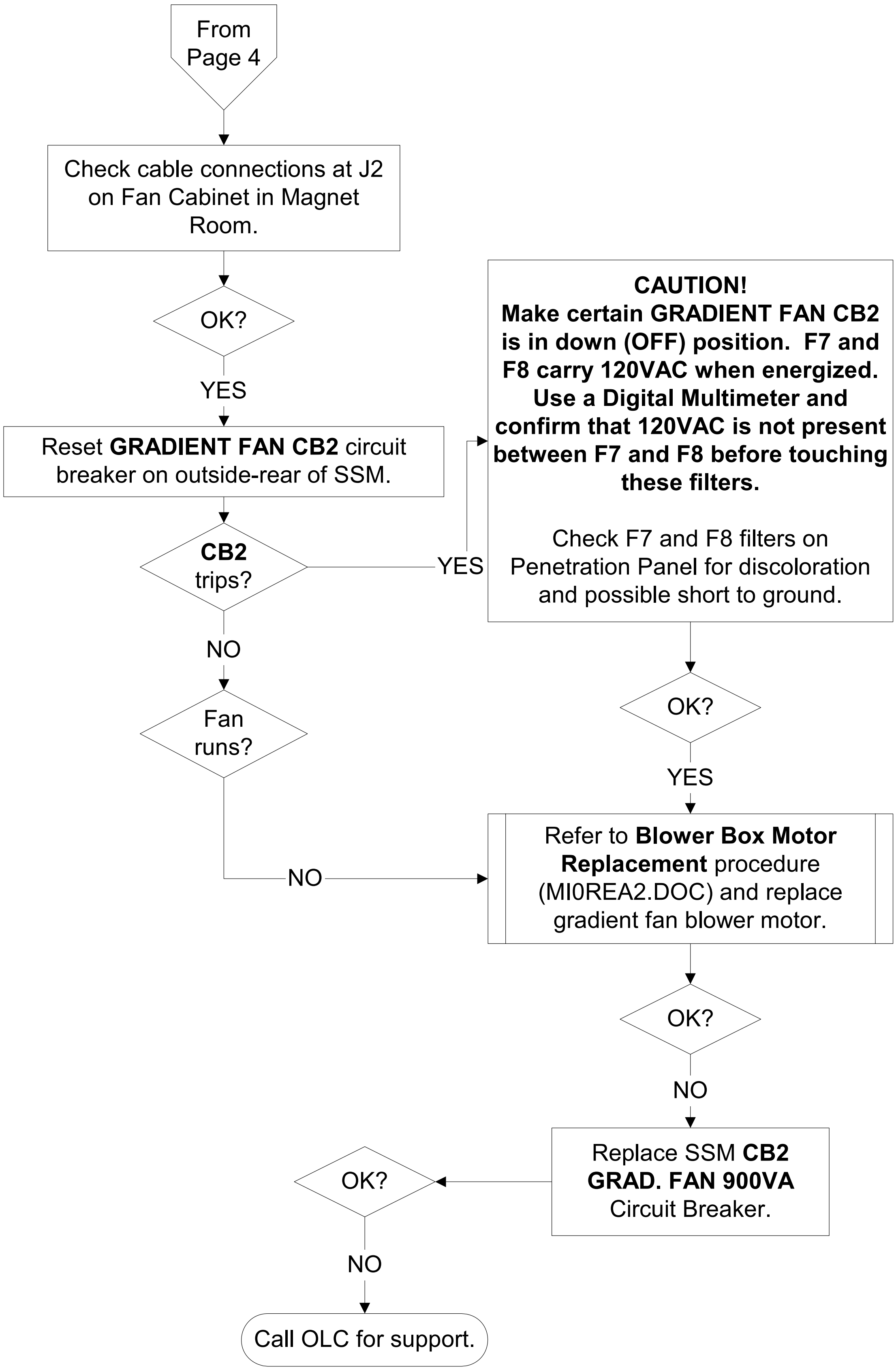
2-2 Cradle Motion Troubleshooting (Continued)



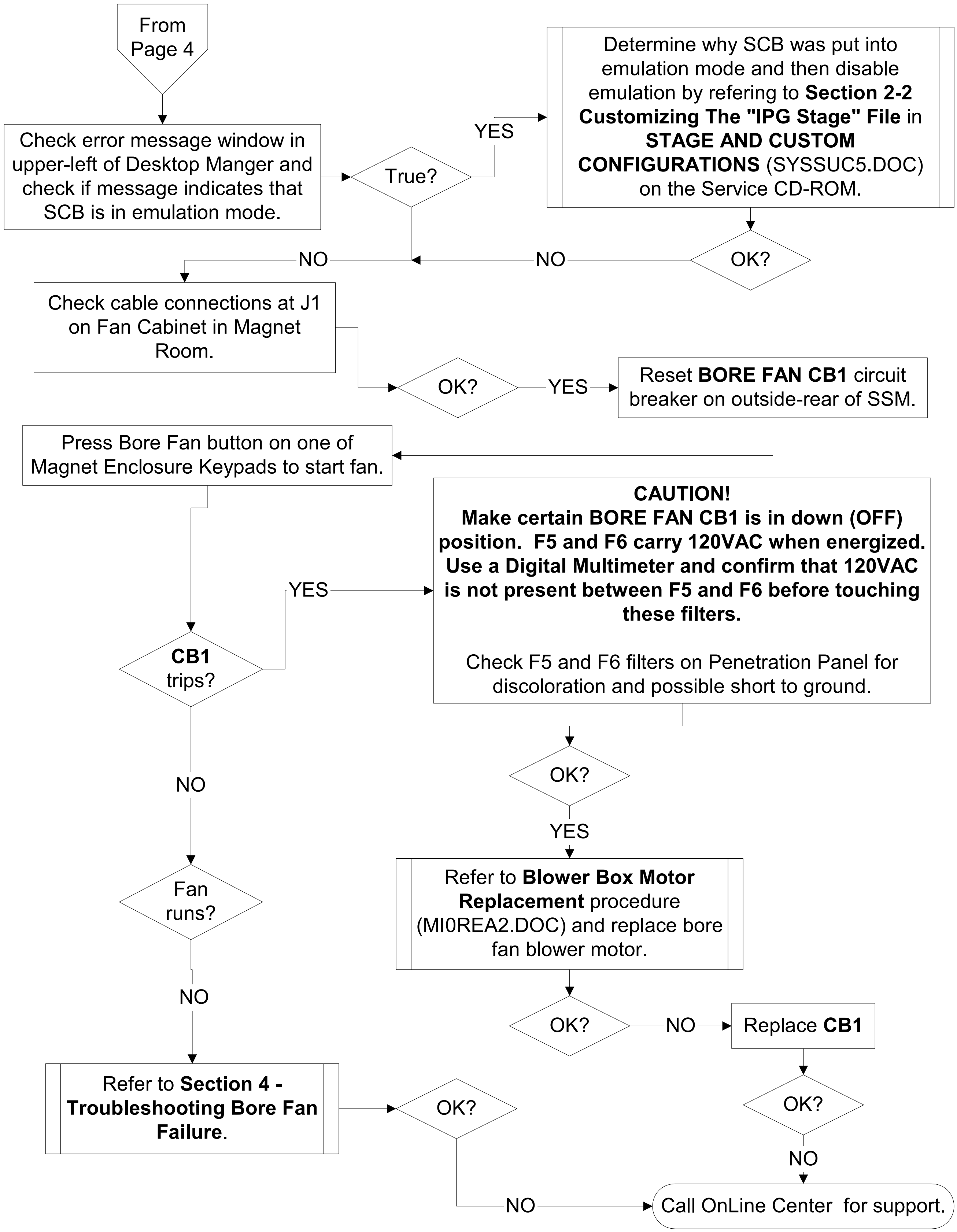
2-3 Dock Motor Troubleshooting



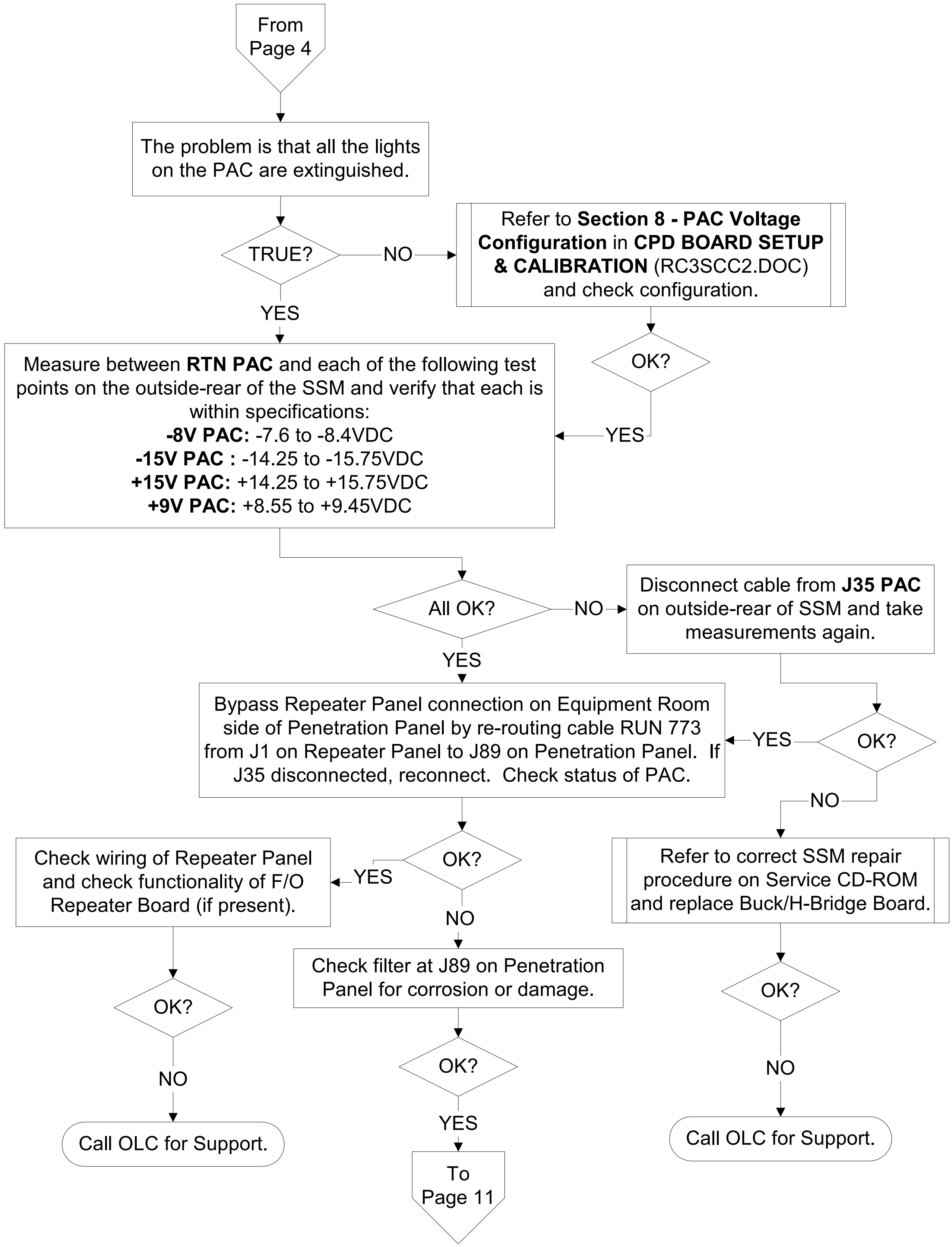
2-4 Gradinet Fan Troubleshooting



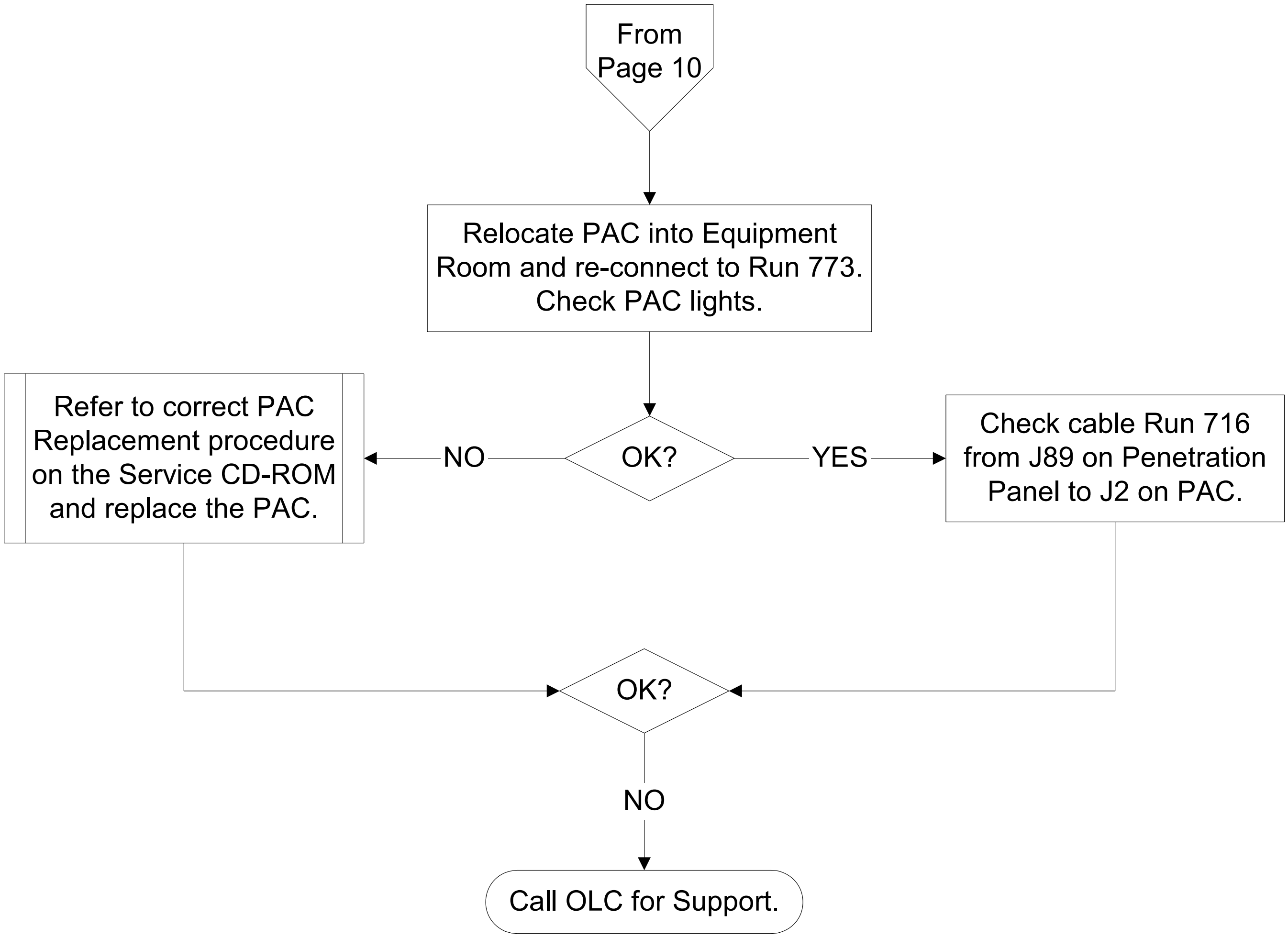
2-5 Bore Fan Troubleshooting



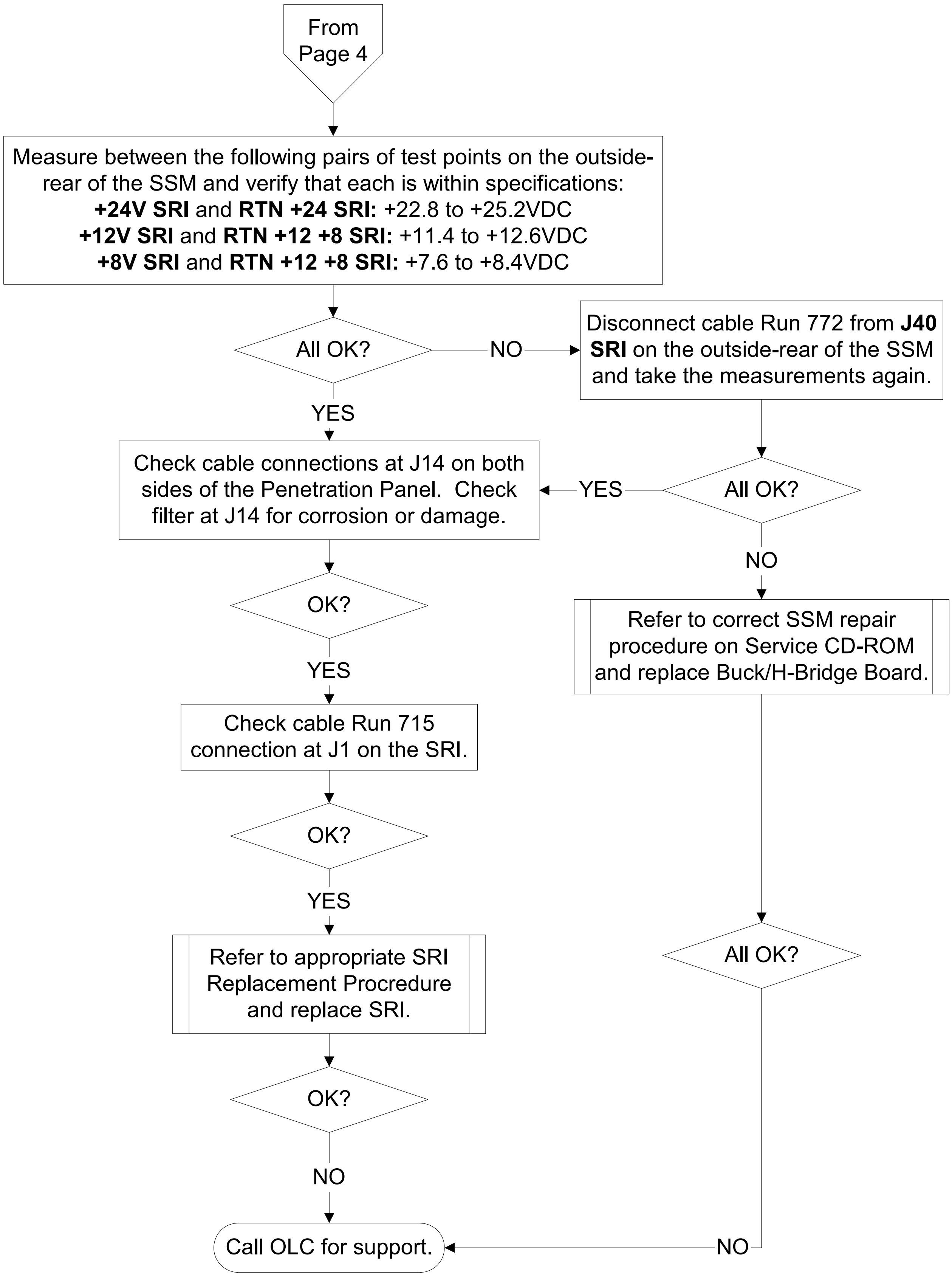
2-6 Physiological Acquistion Controller (PAC) Troubleshooting



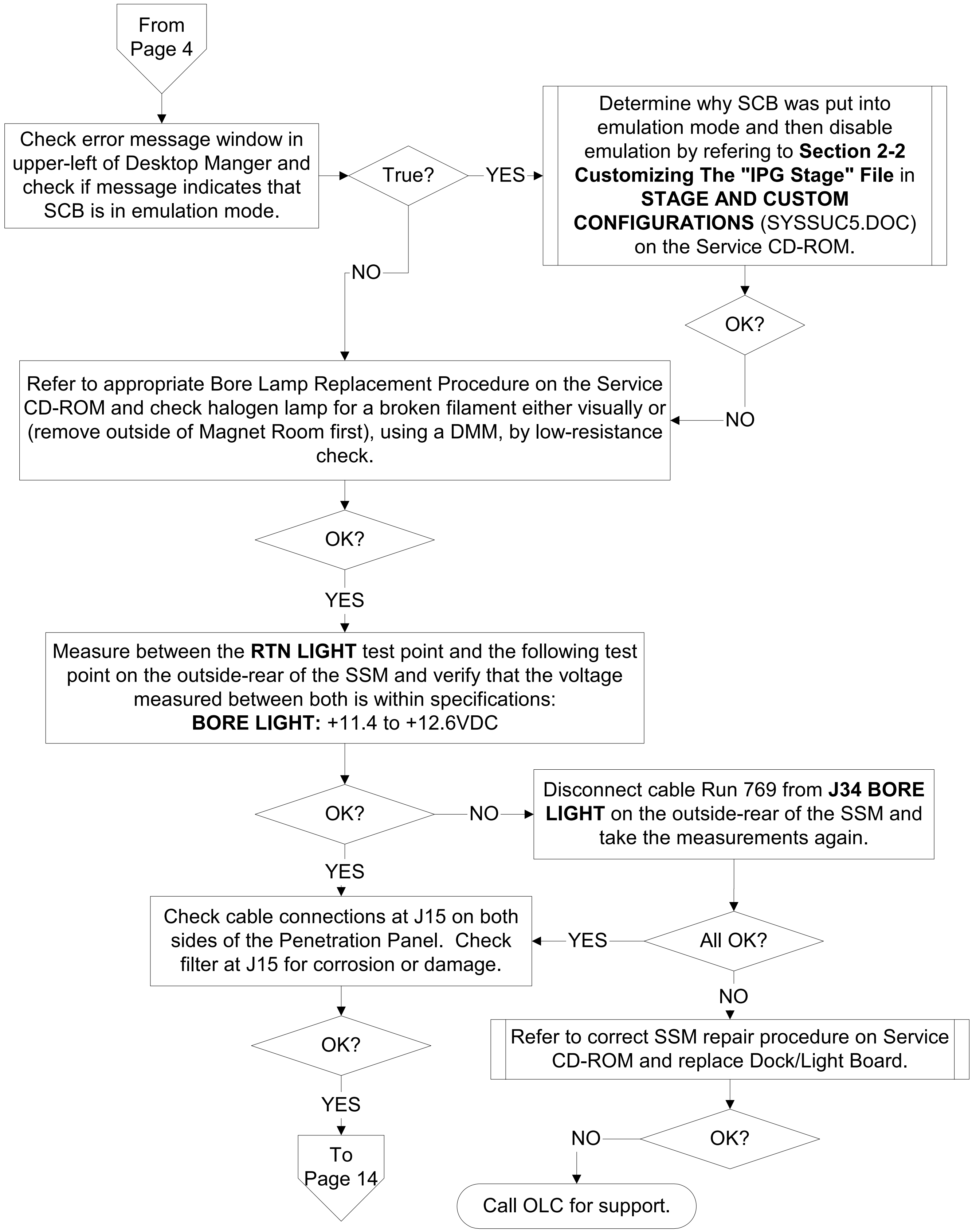
2-6 Physiological Acquistion Controller (PAC) Troubleshooting (Continued)



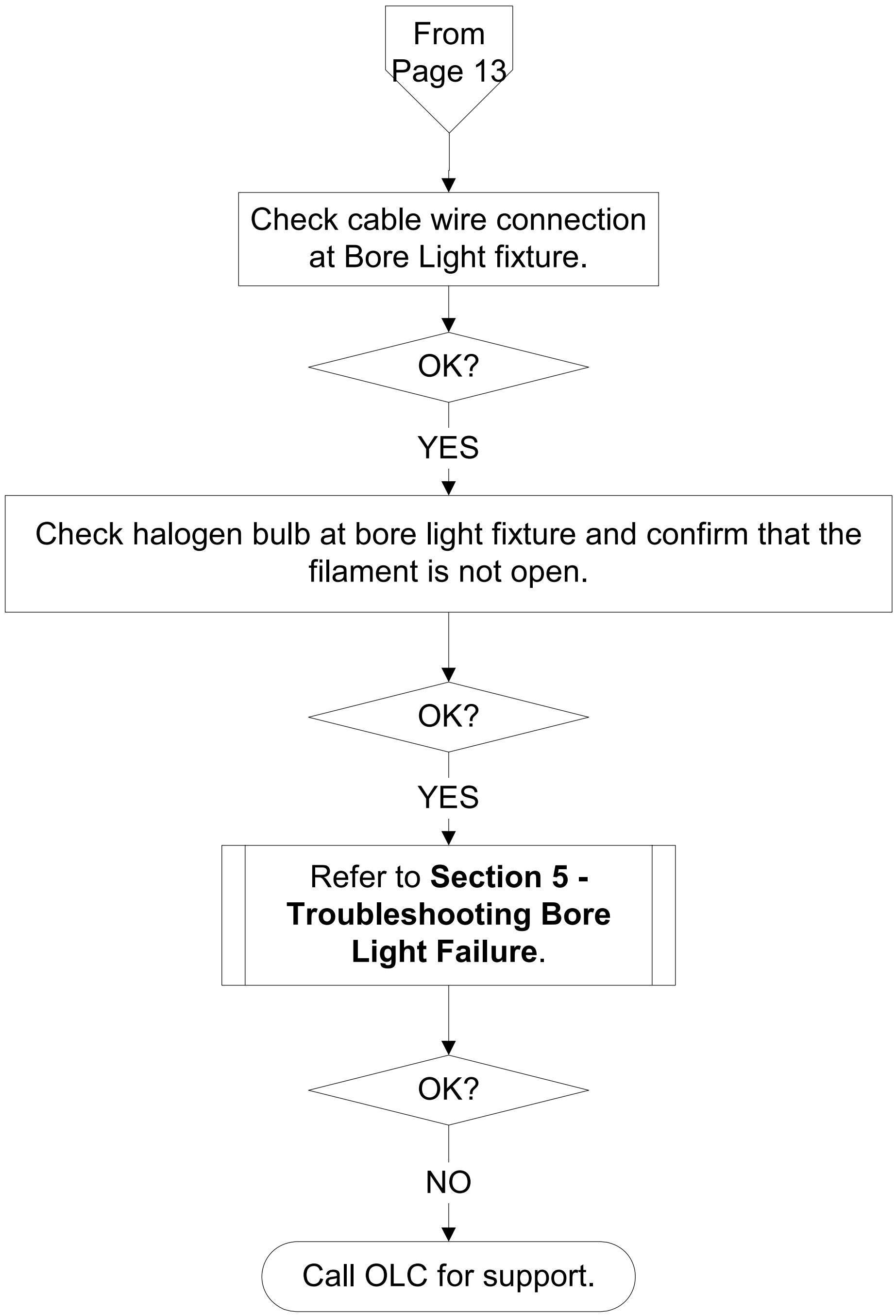
2-7 Scan Room Interface (SRI) Troubleshooting



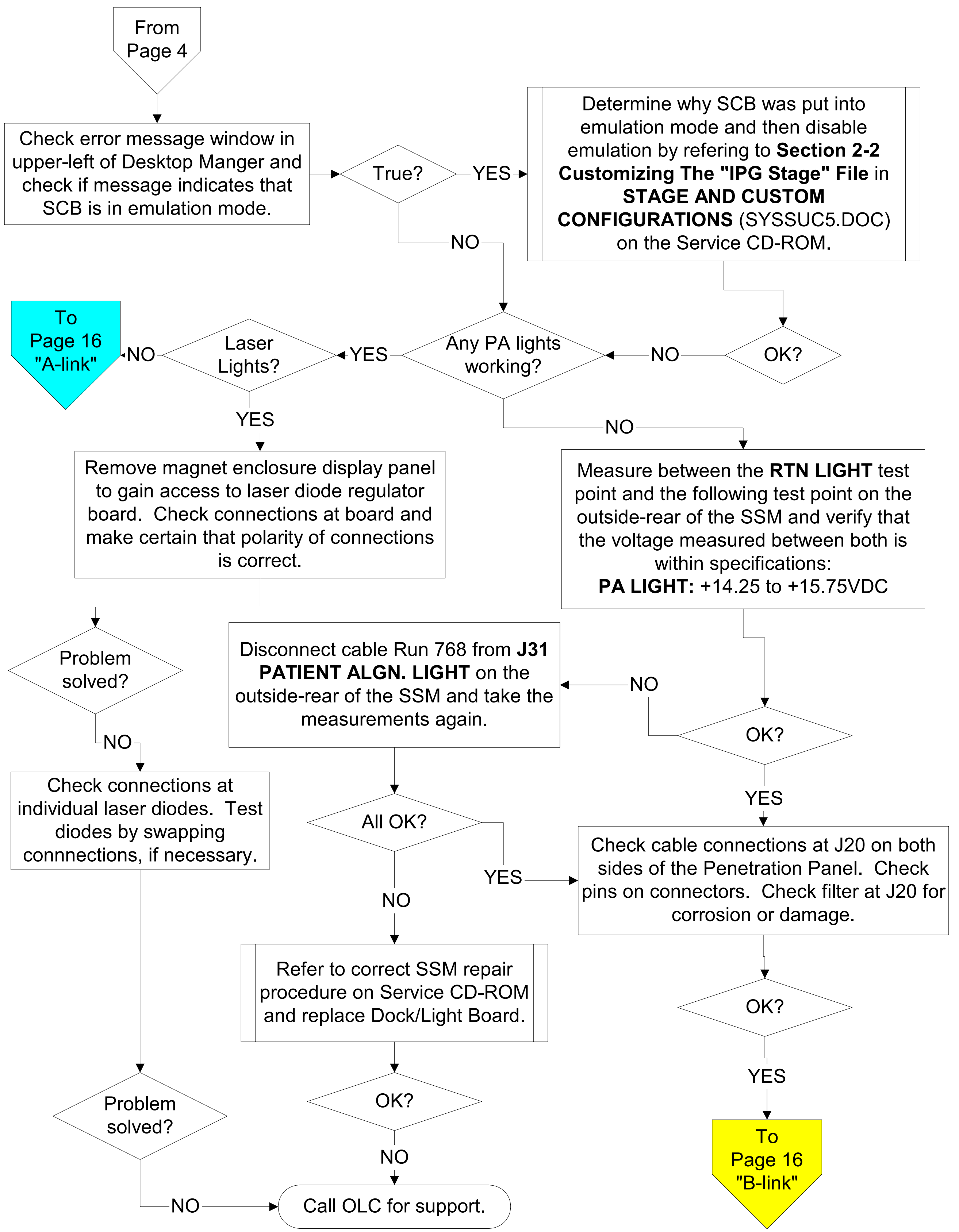
2-8 Bore Light Troubleshooting



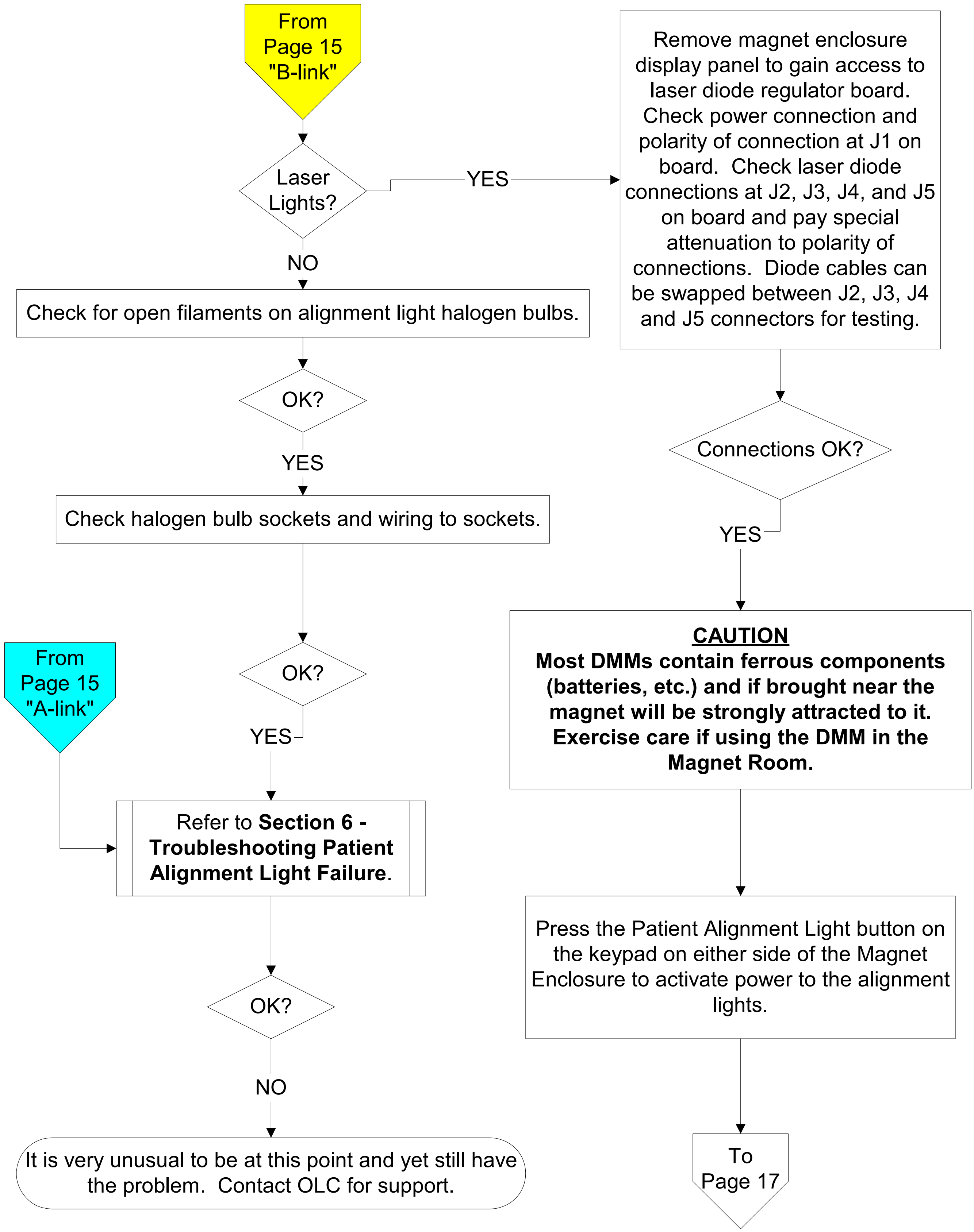
2-8 Bore Light Troubleshooting (Continued)



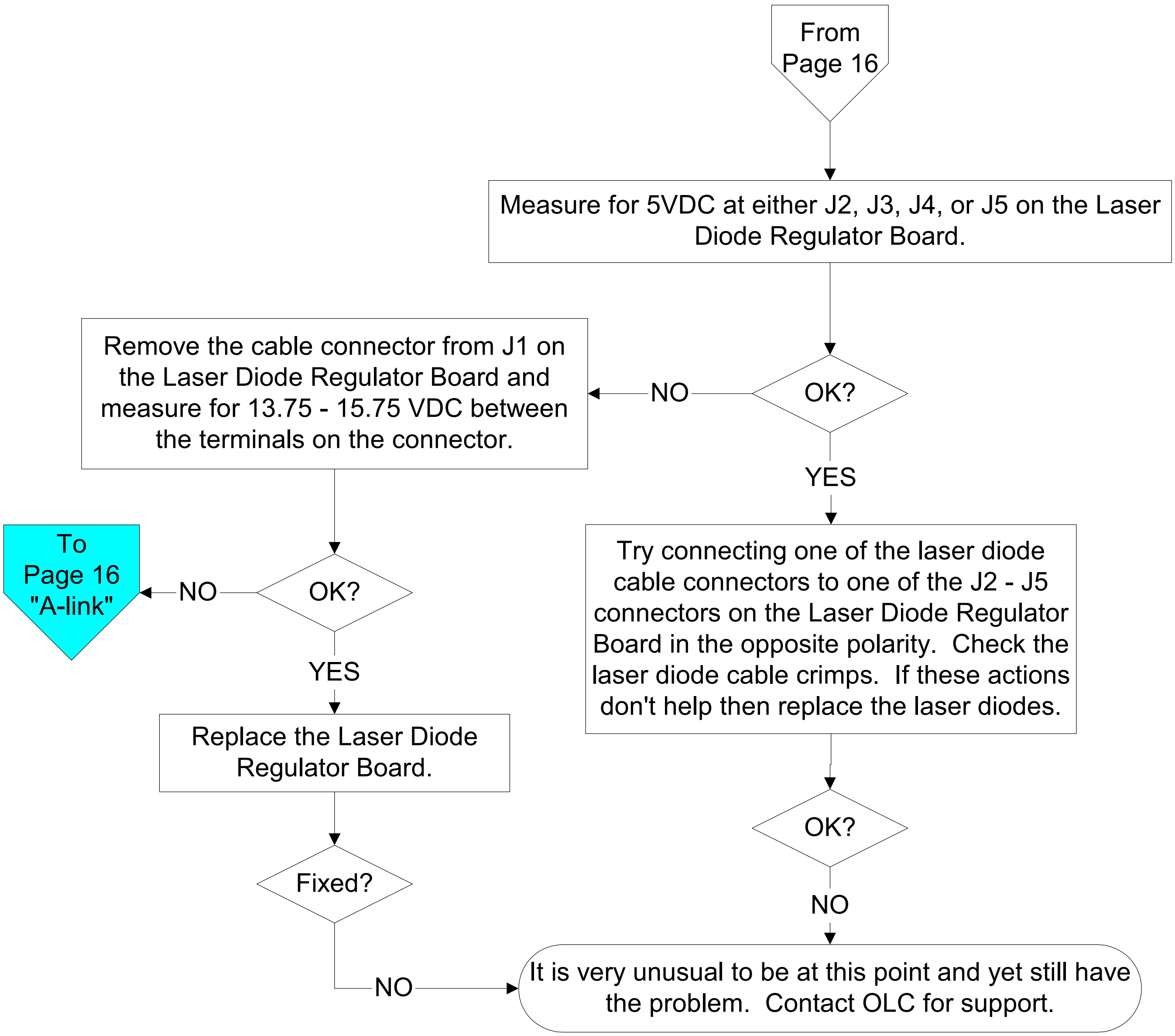
2-9 Patient Alignment Light Troubleshooting



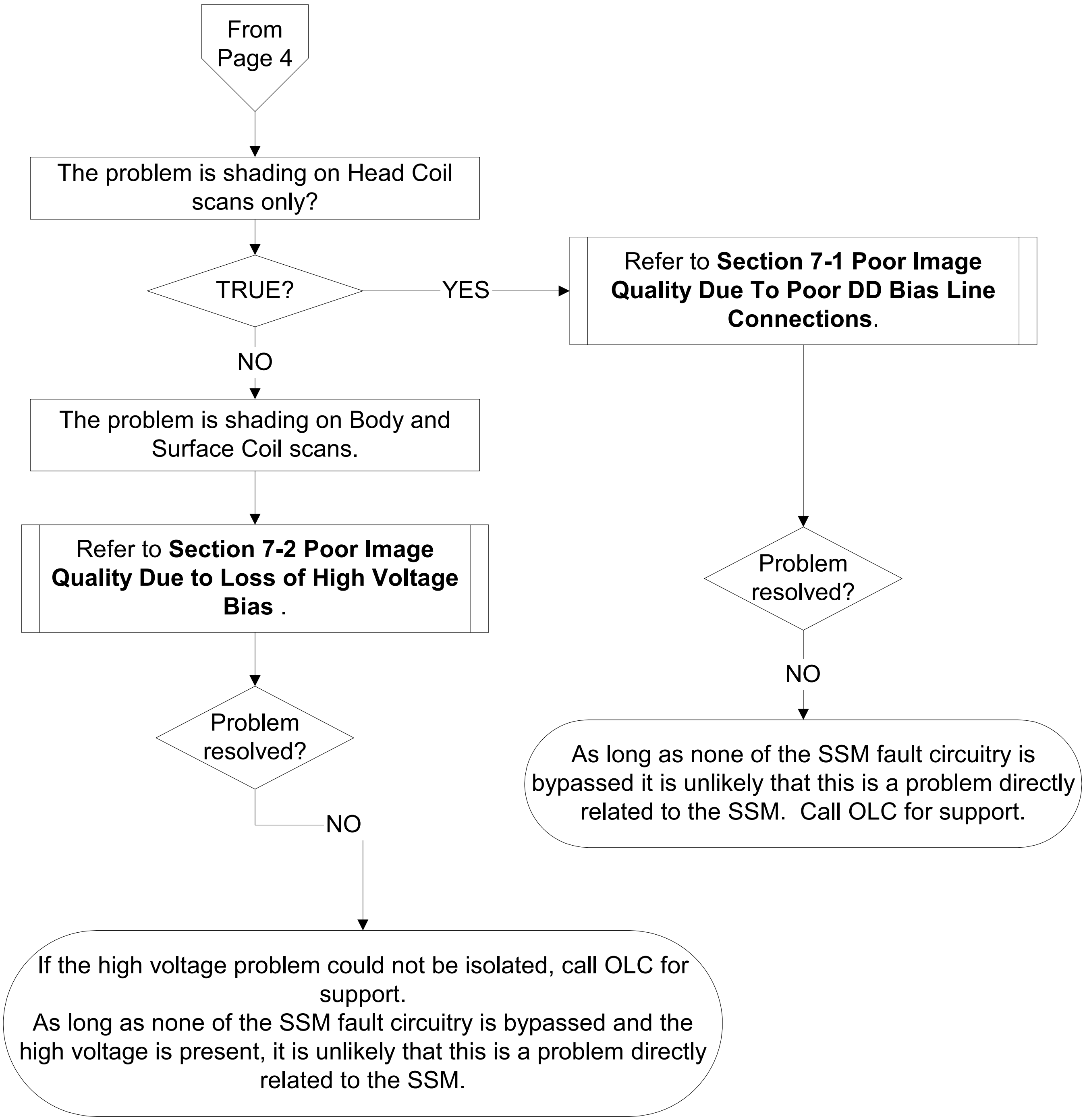
2-9 Patient Alignment Light Troubleshooting (Continued)



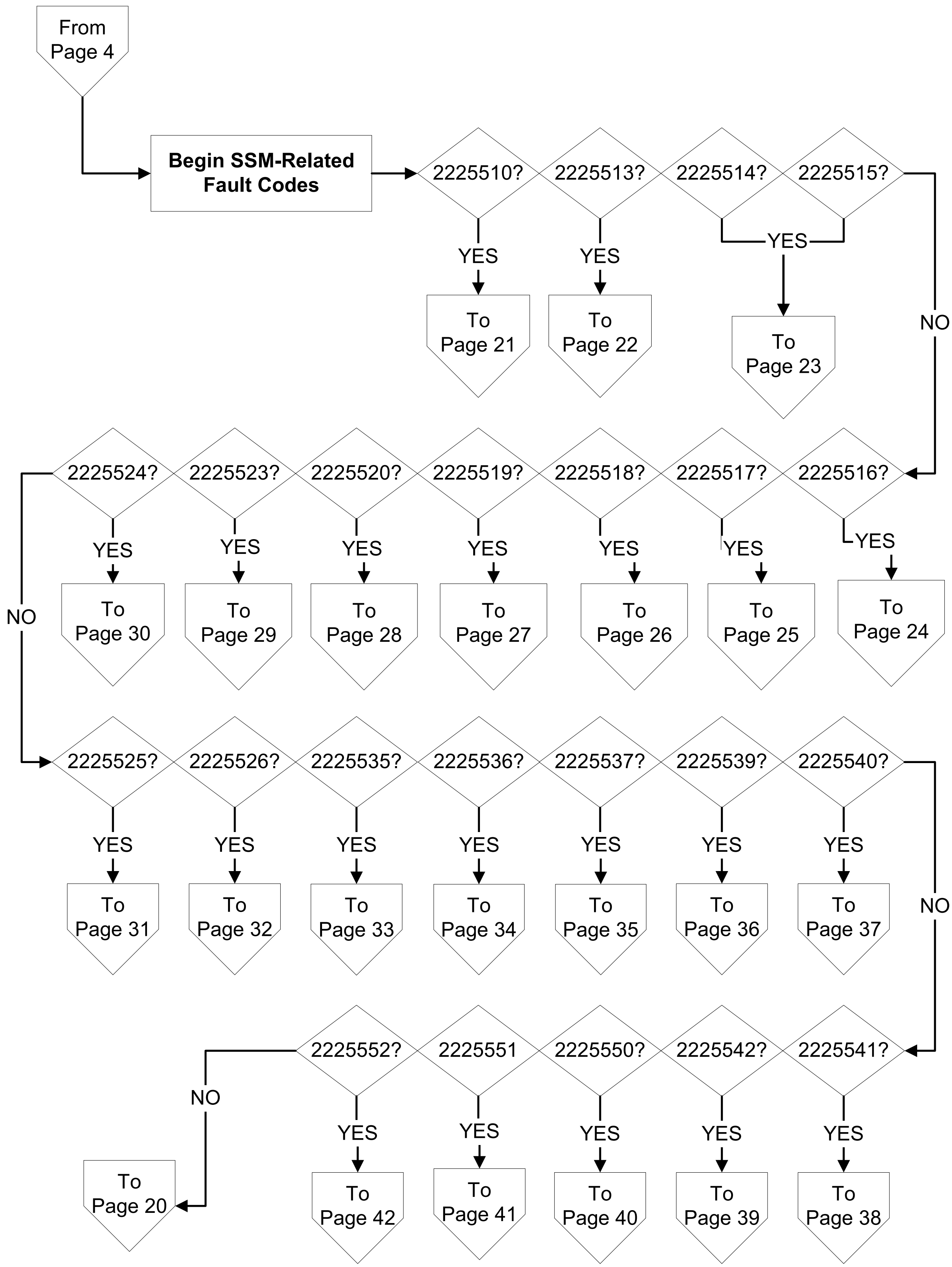
2-9 Patient Alignment Light Troubleshooting (Continued)



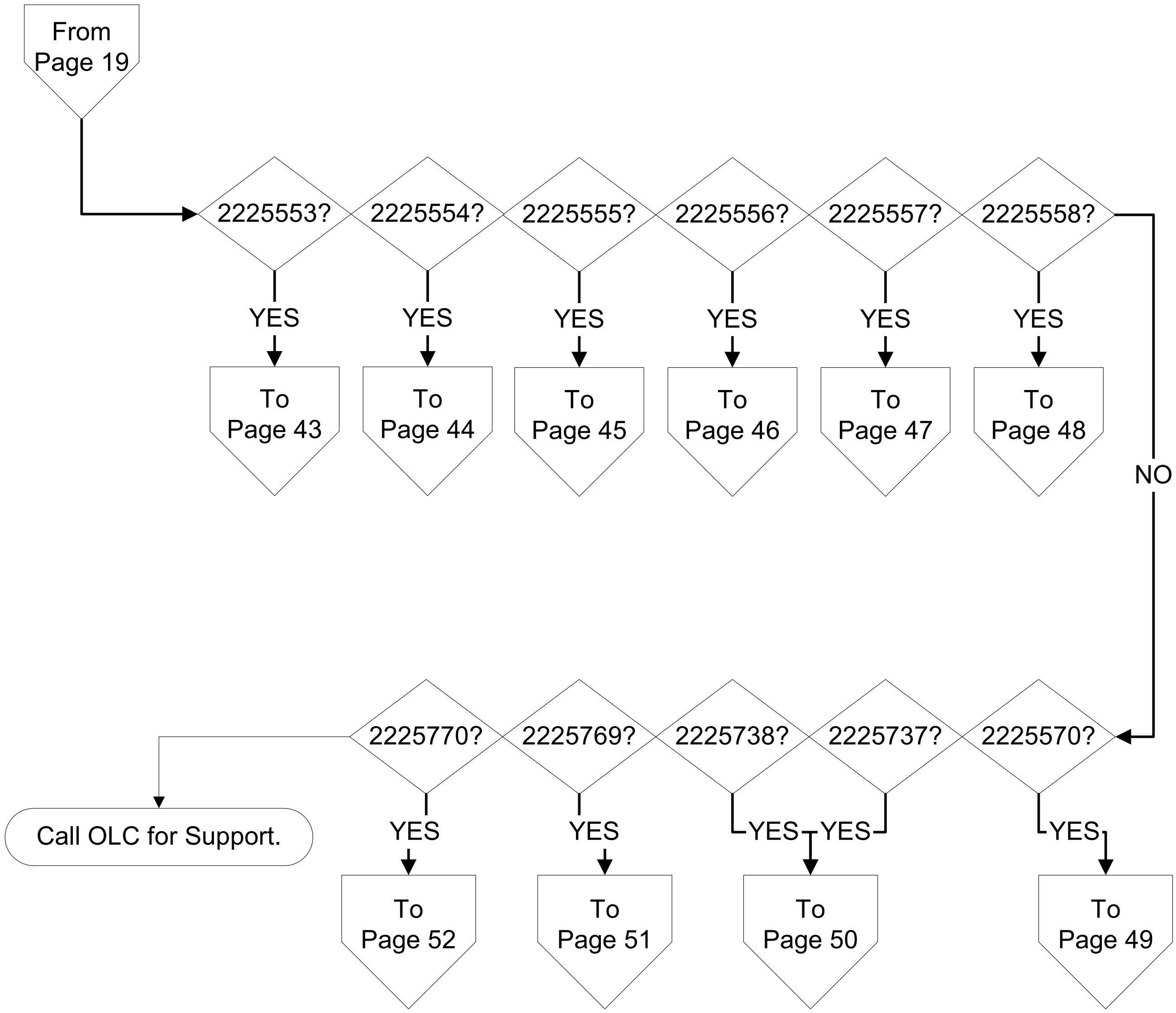
2-10 Troubleshooting Possible Image Quality Problems Related to SSM



2-11 SSM Fault Code Tree



2-11 SSM Fault Code Tree (Continued)



2-12 Error Message # 2225510

From Page 19

Error Message # 2225510:

Dynamic disable and T/R driver response timeout. T/R and dynamic disable communication link failure. Verify that the RF MDS peripheral is connected to the T/R driver assembly.

Check connection of MDS Link fiber optic link cables on rear of SSM:

1. Run 708 gray TX fiber optic cable
2. Run 709 blue RX fiber optic cable

Check that **CB4 SYSTEM SUPPORT MODULE** Circuit Breaker on rear of SSM is in up (ON) position.

Confirm that SSM is receiving AC power. Make certain PDU is not in Emergency Stop (E-Stop) mode.

Cycle power to the SSM and then reset TPS.

It is unusual for failure mode to still exist at this point. May want to check for other related errors in the log. May need to replace the CPD Board. Speak to OnLine Center before taking this action.

END

2-13 Error Message # 2225513

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Error Message # 2225513:

Attention: T/R driver in service, not product mode.

This is not an error message as much as it is an advisory message. It will not inhibit scanning but extensive damage to the Dynamic Disable, Inductive Drive Disable, or T/R Driver hardware can occur if the SSM is left in Service Mode during normal product scanning. This also presents a safety risk to patients. **Do not leave the SSM in Service Mode during normal product scanning.** Check the following DRV FLT Switches on the front of the SSM and make certain that each is in the up (**EN**able) position:

1. **HVON**
2. **MC**
3. **TR**
4. **DD**

OK?

NO

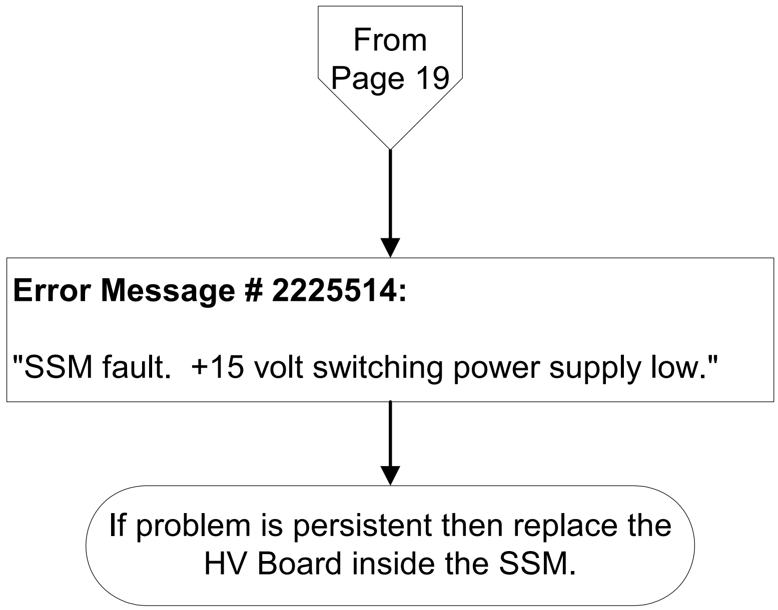
It is highly unusual to reach this point. Check the switches again! If necessary, replace the Front Panel Assembly Board inside the SSM.

OK?

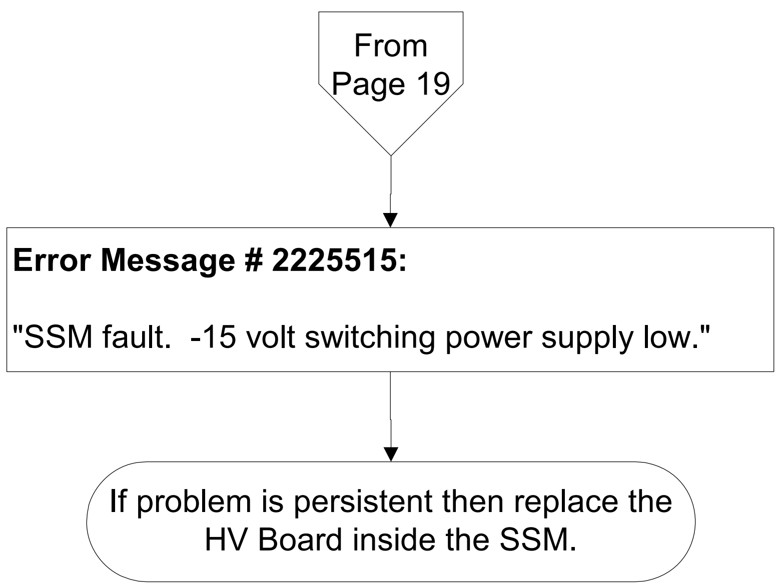
NO

Call OLC for support.

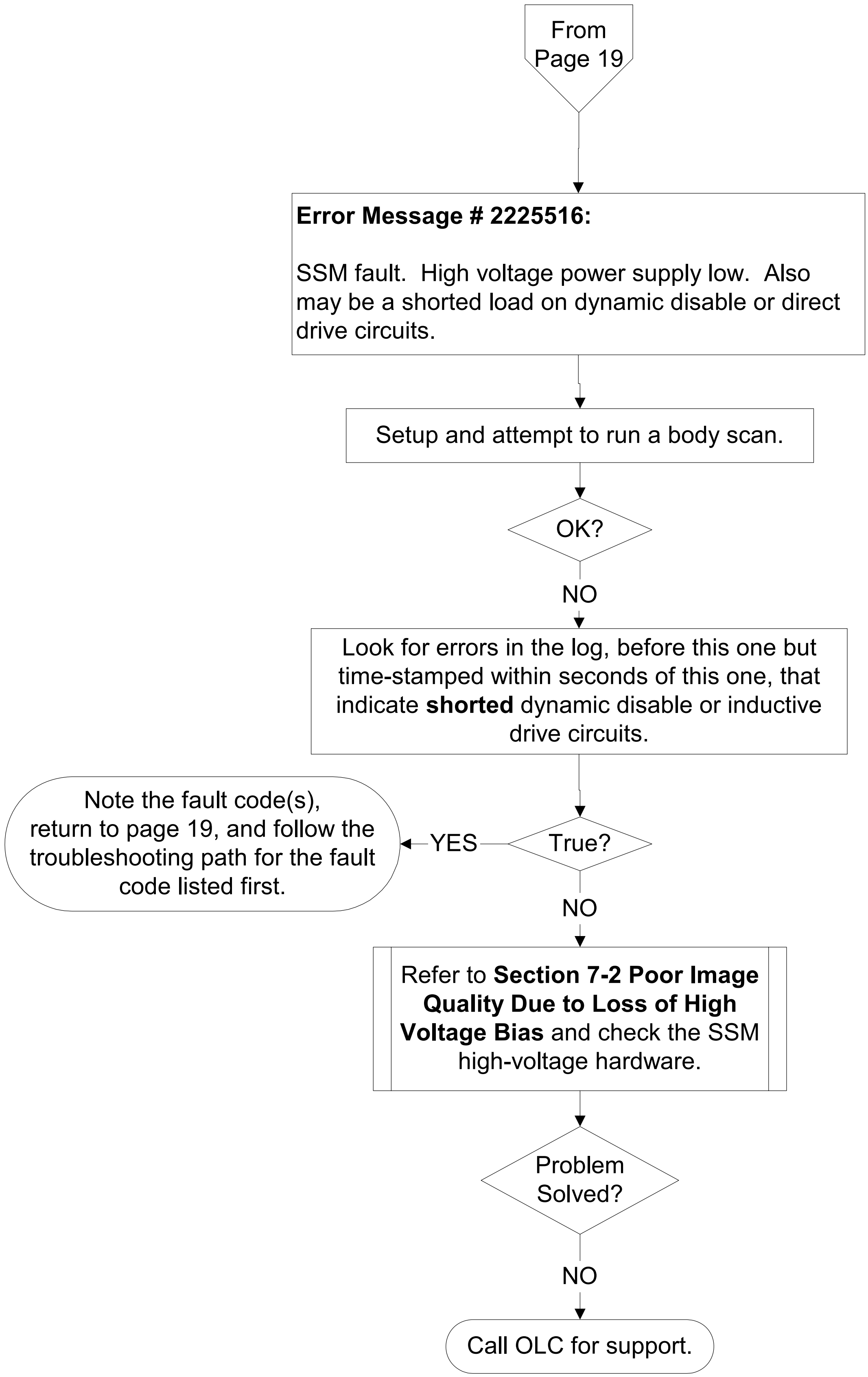
2-14 Error Message # 2225514



2-15 Error Message # 2225515



2-16 Error Message # 2225516



2-17 Error Message # 2225517

From Page 19

Error Message # 2225517:

T/R driver fault, head path open. Possibly a disconnected cable between the driver board and the head T/R network, or an open diode in the head T/R switch.

Check that the head transmission line is connected to the extension coax cable and that the extension coax cable is, in turn, connected to **J3 Head Output** on the rear of the RFI.

OK?

YES

Refer to **SECTION 8 - TROUBLESHOOTING TR DRIVER FAULTS**.

OK?

NO

Look carefully through the error log and make certain that there are no other errors indicating another problem in the log.

Carefully inspect the length of the data cable connecting to **J5 UNBLK EN** on the rear of the SSM and confirm it is not cut and damaged somewhere along the length of the cable.

If necessary, call OLC for support.

2-18 Error Message # 2225518

From Page 19

Error Message # 2225518:

T/R driver fault, head path short. Possibly a shorted diode in the head T/R switch network, or a short circuit load.

Refer to **SECTION 8 - TROUBLESHOOTING TR DRIVER FAULTS**.

OK?

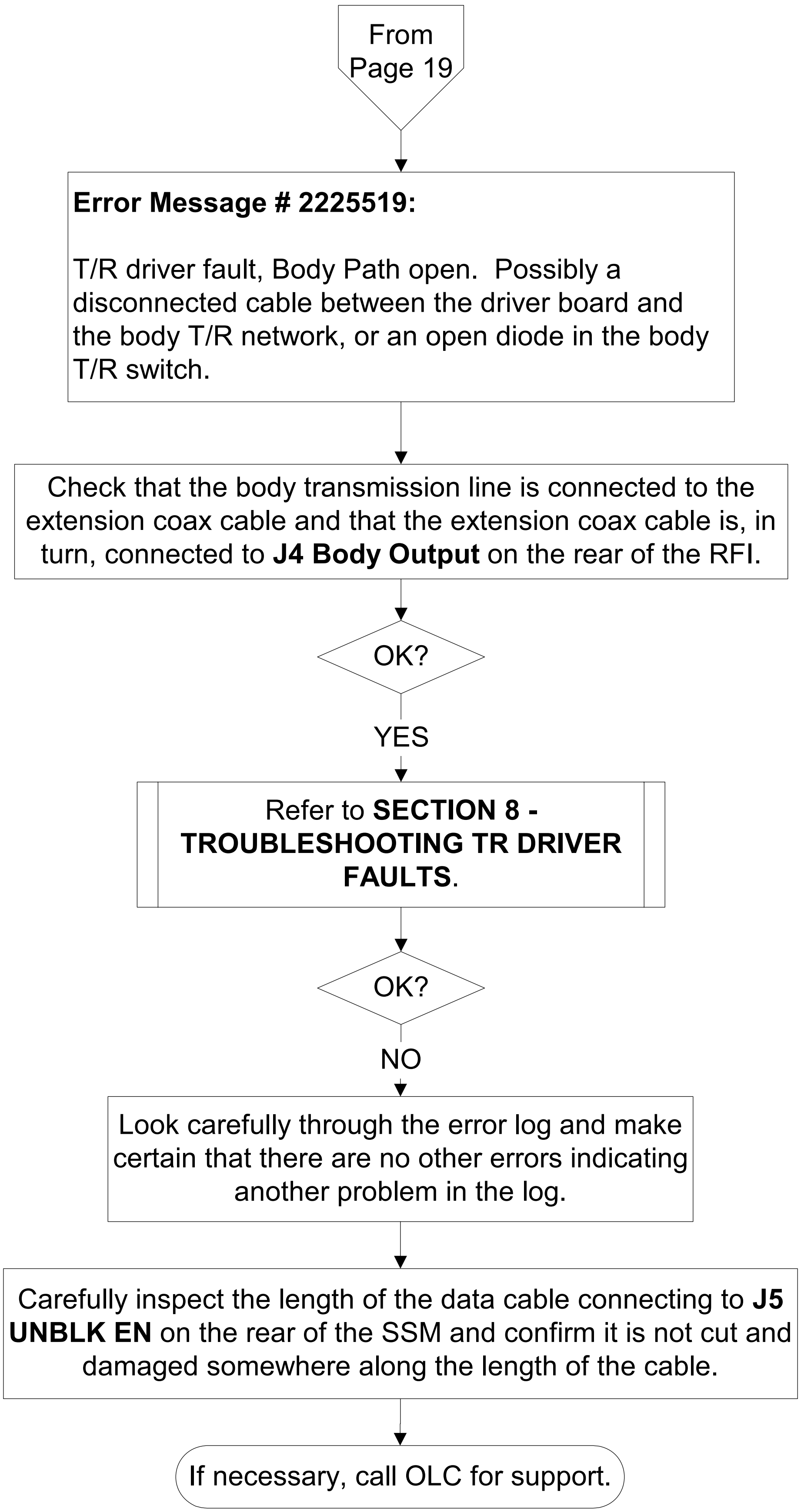
NO

Look carefully through the error log and make certain that there are no other errors indicating another problem in the log.

Carefully inspect the length of the data cable connecting to **J5 UNBLK EN** on the rear of the SSM and confirm it is not cut and damaged somewhere along the length of the cable.

If necessary, call OLC for support.

2-19 Error Message # 2225519



2-20 Error Message # 2225520

From Page 19

Error Message # 2225520:

T/R driver fault, Body Path short. Possibly a shorted diode in the body T/R switch network, or a short circuit load.

Refer to **SECTION 8 - TROUBLESHOOTING TR DRIVER FAULTS**.

OK?

NO

Look carefully through the error log and make certain that there are no other errors indicating another problem in the log.

Carefully inspect the length of the data cable connecting to **J5 UNBLK EN** on the rear of the SSM and confirm it is not cut and damaged somewhere along the length of the cable.

If necessary, call OLC for support.

2-21 Error Message # 2225523

From Page 19

Errror Message # 2225523:

T/R driver fault, spectro path open. Possibly a disconnected cable between the driver board and the spectro T/R network, or an open diode in the spectro T/R switch.

Check that the spectro transmission line is connected to **J45** on the exterior of the MNS Cabinet and to **J83** on the Penetration Panel.

OK?

YES

Refer to **SECTION 8 - TROUBLESHOOTING TR DRIVER FAULTS**.

OK?

NO

Look carefully through the error log and make certain that there are no other errors indicating another problem in the log.

Carefully inspect the length of the data cable connecting to **J5 UNBLK EN** on the rear of the SSM and confirm it is not cut and damaged somewhere along the length of the cable.

If necessary, call OLC for support.

2-22 Error Message # 2225524

From Page 19

Error Message # 2225524:

T/R driver fault, spectro path short. Possibly a shorted diode in the the spectro T/R switch network, or a short circuit load.

Refer to **SECTION 8 - TROUBLESHOOTING TR DRIVER FAULTS**.

OK?

NO

Look carefully through the error log and make certain that there are no other errors indicating another problem in the log.

Carefully inspect the length of the data cable connecting to **J5 UNBLK EN** on the rear of the SSM and confirm it is not cut and damaged somewhere along the length of the cable.

If necessary, call OLC for support.

2-23 Error Message # 2225525

From Page 19

Error Message # 2225525:

Inductive or Direct Drive fault, path open. Driver bias fault. Possibly a disconnected cable between the driver board and the driver switches, or an open diode in the driver switches.

Check that none of the cables in the Inductive Drive Disable Circuit are disconnected (**J44 DD** at the SSM, **J93** at the Equipment Room side of the High-Voltage Filter, **J74** on the Magnet Room side of the HV Filter, and **J6** on the Body Hybrid Splitter).

OK?

YES

Refer to **SECTION 9 - TROUBLESHOOTING DYNAMIC DISABLE AND INDUCTIVE DRIVE DISABLE FAULTS**.

OK?

NO

Look carefully through the error log and make certain that there are no other errors indicating another problem in the log.

Carefully inspect the length of the data cable connecting to **J5 UNBLK EN** on the rear of the SSM and confirm it is not cut and damaged somewhere along the length of the cable.

Call OLC for support.

2-24 Error Message # 2225526

From Page 19

↓

Error Message # 2225526:

Inductive or Direct Drive fault, path short. Drive bias fault. Possibly a shorted diode in the driver switches, or a short circuit load.

↓

Refer to **SECTION 9 - TROUBLESHOOTING DYNAMIC DISABLE AND INDUCTIVE DRIVE DISABLE FAULTS**.

↓

OK?

NO

↓

Look carefully through the error log and make certain that there are no other errors indicating another problem in the log.

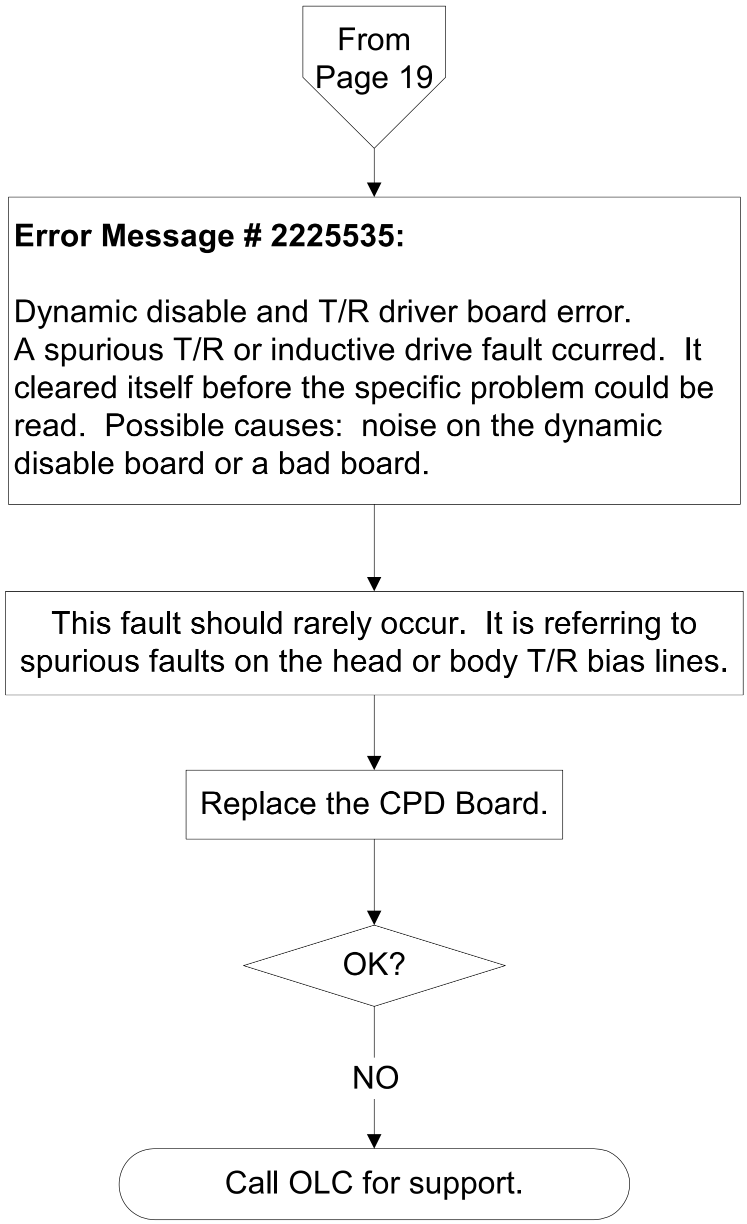
↓

Carefully inspect the length of the data cable connecting to **J5 UNBLK EN** on the rear of the SSM and confirm it is not cut and damaged somewhere along the length of the cable.

↓

If necessary, call OLC for support.

2-25 Error Message # 2225535



2-26 Error Message # 2225536

From Page 19

Error Message # 2225536:

Dynamic disable and T/R driver board error.
A spurious dynamic disable fault occurred. It cleared itself before the specific problem could be read. Possible causes: noise on the dynamic disable board or a bad board.

This fault is referring to spurious faults on one of the dynamic disable bias lines. This fault tends to be intermittent or only occur with certain sequences.

Try each of the suggestions below one at a time until the problem is solved:

- Try using wire or wire-braid to ground the shell of each of the J72, J73, J75, and J76 cable connectors that connect to the HV Filter to the Penetration Panel. The HV Filter is designed to isolate these from the Pen Panel.

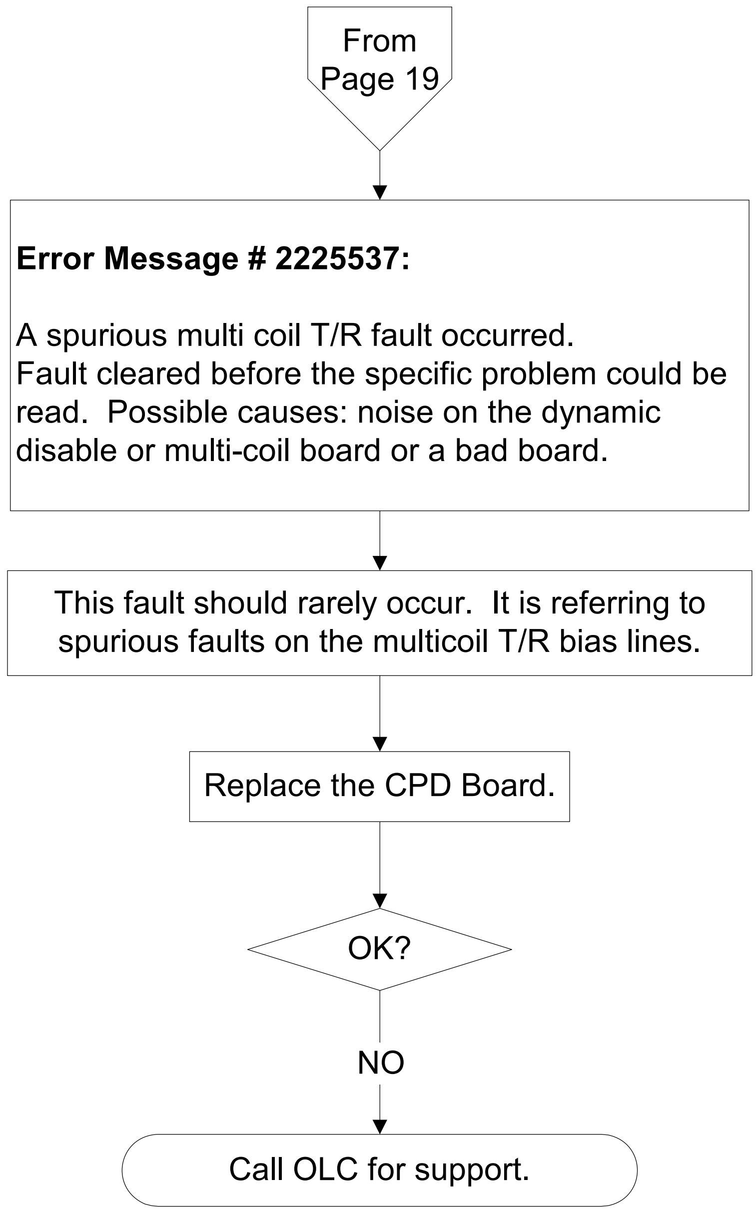
- Try swapping coax cables J72 (Run 778) with J75 (Run 781). Push the CAL button on the front of the SSM.

Problem solved?

NO

Write a CQA explaining specifics of issue (Under what conditions does fault occur? What has been done?) and contact OLC for support

2-27 Error Message # 2225537



2-28 Error Message # 2225539

From Page 19

Error Message # 2225539:

Dynamic disable fault, Front of Coil open.
Dynamic Disable bias fault on front drivers. Possibly a disconnected cable between the driver board and the dynamic disable switch boxes, or an open diode in dynamic disable switch boxes on FRONT of BRM.

Check that none of the cables in the Body 1 circuit are disconnected (**J42 Body 1** at the SSM, **J92** at the Equipment Room side of the High-Voltage Filter, **J72** and **J73** on the Magnet Room side of the HV Filter, and **J3/J5** and **J4/J6** at the front of the RF Body Coil.

OK?

YES

Refer to **SECTION 9 - TROUBLESHOOTING DYNAMIC DISABLE AND INDUCTIVE DRIVE DISABLE FAULTS**.

OK?

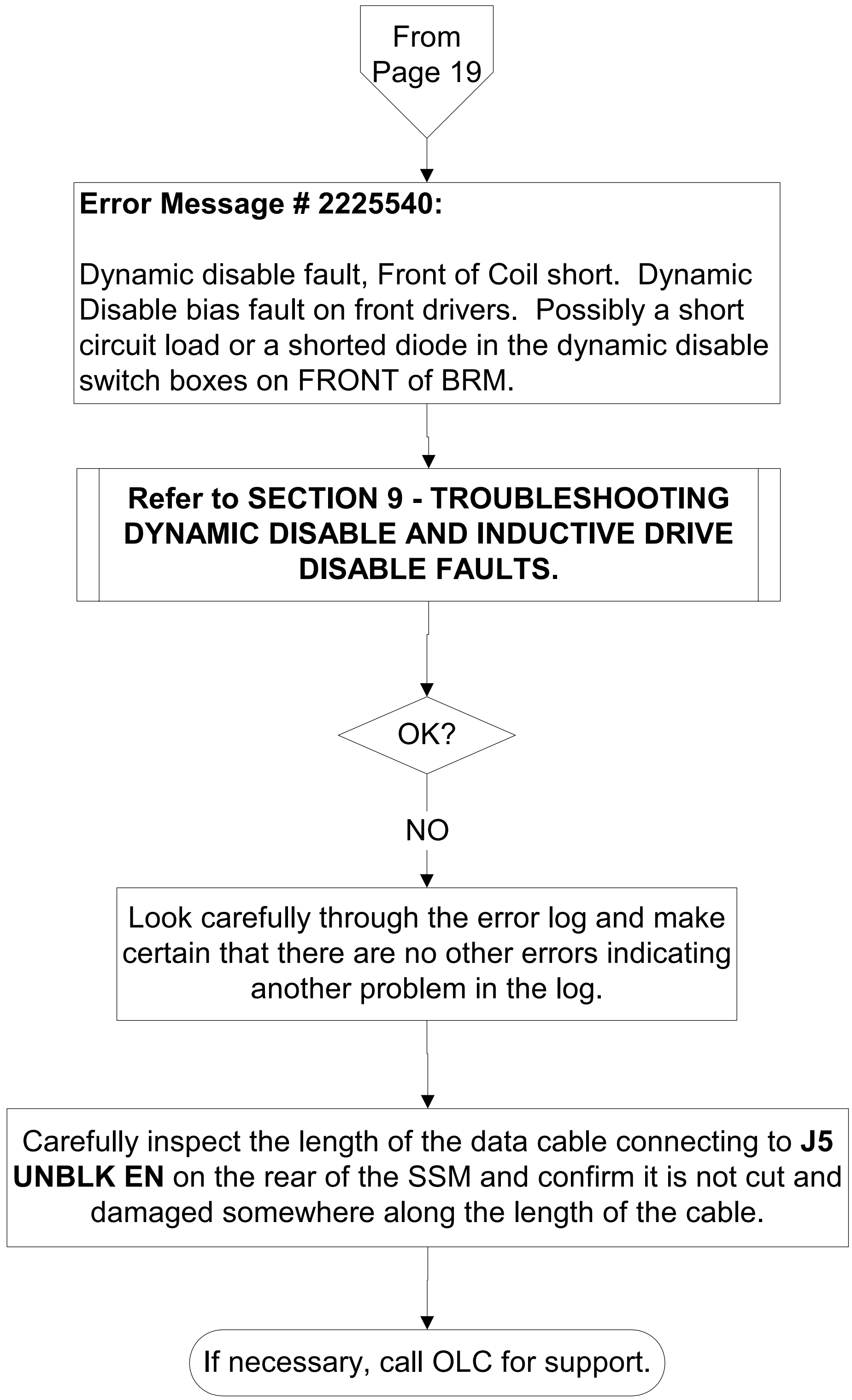
NO

Look carefully through the error log and make certain that there are no other errors indicating another problem in the log.

Carefully inspect the length of the data cable connecting to **J5 UNBLK EN** on the rear of the SSM and confirm it is not cut and damaged somewhere along the length of the cable.

Call OLC for support.

2-29 Error Message # 2225540



2-30 Error Message # 2225541

From Page 19

Error Message # 2225541:

Dynamic disable fault, Rear of Coil open.
Dynamic Disable bias fault on back drivers. Possibly a disconnected cable between the driver board and the dynamic disable switch boxes, or an open diode in dynamic disable switch boxes on BACK of BRM.

Check that none of the cables in the Body 2 circuit are disconnected (**J43 Body 2** at the SSM, **J94** at the Equipment Room side of the High-Voltage Filter, **J75** and **J76** on the Magnet Room side of the HV Filter, and **J3/J5** and **J4/J6** at the rear of the RF Body Coil.

OK?

YES

Refer to SECTION 9 - TROUBLESHOOTING DYNAMIC DISABLE AND INDUCTIVE DRIVE DISABLE FAULTS.

OK?

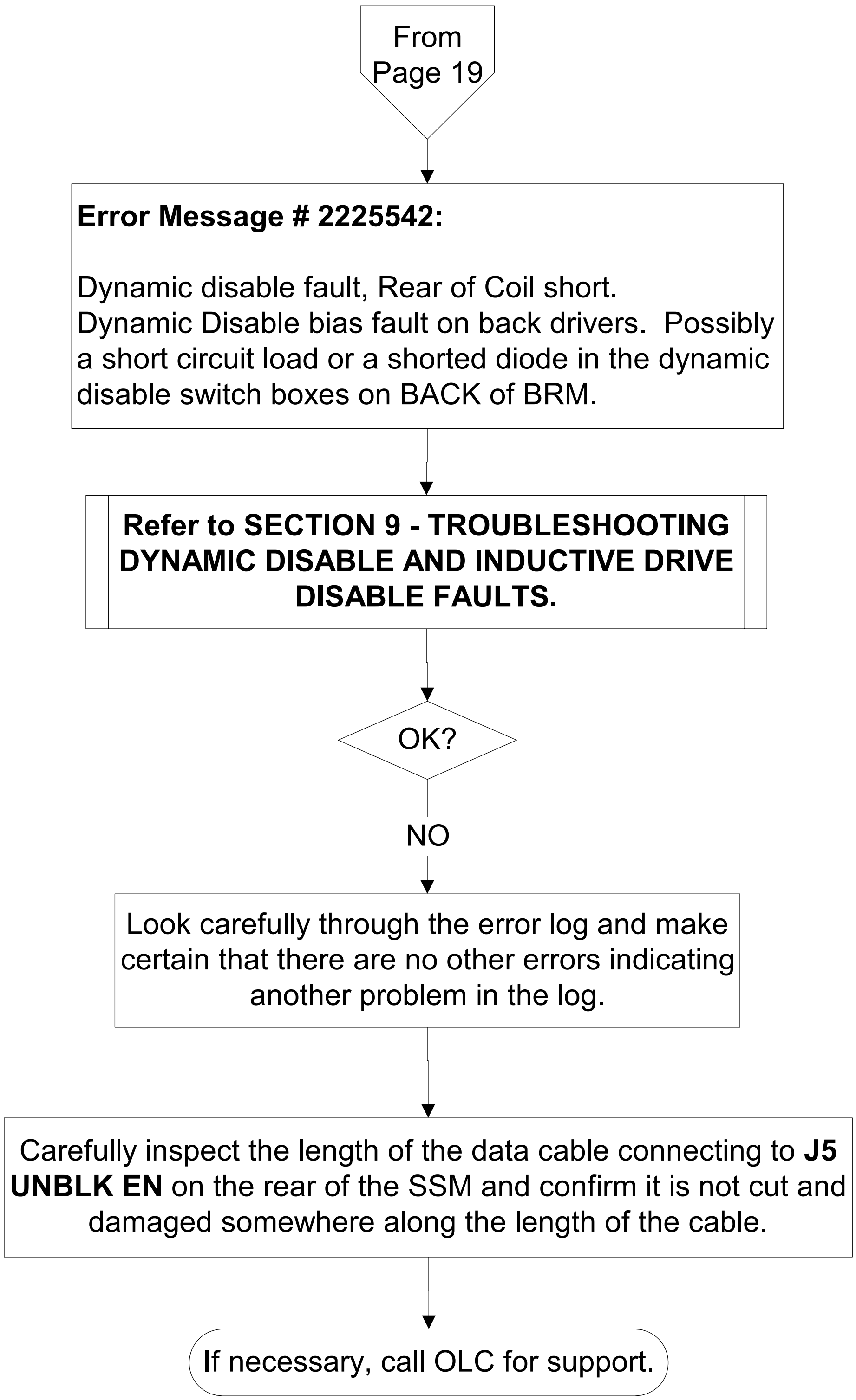
NO

Look carefully through the error log and make certain that there are no other errors indicating another problem in the log.

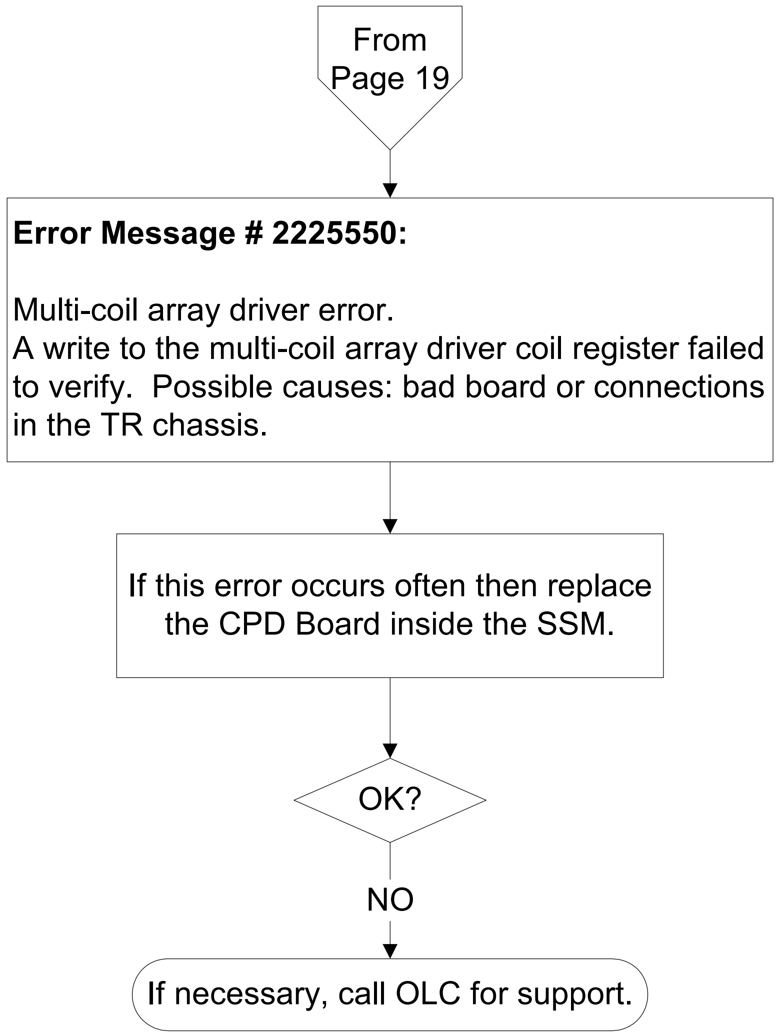
Carefully inspect the length of the data cable connecting to **J5 UNBLK EN** on the rear of the SSM and confirm it is not cut and damaged somewhere along the length of the cable.

Call OLC for support.

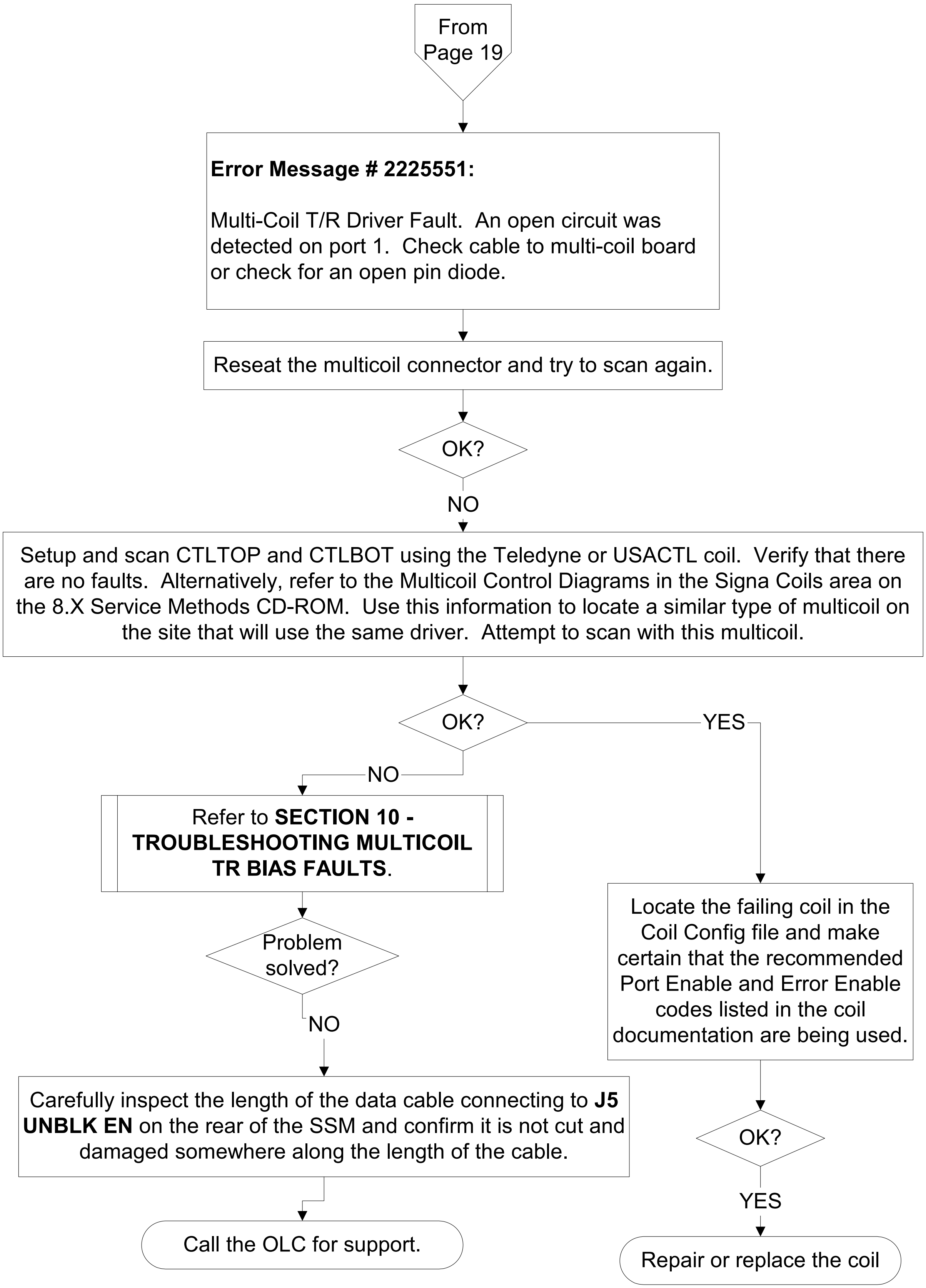
2-31 Error Message # 2225542



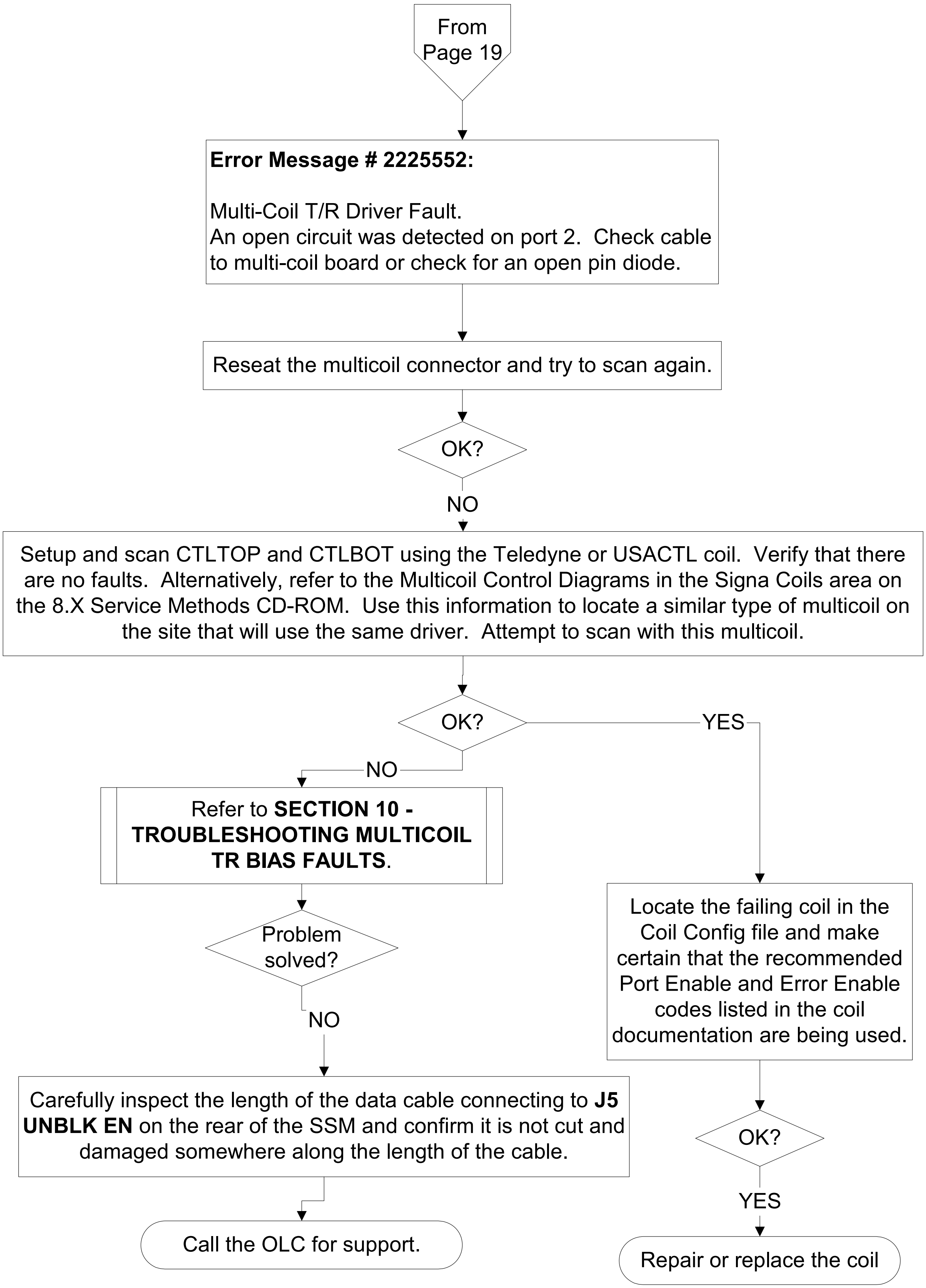
2-32 Error Message # 2225550



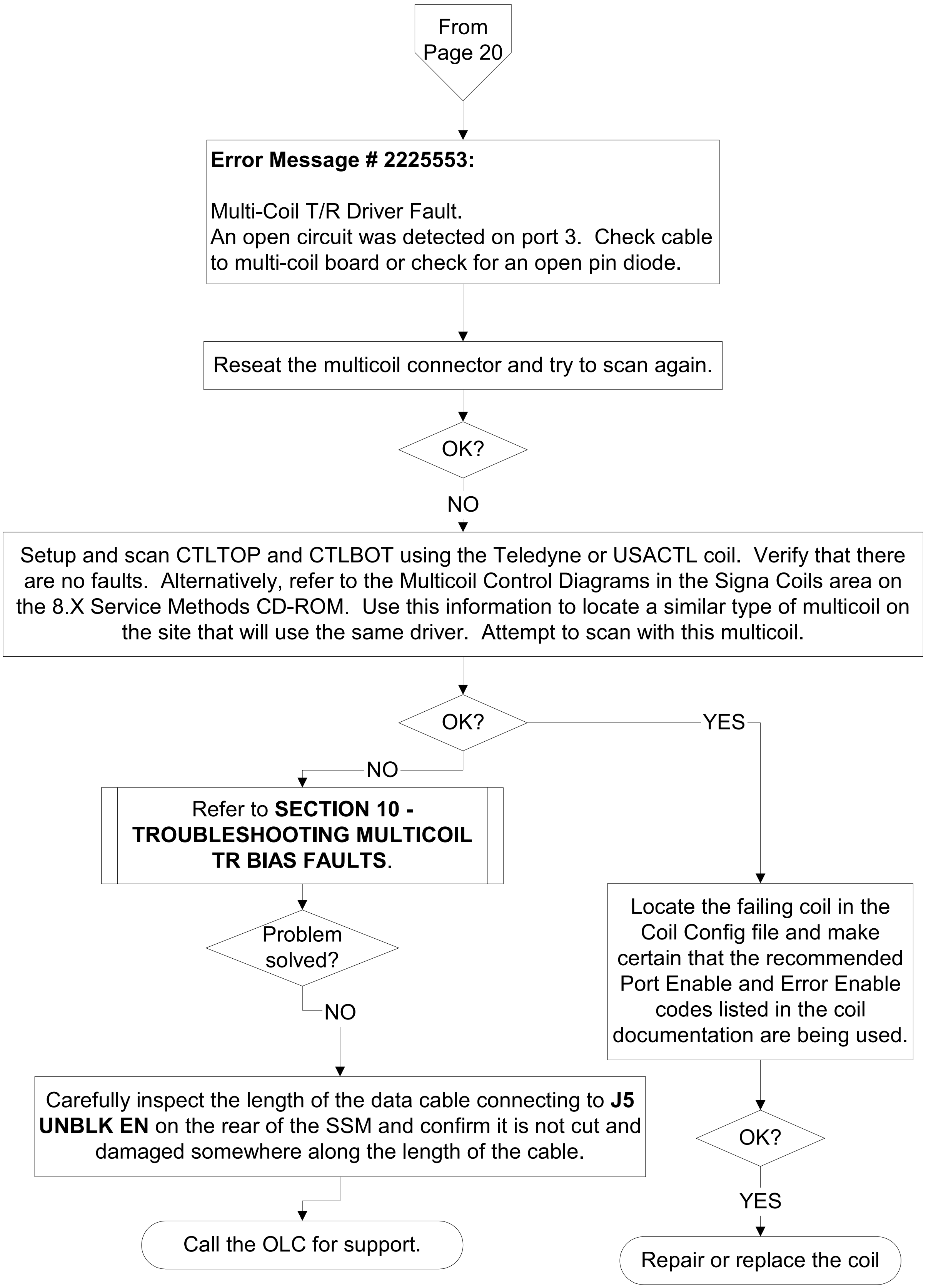
2-33 Error Message # 2225551



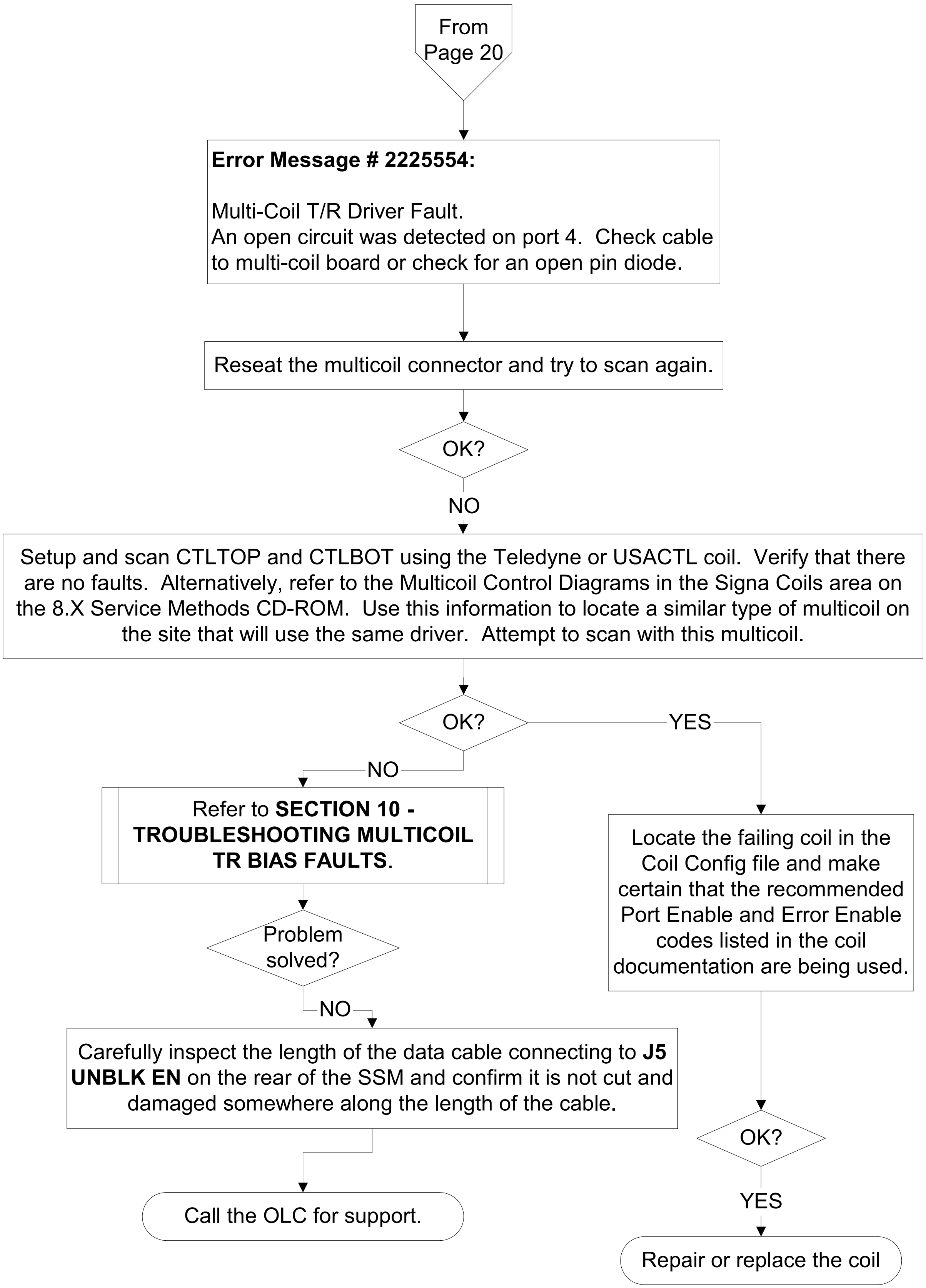
2-34 Error Message # 2225552



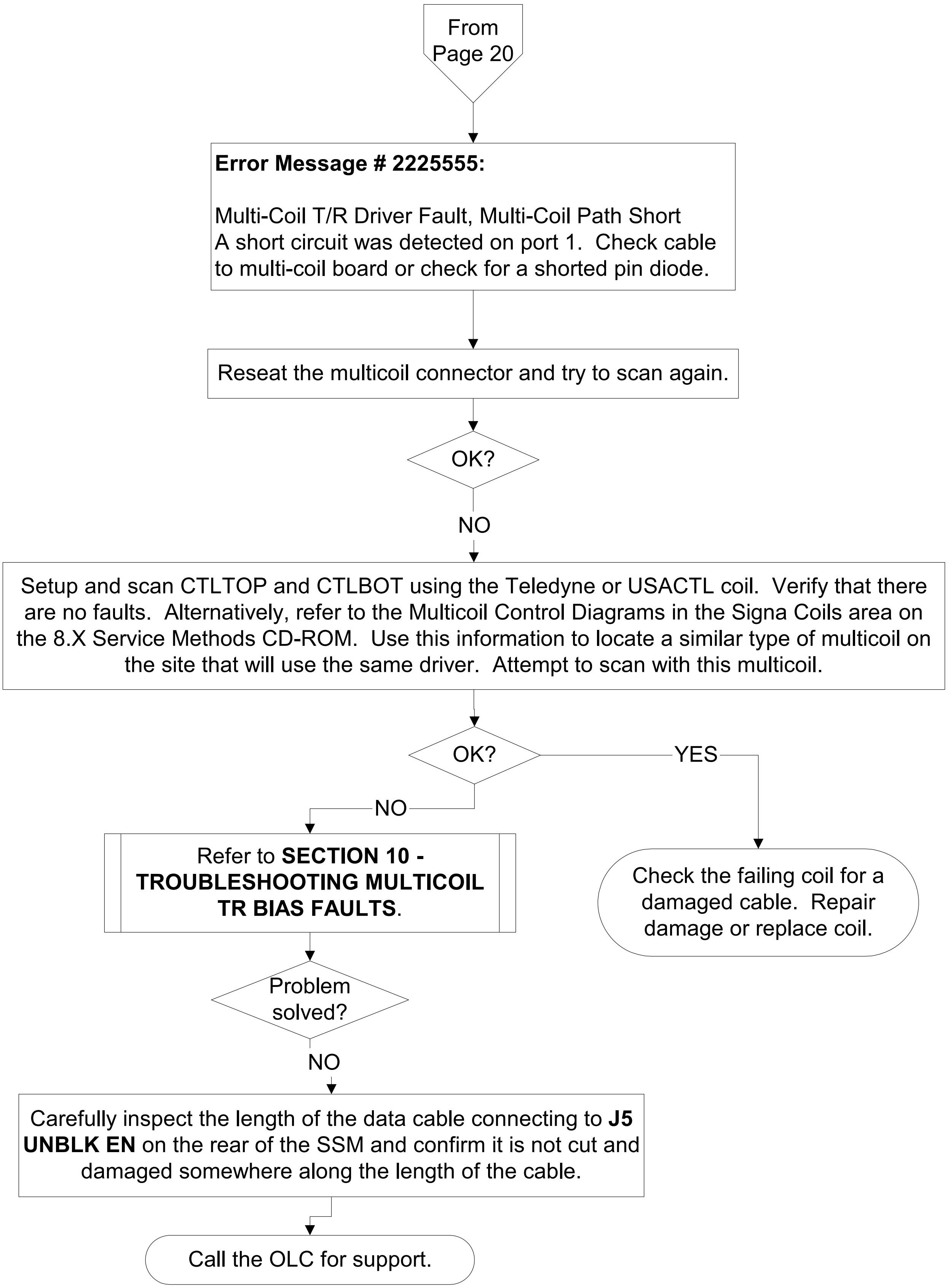
2-35 Error Message # 2225553



2-36 Error Message # 2225554



2-37 Error Message # 2225555



2-38 Error Message # 2225556

From Page 20

Error Message # 2225556:

Multi-Coil T/R Driver Fault, Multi-Coil Path Short.
A short circuit was detected on port 2. Check cable to multi-coil board or check for a shorted pin diode.

Reseat the multicoil connector and try to scan again.

OK?

NO

Setup and scan CTLTOP and CTLBOT using the Teledyne or USACTL coil. Verify that there are no faults. Alternatively, refer to the Multicoil Control Diagrams in the Signa Coils area on the 8.X Service Methods CD-ROM. Use this information to locate a similar type of multicoil on the site that will use the same driver. Attempt to scan with this multicoil.

OK?

YES

Check the failing coil for a damaged cable. Repair damage or replace coil.

NO

Refer to **SECTION 10 - TROUBLESHOOTING MULTICOIL TR BIAS FAULTS**.

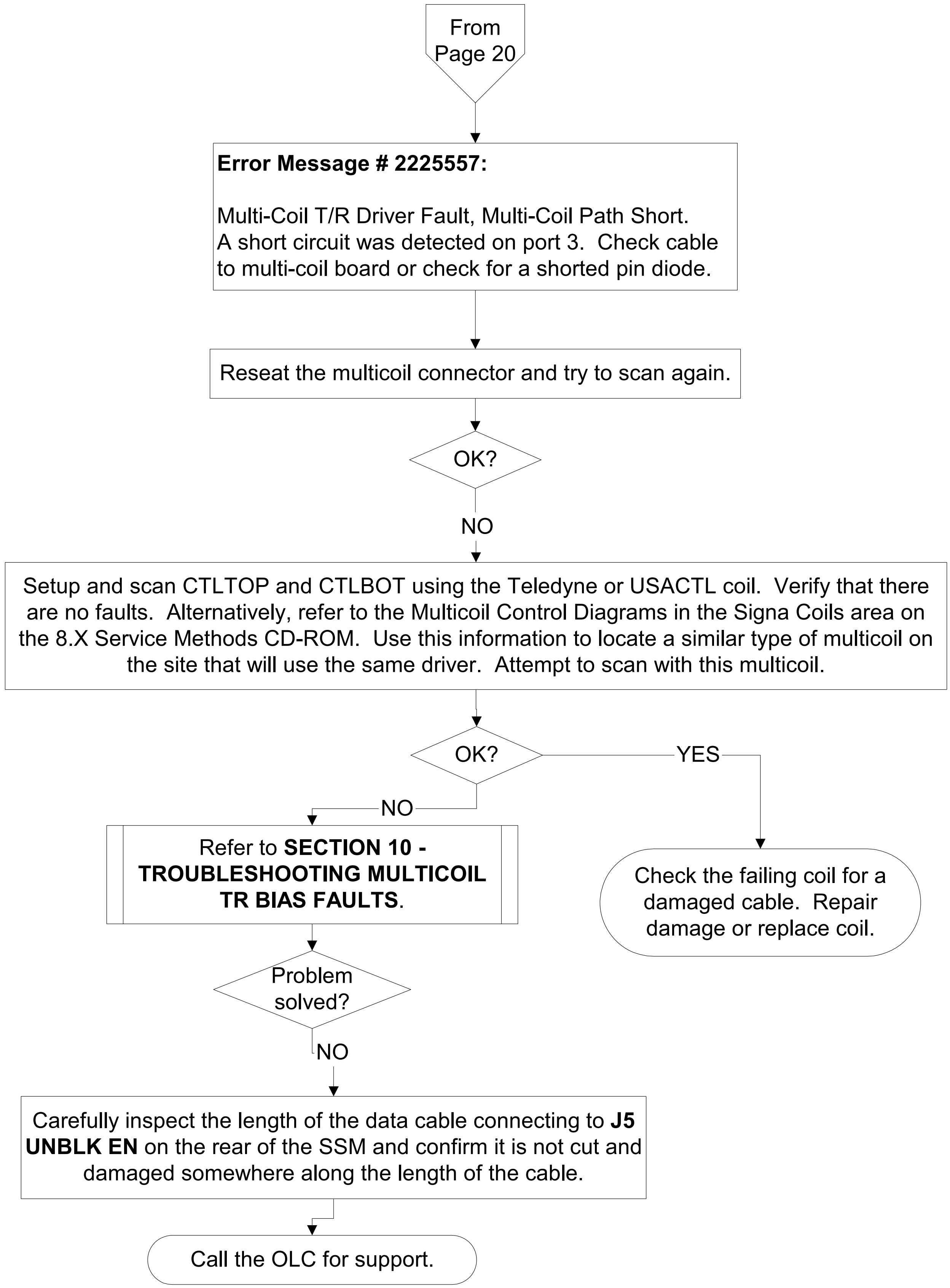
Problem solved?

NO

Carefully inspect the length of the data cable connecting to **J5 UNBLK EN** on the rear of the SSM and confirm it is not cut and damaged somewhere along the length of the cable.

Call the OLC for support.

2-39 Error Message # 2225557



2-40 Error Message # 2225558

From Page 20

Error Message # 2225558:

Multi-Coil T/R Driver Fault, Multi-Coil Path Short.
A short circuit was detected on port 4. Check cable to multi-coil board or check for a shorted pin diode.

Reseat the multicoil connector and try to scan again.

OK?

NO

Setup and scan CTLTOP and CTLBOT using the Teledyne or USACTL coil. Verify that there are no faults. Alternatively, refer to the Multicoil Control Diagrams in the Signa Coils area on the 8.X Service Methods CD-ROM. Use this information to locate a similar type of multicoil on the site that will use the same driver. Attempt to scan with this multicoil.

OK?

YES

Check the failing coil for a damaged cable. Repair damage or replace coil.

NO

Refer to **SECTION 10 - TROUBLESHOOTING MULTICOIL TR BIAS FAULTS**.

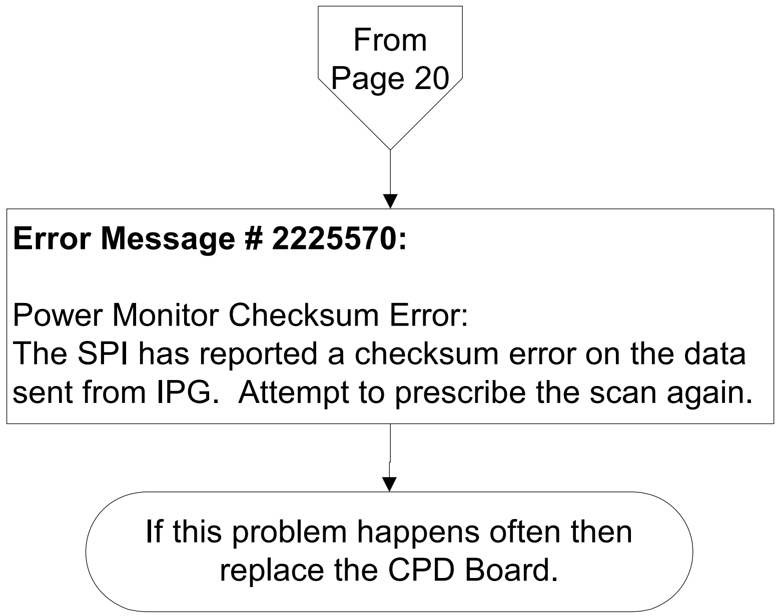
Problem solved?

NO

Carefully inspect the length of the data cable connecting to **J5 UNBLK EN** on the rear of the SSM and confirm it is not cut and damaged somewhere along the length of the cable.

Call the OLC for support.

2-41 Error Message # 2225570



2-42 Error Messages 2225737 & 2225738

From Page 20

Error Message # 2225737:

Power Monitor 1 Fault.
RF Transmit power has reached safety limit allowed for scan parameters and patient weight.
Check patient weight entered.
amplitude: XX
pulse width: XX
duty cycle: XX
<amplitude, pulse width, duty cycle cause here>
This could also occur when power cycling SSM.

Error Message # 2225738:

Power Monitor 2 Fault.
RF Transmit power has reached safety limit allowed for scan parameters and patient weight.
Check patient weight entered.
amplitude: XX
pulse width: XX
duty cycle: XX
<amplitude, pulse width, duty cycle cause here>
This could also occur when power cycling SSM.

Recommended Actions:

- If both monitors are faulting together then it is likely that the power monitor hardware is functioning properly.
- Check patient weight value entered by customer for scan and make sure it has not been significantly under-reported.
- Have customer pad patient extremities away from sides of tube so that there is at least 0.5 inches (1.27 cm) distance between patient and side of tube.
- If site has Multi-nuclear Spectroscopy (MNS) Hardware make certain that MNS Cabinet cabling is connected and MNS Cabinet is powered on.
- Check TG from customer scans or SPT and make sure it has not increased from what it is normally due to a system hardware problem. Investigate the cause of any substantial increases in TG.
- If power monitor faults only happen when using head coil then replace white head coil transmit cable (connects to female BNC outlined with blue circle on QD Box) and check Head Coil, Head TR Nework for damage.
- Check Config File Mgr and make certain Start TA, Coil Loss, and Cable Loss values for the coil in use when the fault happens are correct.
- Check RF output power and make sure it is within specifications. Calibrate the RFI if the RF output power is not in specifications.
- Run **proprietary** power monitor functional checks and use the field laptop and the MONS1.EXE software to check that the power monitors (APM Board inside SSM) are in good working order. During the functional check verify that there is not excessive RF leakage from RFI Body Sense ports A and B to Head Sense ports A and B. Replace the RFI if RF leakage is excessive.

2-43 Error Message # 2225769

From Page 20

Error Message # 2225769:

Power Monitor may have tripped due to the Spectro RF Amp being powered off. Check Spectro RF Amp power, or reconfigure as a non-Spectro system and reboot the system.

Above message self-explanatory. Perform actions and try scan again. If necessary, call OLC for support.

2-44 Error Message # 2225770

From Page 20

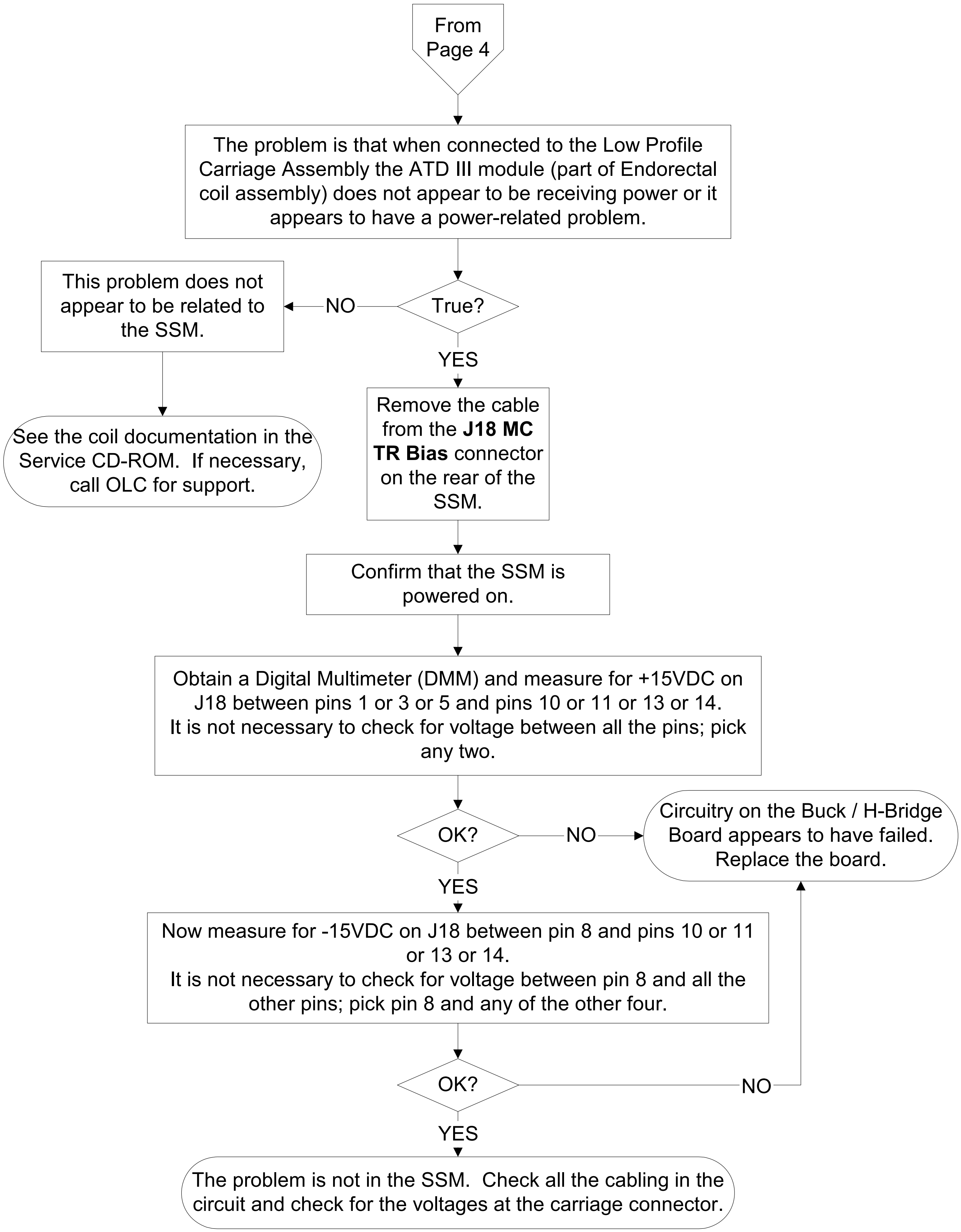
Error Message # 2225770:

Possible power monitor hardware failure: power sensed with monitor blanked, DC rail to power monitor, or +15V sense failed. Spectro RF may be powered off. Check Spectro RF Amp power, or reconfigure as a non-Spectro system and reboot the system.

Make certain that actions described above have been performed.

If problem persists then replace the APM Board.

2-45 ATD III Troubleshooting



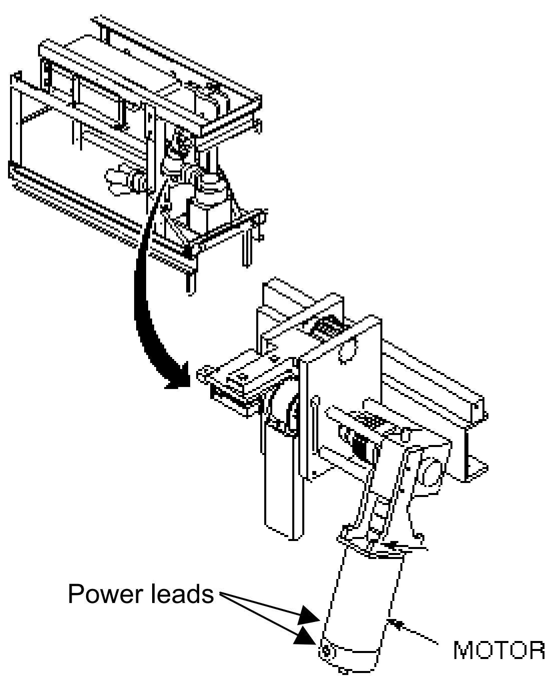
3 – TROUBLESHOOTING LONGITUDINAL DRIVE FAILURES

This section is divided into two parts. **Section 3-1 Troubleshooting Lack of Carriage Motion** should be performed if, when the FAST IN buttons on the Magnet Enclosure are pressed, the Longitudinal Drive Motor cannot be heard rotating and, as a result, the Carriage does not move. If the problem was not isolated in the previous section or if the motor rotates but carriage motion is erratic or poor then also perform the actions in **Section 3-2 Troubleshooting Carriage Motion Problems**. This section will assist the FE with determining if carriage motion / longitudinal drive motor rotation problems are due to components within the SSM. This process requires two Field Engineers. The FEs may also want to reference the System Support Module block diagrams located in the TPS/ISE tab of **Signa Release 8.X Block Diagrams & Supplemental Schematics (Direction 2153389)** from the Service CD-ROM during this process.

3-1 Troubleshooting Lack of Carriage Motion

Perform the steps in this section if the Longitudinal Drive Motor cannot be heard rotating when the FAST IN buttons on the Magnet Enclosure are pressed and, as a result, the Carriage does not move. Skip this section if the Carriage moves but the motion is poor. If in doubt, perform both sections.

1. Remove the right cover from the rear pedestal to expose the longitudinal drive motor. See Illustration 3-1.



RIGHT COVER REMOVED TO EXPOSE LONGITUDINAL DRIVE MOTOR
ILLUSTRATION 3-1

2. Confirm that the two spade lead power connectors are firmly attached to the spade lugs on either side of the longitudinal motor. See Illustration 3-1.

3-1 Troubleshooting Lack of Carriage Motion (Continued)

3. Verify that the sub-D connector labeled Run 318 J11 is firmly connected to J11 at the magnet room side of the penetration panel.

4. Verify that the sub-D connector labeled Run 770 J11 is firmly connected to J11 at the equipment room side of the penetration panel.

5. Verify that the sub-D connector labeled Run 770 J41 is firmly connected to the **J41 LONG AMP** connector on the rear of the SSM.

6. Locate the SRI Module.

    a. **LCC Magnet with old-style Horizon covers:** Slide front cover forward to expose the SRI Module mounted on the front of magnet face.

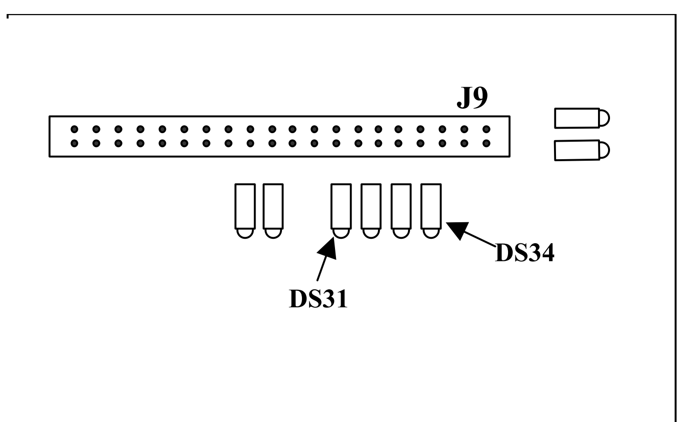
    b. **LCC Magnet with Wide-Open Covers:** Remove side skirt from magnet and slide sled fixture containing SRI Module out from under magnet.

    c. **All other magnet types:** Remove right side cover from magnet to expose SRI Module mounted on cover slide rail.

7. Loosen 5 screws on the end of the SRI Module slide cover and carefully remove any copper tape securing the cover to the module.

8. Once the copper tape is removed, slide cover to left in order to expose the main circuit board in the SRI Module.

9. Refer to Illustration 3-2. Look in the upper-right corner of the SRI circuit board and locate LED **DS31** (UPLIM). Verify that with the table in the full-up position this LED is illuminated. **APPENDIX B – IDENTIFICATION OF SRI LEDS** contains more information concerning the SRI LEDS.



LOCATION OF LEDS INSIDE SRI
ILLUSTRATION 3-2

3-1 Troubleshooting Lack of Carriage Motion (Continued)

NOTE

The DOCKSTP signal, which corresponds to LED **DS34**, is no longer monitored by the Horizon LX SRI.

10. If LED **DS31** is not illuminated:

    a. Check the cable connections on either side of Run 421 between J1 on the Dock Limit Switch Box and J3 on the SRI.

    b. Check cable connection between cable from Dock Assembly and J2 on the Dock Limit Switch Box.

DANGER!!

FERROUS MATERIAL HAZARD!! THE DOCK ASSEMBLY CONTAINS A MOTOR MADE WITH FERROUS COMPONENTS. DO NOT DISASSEMBLE THE DOCK ASSEMBLY INSIDE THE MAGNET ROOM. INSTEAD, REFER TO SECTION 3 – DOCK COMPONENTS IN THE SRI, DOCK HARDWARE (ME1REA1.DOC) DOCUMENT AND REMOVE THE DOCK FROM THE ROOM BEFORE DISASSEMBLY AS DESCRIBED IN THE PROCEDURE.

    c. Check the wiring to and the functionality of the Up-Limit Switch inside the dock. This switch contacts with a plunger on the front of the table and allows the system to tell when the table is in the full-up position.

11. Remove the cover from the Dock Limit Switch Box to expose the Dock Limit Switch Board.

12. Locate LED DS15 (8 VDC supply) in the upper-left side of the Dock Limit Switch Board and confirm it is illuminated. This LED will be illuminated if the Dock Limit Switch Board is receiving 8 VDC power from the SRI.

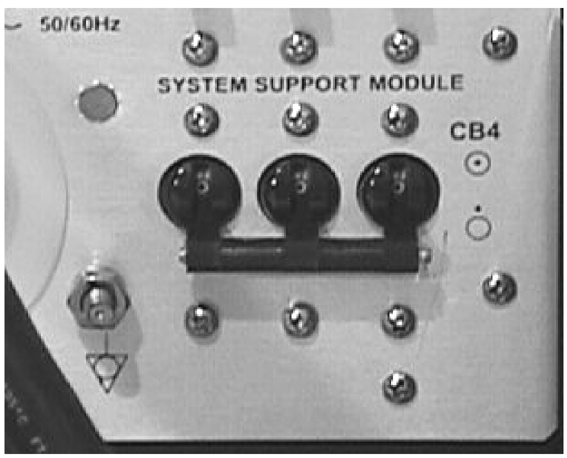
3-2 Troubleshooting Carriage Motion Problems

Perform the steps in this section if the steps in **Section 3-1 Troubleshooting Lack of Carriage Motion** were already performed or if the Carriage moves but has poor motion. This section assumes that the FE has already verified that the Drive Belt was correctly routed and that the Longitudinal Motor Drive hardware in the Rear Pedestal is correctly adjusted and functioning properly.

1. Locate the **CB4 SYSTEM SUPPORT MODULE** Circuit Breaker in the lower-right corner rear of the SSM and place it in the down (OFF) position. See Illustration 3-3.

CAUTION

Placing the CB4 System support module circuit breaker in the down (OFF) position will remove power from high voltage Dynamic Disable and Inductive Drive Disable Circuits but does not remove 120 VAC power from the RF Amplifiers, Spectroscopy RF Amplifier, Front Service Outlets, or Gradient Blower Fan.



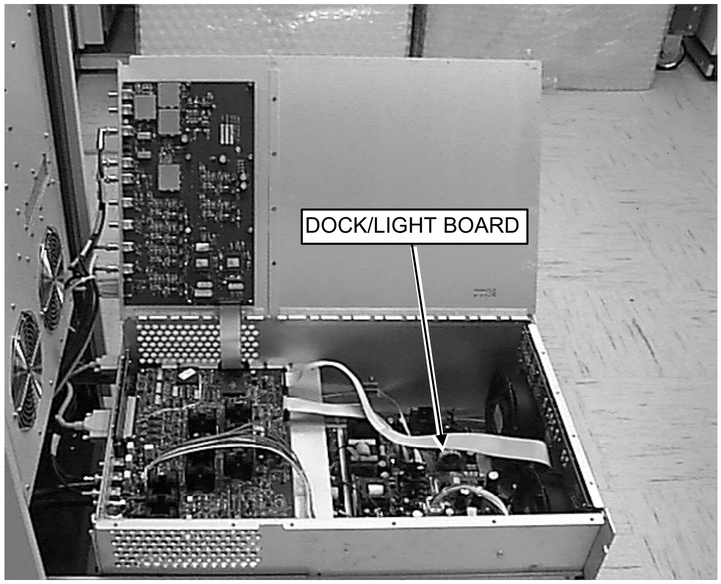
CB4 SYSTEM SUPPORT MODULE CIRCUIT BREAKER IN DOWN (OFF) POSITION
ILLUSTRATION 3-3

1. Refer to **APPENDIX A – REMOVING AND REPLACING SSM CHASSIS LID** and remove SSM lid.

2. Obtain Digital Multimeter (DMM) with clip-on test leads and configure to measure at least 50 VDC.

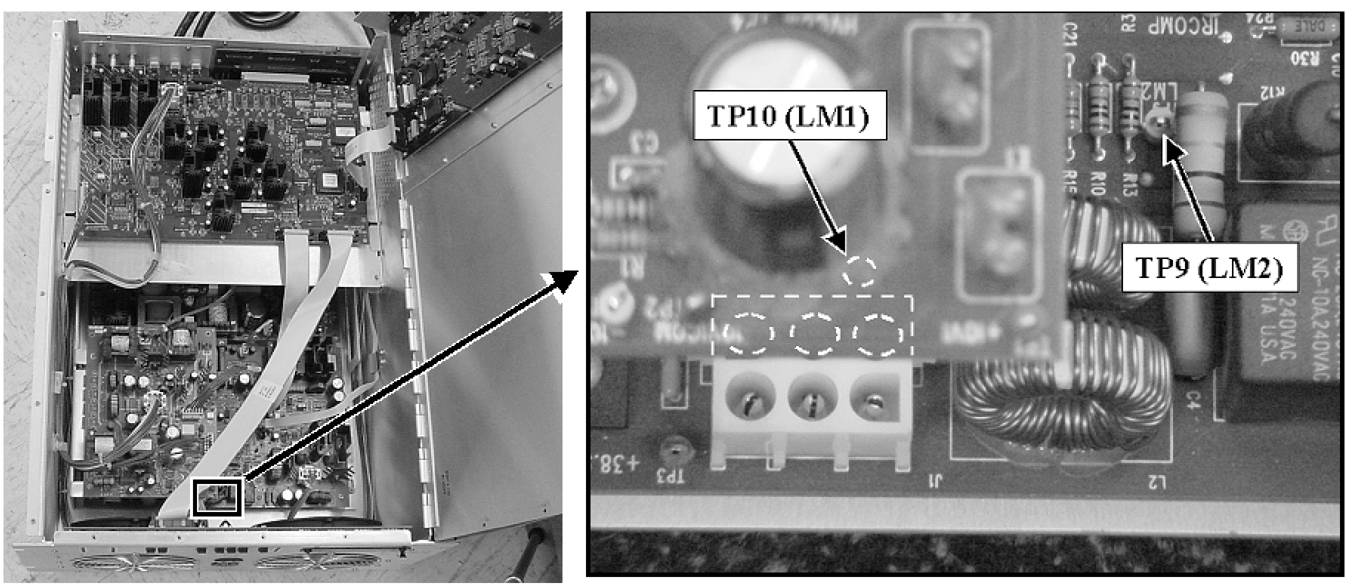
3-2 Troubleshooting Carriage Motion Problems (Continued)

3. Refer to Illustration 3-4 and locate Dock/Light Board inside bottom of SSM. This board is mounted on the bottom of the SSM chassis in front of the fans.



LOCATION OF DOCK/LIGHT BOARD INSIDE SSM
ILLUSTRATION 3-4

4. Clip DMM test leads to **TP9 (LM2)** and **TP10 (LM1)** test points on front of Dock/Light Board. See Illustration 3-5. Notice that TP10 (LM1) is located underneath the HV Board. Lead polarity is not important. The voltage measured between **TP9** (**L**ong**M**tr**2**) and **TP10** (**L**ong**M**tr**1**) is the Longitudinal Drive Motor voltage.



LOCATION OF TP9 AND TP10 TEST POINTS ON DOCK/LIGHT BOARD
ILLUSTRATION 3-5

5. Close SSM lid and let it rest on test leads.

3-2 Troubleshooting Carriage Motion Problems (Continued)

WARNING!

DO NOT LEAVE THE SSM LID OPEN IN HEAD OR SURFACE MODE WITH POWER ON. THE SSM WILL OVERHEAT AND COMPONENT DAMAGE WILL RESULT. IN HEAD OR SURFACE MODE THE LID MUST BE LEFT IN THE CLOSED POSITION WHEN POWER IS APPLIED TO INSURE ADEQUATE AIR FLOW THROUGH THE SSM.

6. Apply power to the SSM.

7. Have someone in the Magnet Room push the IN FAST button on the magnet enclosure. Watch the DMM display and note the voltage. In normal operation this voltage should range between 17 and 20 VDC. If the cradle isn't moving then this voltage should ramp up to about 30 volts and then, then the system determines that there is no movement, it will ramp down to 0 (zero) volts.

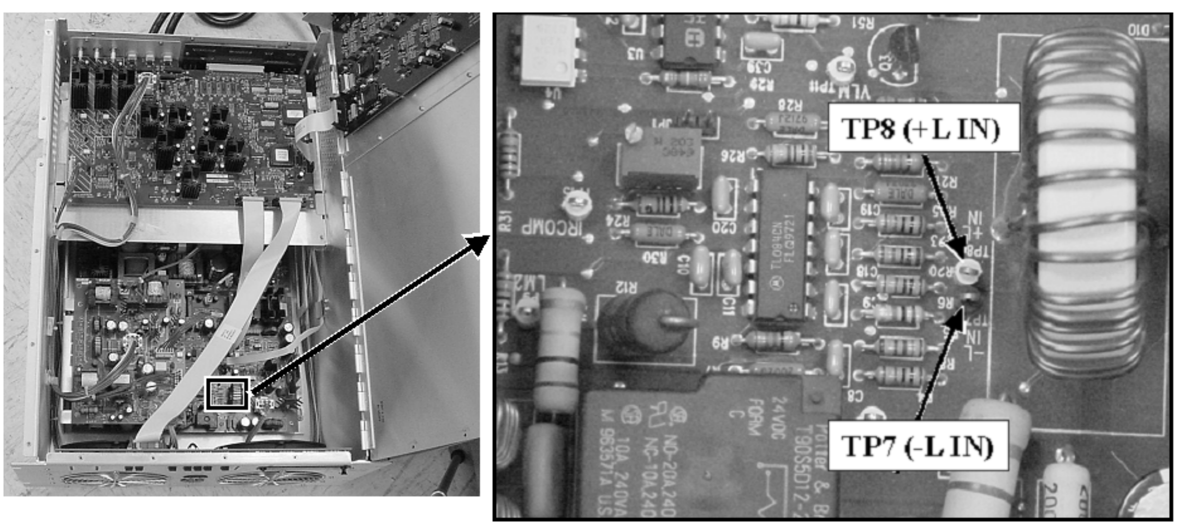
8. **If the voltage is not present or is very low:**

   a. Remove the cable connected to **J41 LONG AMP** on the rear of the SSM and set it aside.

   b. Measure the voltage again while driving the carriage into the bore.

9. **If the voltage across TP9 and TP10 is still not present:**

   c. Carefully remove test leads from **TP9** and **TP10** and connect them to **TP7 (-L IN)** and **TP8 (+L IN)** on the Dock/Light Board. See Illustration 3-6. Again, lead polarity is not important. Test points **TP7** and **TP8** allow convenient measurement of the MOTORCMD signal from the SRI which drives the Longitudinal Drive Motor.



LOCATION OF TP7 AND TP8 ON FRONT OF DOCK / LIGHT BOARD
ILLUSTRATION 3-6

3-2 Troubleshooting Carriage Motion Problems (Continued)

d. While attempting to drive the carriage into the bore verify that the voltage on the DMM display “spikes” up to 9VDC and then to 0 (zero) VDC. In normal operation this voltage “ramps” up to a steady 6 VDC when the carriage starts moving and then, when the carriage movement ends, the voltage “ramps” down from a steady 6 VDC to 0 (zero) VDC.

   i. **If the voltage across TP7 and TP8 is not seen or is seen but is very low:**

   ii. Check for poor cable connections from **J40 SRI POWER** on the rear of the SSM to **J14** on the Penetration Panel to **J1** on the SRI .

   iii. If these all appear OK then replace the SRI.

   iv. If the voltage is seen across TP7 and TP8 then replace the Dock/Light Board.

10. If the voltage across TP9 and TP10 is present but low then replace the Dock/Light Board.

11. **If the voltage across TP9 and TP10 is present and in range** then either the K1 relay isn’t working or one of the circuit load (longitudinal drive motor, pen panel filters, cable connectors, cables) components are possibly open-circuited.

DANGER!!

THE MOTOR IS MADE OUT OF FERROUS COMPONENTS AND WILL BE ATTRACTED TO THE MAGNET IF REMOVED FROM THE REAR PEDESTAL ASSEMBLY. REFER TO SECTION 3 – LONGITUDINAL DRIVE MOTOR REPLACEMENT IN THE APPROPRIATE REAR PEDESTAL HARDWARE REPLACEMENT PROCEDURE (CX OR NON-CX PROCEDURE) BEFORE ATTEMPTING TO REMOVE THE MOTOR FROM THE REAR PEDESTAL.

a. The longitudinal drive motor can be tested outside the magnet room by removing it from the rear pedestal and then connecting it to the cable that is normally connected to **J11** (RUN 770) on the equipment room side of the penetration panel. See the **DANGER** message above and refer to documented removal process to remove the motor from the rear pedestal.

b. Have someone in the Magnet Room push the IN FAST button on the magnet enclosure. Measure the voltage across the spade lug connections on the motor.

   i. If the voltage ramps up to 30 volts but the motor is still not rotating then check the motor brushes and, if necessary, replace the motor.

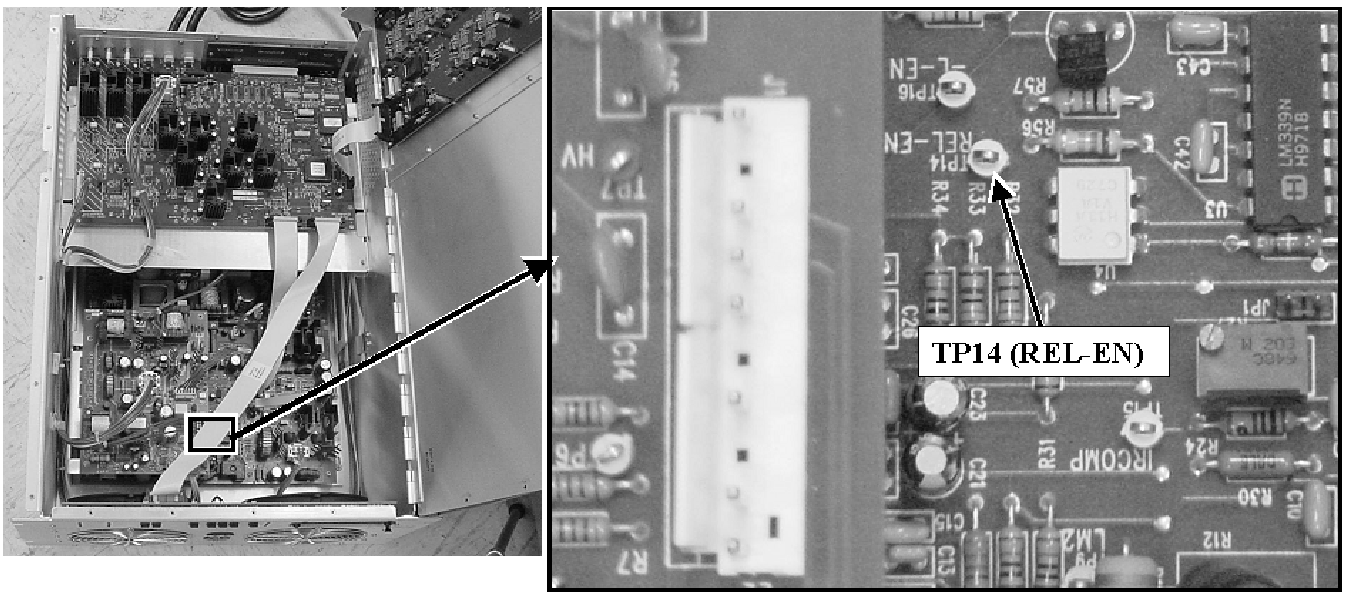
   ii. If the motor now works then inspect the penetration panel filter and sub-D connectors at **J11** on the Equipment and Magnet Room sides of the Penetration Panel for corrosion or obvious signs of damage.

3-2 Troubleshooting Carriage Motion Problems (Continued)

WARNING!

DO NOT LEAVE THE SSM LID OPEN IN HEAD OR SURFACE MODE WITH POWER ON. THE SSM WILL OVERHEAT AND COMPONENT DAMAGE WILL RESULT. IN HEAD OR SURFACE MODE THE LID MUST BE LEFT IN THE CLOSED POSITION WHEN POWER IS APPLIED TO INSURE ADEQUATE AIR FLOW THROUGH THE SSM.

c. If the 30 volt ramp is still not seen when someone is pressing the IN FAST button then open the SSM top cover and locate the **TP14 (REL-EN)** on the Dock / Light Board. See Illustration 3-7. Connect the DMM test leads from **TP14 (REL-EN)** to chassis ground. Polarity is not important.



LOCATION OF TP14 ON FRONT OF DOCK / LIGHT BOARD
ILLUSTRATION 3-7

d. Have someone press one of the FAST IN buttons on either Magnet Enclosure Keypad. The voltage should read about 30.9 volts when the IN FAST button is not pressed and should change to 10.9 volts when the IN FAST button is pressed.

   i. **If no voltage change was seen** then the Out Enable signal from SRI may not be reaching SSM. Check the continuity of the line between the SRI and the SSM. Pins 3 and 4 on the J7 cable carry the differential signals between the SRI and SSM. Otherwise, replace the SRI.

   ii. **If change was seen** then Dock/Light Board may have failed. Replace Dock/Light Board.

12. Remove the test leads and perform actions in **Section A-2 Replacing SSM Chassis Lid**.

3-2 Troubleshooting Carriage Motion Problems (Continued)

13. Perform one head and body scan, if possible.

14. Return to troubleshooting flowchart, if necessary.

4 – TROUBLESHOOTING BORE FAN FAILURE

This section is divided into two parts. **Section 4-1 Checking Motor AC Voltage** provides the FE with a process for determining that the motor is receiving power. **Section 4-2 Checking Functionality of SSM Bore Fan Switching Circuitry** provides the FE with a process for determining that the failure is not within the SSM.

4-1 Checking Motor AC Voltage

1. Confirm that the Gradient Fan runs normally.

2. Refer to Illustration 4-1.

    a. Place the **CB1 BORE VENT FAN** Circuit Breaker in the down (OFF) position.

    b. Place the **CB2 GRADIENT FAN** Circuit Breaker in the down (OFF) position.

    c. Remove the cable connected to **J37 BORE VENT FAN 300VA** and set this cable aside.

    d. Remove the cable connected to **J47 GRAD. FAN 900VA** and set this cable aside.

BORE VENT CB1 CIRCUIT BREAKER LOCATION
ILLUSTRATION 4-1

3. Locate the Bore Vent cable labeled MR1-A7-J37 removed in the last step and connect this cable instead to the **J47 GRAD. FAN 900VA** connector on the rear of the SSM.

4. Place the **CB2 GRADIENT FAN** Circuit Breaker in the up (ON) position.

5. Confirm that the bore vent fan is running normally.

6. **If the fan is not running normally** then place the **CB2 GRADIENT FAN** Circuit Breaker in the down (OFF) position.

    a. Remove the Bore Vent Cable labeled MR1-A7-J37 from the **J47 GRAD. FAN 900VA** connector on the rear of the SSM.

    b. Re-connect the Bore Vent Cable labeled MR1-A7-J37 to the **J37 BORE VENT FAN 300VA** connector on the rear of the SSM.

4-1 Checking Motor AC Voltage (Continued)

c. Re-connect the Gradient Fan Cable labeled MR1-A7-J47 to the **J47 GRAD. FAN 900VA** connector on the rear of the SSM.

d. Refer to the **BLOWER BOX MOTOR REPLACEMENT** procedure (MI0REA2.DOC) and replace the bore fan motor.

e. After finished with the Bore Fan Motor replacement, place the **CB1 BORE FAN VENT 300VA** Circuit Breaker on the rear of the SSM in the up (ON) position. Confirm that the Bore Fan is running.

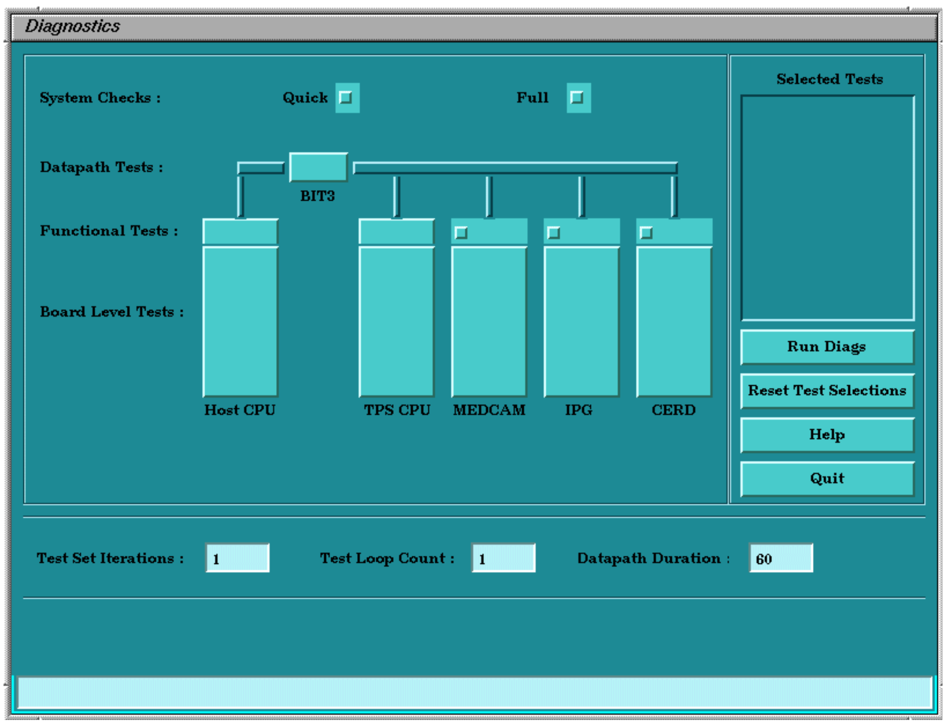
f. Return, if necessary, to the troubleshooting flowchart.

7. **If the fan is running normally** then proceed to the next section.

4-2 Checking Functionality of SSM Bore Fan Switching Circuitry

This section assumes that all the checks described in **Section 4-1 Checking Motor AC Voltage** have been completed and that the Bore Fan Motor is known to be functional.

1. On the Service Desktop, select **[Diagnostics], [Diagnostics Main Menu],** then **[Start]**. Wait for the Diagnostics Main Menu to appear as shown in Illustration 4-2.

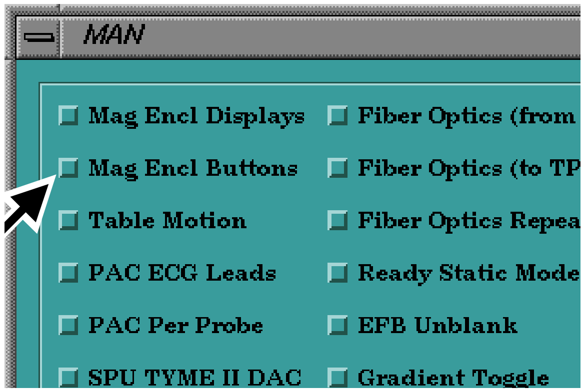


DIAGNOSTICS MAIN MENU
ILLUSTRATION 4-2

2. Select IPG Functional Tests and select MANUAL TESTS from the menu.

4-2 Checking Functionality of SSM Bore Fan Switching Circuitry (Continued)

3. Select **Mag Encl Buttons** from the first column in the menu as shown in Illustration 4-3.



MANUAL DIAGNOSTICS MENU
ILLUSTRATION 4-3

4. Wait about 10 seconds and then enter the Magnet Room.

   a. Press the buttons on the right and left Magnet Enclosure Keypads that toggle power to the Bore Fan Motor on and off.

   b. Confirm that each time a button is pressed a unique number and a number showing the total of number of times a button was pressed are displayed in the Scan Time and Table Motion Display Windows, respectively.

5. If one or both of the buttons don’t respond then check the functionality of the SRI and the keypad(s).

6. If the test was successful then cancel out of the test and exit the diagnostics.

7. Completely remove power from the SSM by placing the System Support Module circuit breaker on the front of the PDU in the OFF position and then Locking and Tagging out the panel per Safety requirements.

DANGER!!

AVOID SERIOUS INJURY OR DEATH BY ELECTROCUTION. LOCK AND TAGOUT THE SYSTEM SUPPORT MODULE CIRCUIT BREAKER AT THE PDU AS THE SSM CIRCUITS TO BE CHECKED NEXT WILL NOT BE DE-ENERGIZED BY PLACING ANY OF THE CIRCUIT BREAKERS ON THE REAR OF THE SSM IN THE OFF POSITION.

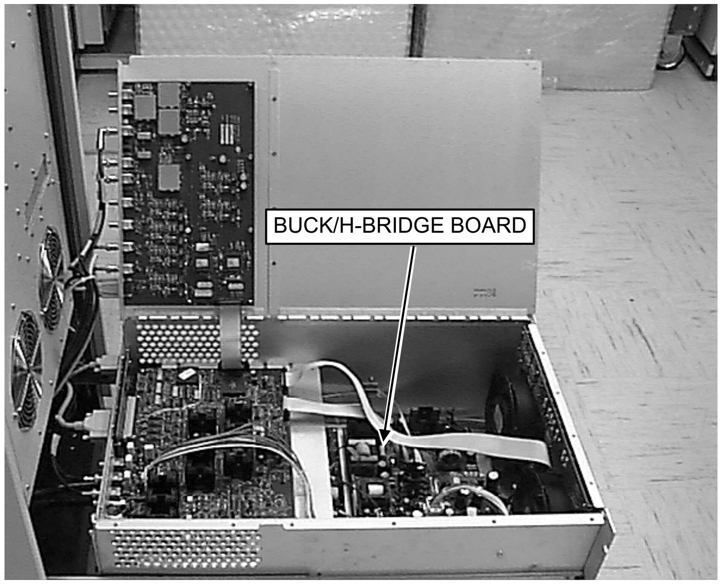
8. Refer to **APPENDIX A – REMOVING AND REPLACING SSM CHASSIS LID** and remove SSM lid.

9. Obtain Digital Multimeter (DMM) with test leads.

10. Wait 5 minutes for the SSM power supplies to discharge before opening the SSM lid.

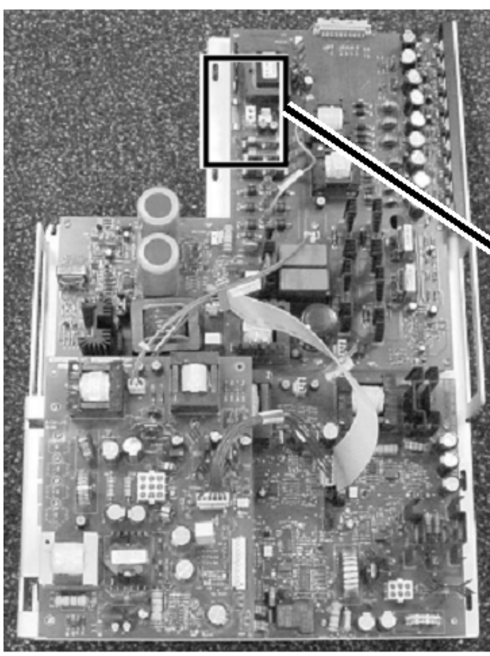
4-2 Checking Functionality of SSM Bore Fan Switching Circuitry (Continued)

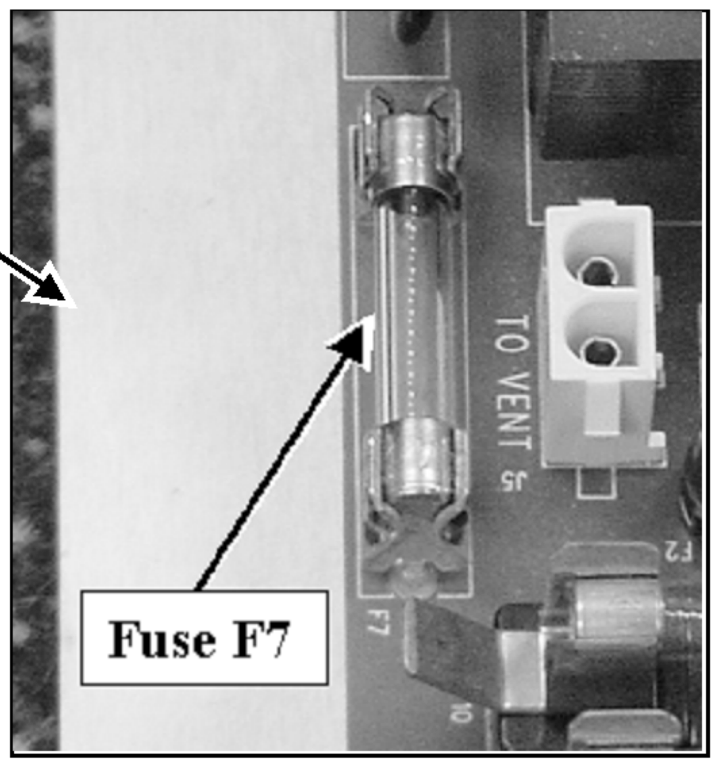
11. Refer to Illustration 4-4 and locate Buck/H-Bridge Board inside bottom of the SSM. This board is mounted on the bottom of the SSM chassis behind the Dock/Light Board.



LOCATION OF BUCK/H-BRIDGE BOARD INSIDE SSM
ILLUSTRATION 4-4

12. Remove the aluminum plate that the CPD Board is affixed so as to gain access to fuse F7 on the Buck/H-Bridge Board. See Illustration 4-5.





FUSE F7 LOCATED ON REAR OF BUCK / H-BRIDGE BOARD
ILLUSTRATION 4-5

4-2 Checking Functionality of SSM Bore Fan Switching Circuitry (Continued)

13. See Illustration 4-5 and locate 4 amp fuse F7 on the Buck/H-Bridge Board. Use the DMM to check the continuity of the fuse and verify that it is not open.

14. Verify the electrical connection between CB4 SYSTEM SUPPORT MODULE Circuit Breaker and the white J5 connector seen next to fuse F7 in Illustration 4-6 on the Buck/H-Bridge Board.

15. Use the DMM to check the continuity of each leg of the CB1 circuit breaker.

16. If the fuse is open and the Bore Fan is known to be in good working order then replace either the Buck / H-Bridge Board or the SSM.

17. If the CB1 circuit breaker is found to be faulty then replace either the circuit breaker or the SSM.

18. When finished, confirm that all the test leads have been removed and all the SSM cabling has been reconnected.

19. Re-install the CPD Board assembly into the SSM and then close the SSM lid.

20. Remove the Lockout / Tagout hardware from the System Support Module circuit breaker on the front of the PDU and restore power to the SSM.

21. Perform actions in **Section A-2 Replacing SSM Chassis Lid**.

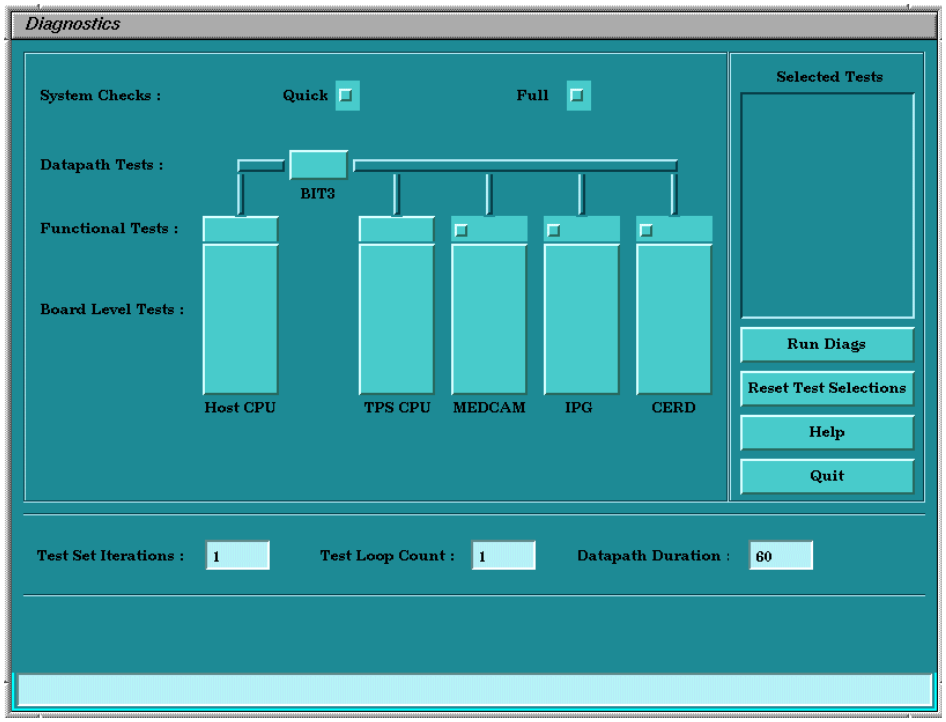
22. Perform one head and body scan, if possible.

23. Return to troubleshooting flowchart, if necessary.

5 – TROUBLESHOOTING BORE LIGHT FAILURE

This section assumes that the FE has already checked the halogen lamp filament and confirmed that the lamp is in good working order.

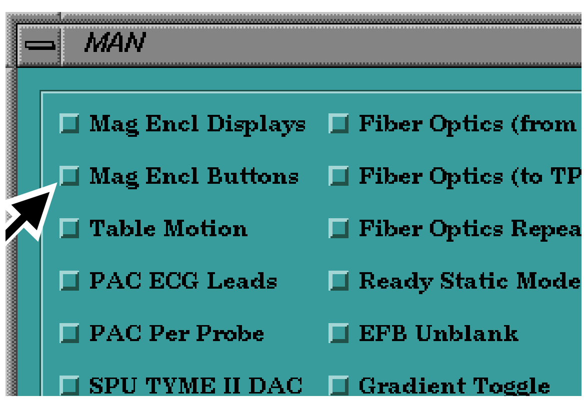
1. On the Service Desktop, select **[Diagnostics], [Diagnostics Main Menu],** then **[Start]**. Wait for the Diagnostics Main Menu to appear as shown in Illustration 5-1.



DIAGNOSTICS MAIN MENU
ILLUSTRATION 5-1

2. Select IPG Functional Tests and select MANUAL TESTS from the menu.

3. Select **Mag Encl Buttons** from the first column in the menu as shown in Illustration 5-2.



MANUAL DIAGNOSTICS MENU
ILLUSTRATION 5-2

4. Wait about 10 seconds and then enter the Magnet Room.

5 – TROUBLESHOOTING BORE LIGHT FAILURE (CONTINUED)

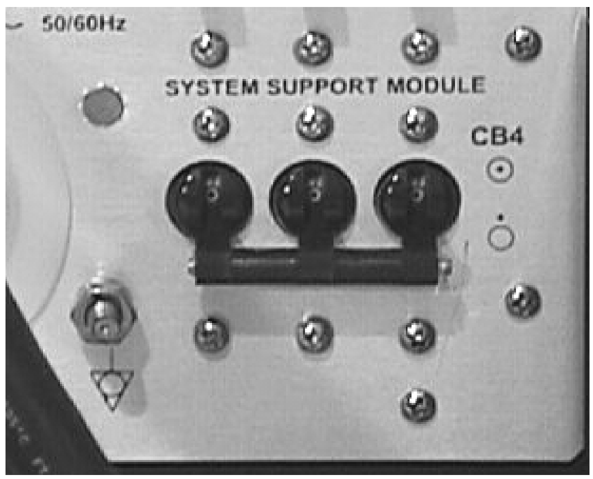
a. Press the buttons on the right and left Magnet Enclosure Keypads that toggle power to the Bore Light on and off.

b. Confirm that each time a button is pressed a unique number and a number showing the total of number of times a button was pressed are displayed in the Scan Time and Table Motion Display Windows, respectively.

5. If one or both of the buttons don't respond then check the functionality of the SRI and the keypad(s).

6. If the test was successful then cancel out of the test and exit the diagnostics.

7. Locate the **CB4 SYSTEM SUPPORT MODULE** Circuit Breaker in the lower-right corner rear of the SSM and place it in the down (OFF) position. See Illustration 5-3.



CB4 SYSTEM SUPPORT MODULE CIRCUIT BREAKER IN DOWN (OFF) POSITION
ILLUSTRATION 5-3

CAUTION

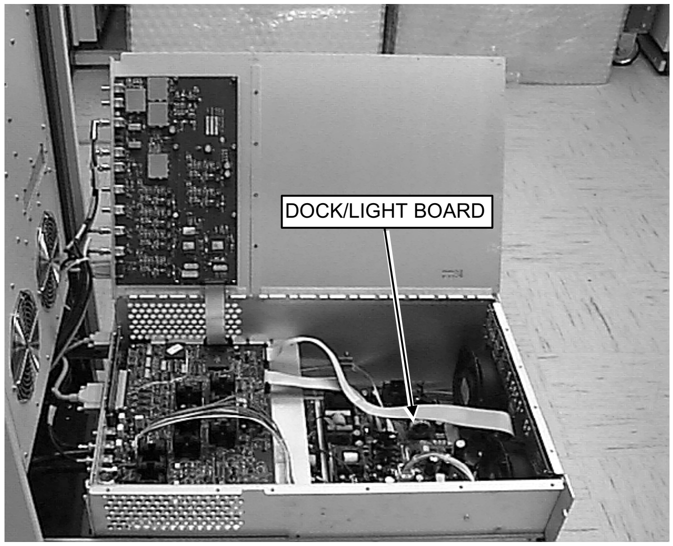
Placing the CB4 System support module circuit breaker in the down (OFF) position will remove power from high voltage Dynamic Disable and Inductive Drive Disable Circuits but does not remove AC power from the NB AMP receptacle, SPECTRO AMP receptacle, Front Service Outlets, or Gradient Blower Fan.

8. Refer to **APPENDIX A – REMOVING AND REPLACING SSM CHASSIS LID** and remove SSM lid.

9. Obtain Digital Multimeter (DMM) with clip-on test leads and configure to measure 5VDC.

5 – TROUBLESHOOTING BORE LIGHT FAILURE (CONTINUED)

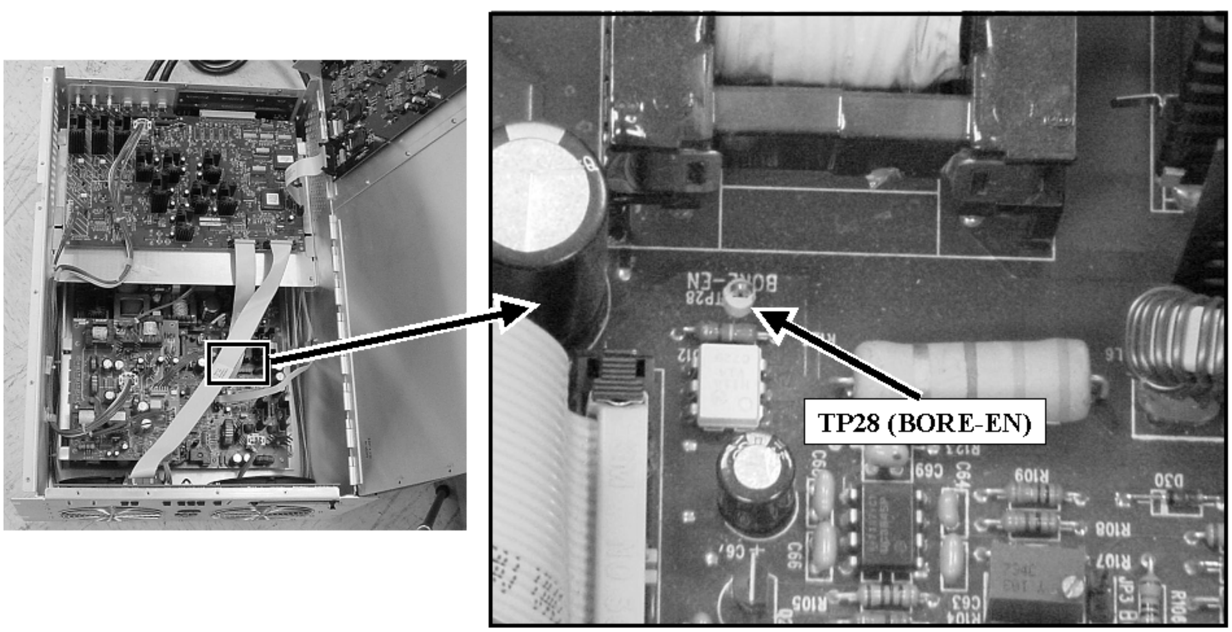
10. Refer to Illustration 5-4 and locate Dock/Light Board inside bottom of the SSM. This board is mounted on the bottom of the SSM chassis behind the fans.



LOCATION OF DOCK/LIGHT BOARD INSIDE SSM
ILLUSTRATION 5-4

11. Reseat the ribbon cable interconnecting connector J4 on the Dock/Light Board to J9 on the CPD Board.

12. Refer to Illustration 5-5 and locate **TP28 (BORE-EN)** on the Dock/Light Board. Connect one DMM test lead to TP28 and the other side to chassis ground. Lead polarity is not important.



LOCATION OF TP28 ON THE DOCK/LIGHT BOARD
ILLUSTRATION 5-5

5 – TROUBLESHOOTING BORE LIGHT FAILURE (CONTINUED)

13. Connect other DMM test lead to SSM chassis ground.

14. Close SSM lid and let it rest on test leads.

15. Locate the **CB4 SYSTEM SUPPORT MODULE** Circuit Breaker in the lower-right corner rear of the SSM and place it in the up (ON) position.

WARNING!

DO NOT LEAVE THE SSM LID OPEN IN HEAD OR SURFACE MODE WITH POWER ON. THE SSM WILL OVERHEAT AND COMPONENT DAMAGE WILL RESULT. IN HEAD OR SURFACE MODE THE LID MUST BE LEFT IN THE CLOSED POSITION WHEN POWER IS APPLIED TO INSURE ADEQUATE AIR FLOW THROUGH THE SSM.

16. Have someone push one of the Bore Light Buttons on either of the keypads on the front of the Magnet Enclosure in the Magnet Room. The DMM should indicate a 5 volt spike.

    a. **If the voltage is 0 (zero) VDC or close to this** then replace the CPD Board.

    b. **If the voltage is 5 VDC or close to this** then the signal is reaching the SSM. Replace the Dock/Light Board.

17. Power off the SSM.

18. Carefully remove the DMM test leads from inside the SSM.

19. Perform actions in **Section A-2 Replacing SSM Chassis Lid**.

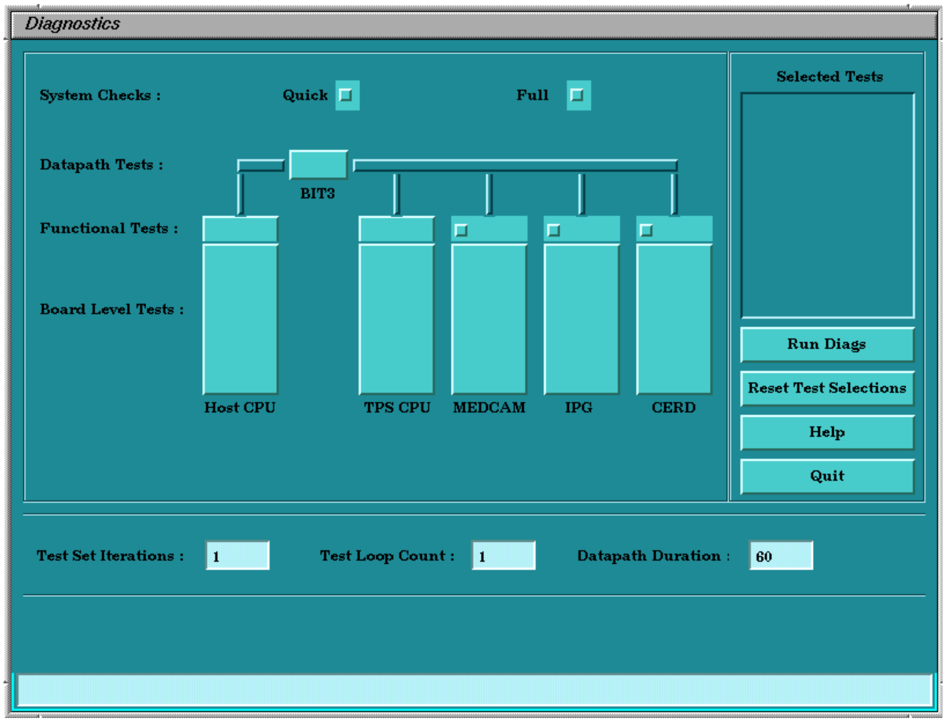
20. Perform one head and body scan, if possible.

21. If necessary, return to the troubleshooting flowchart.

6 – TROUBLESHOOTING PATIENT ALIGNMENT LIGHT FAILURE

This section assumes that the FE has already checked the halogen lamp filament and confirmed that the lamp is in good working order.

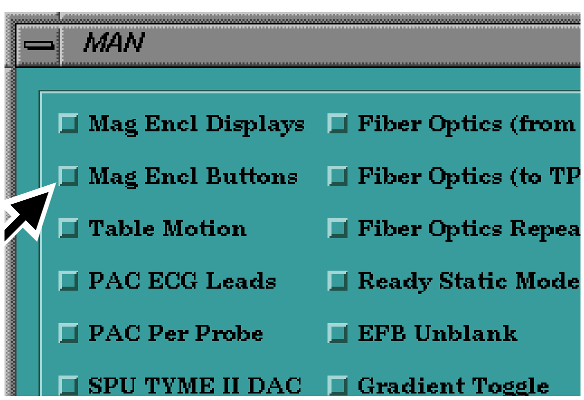
1. On the Service Desktop, select **[Diagnostics], [Diagnostics Main Menu],** then **[Start]**. Wait for the Diagnostics Main Menu to appear as shown in Illustration 6-1.



DIAGNOSTICS MAIN MENU
ILLUSTRATION 6-1

2. Select IPG Functional Tests and select MANUAL TESTS from the menu.

3. Select **Mag Encl Buttons** from the first column in the menu as shown in Illustration 6-2.



MANUAL DIAGNOSTICS MENU
ILLUSTRATION 6-2

4. Wait about 10 seconds and then enter the Magnet Room.

6 – TROUBLESHOOTING PATIENT ALIGNMENT LIGHT FAILURE (CONTINUED)

a. Press the buttons on the right and left Magnet Enclosure Keypads that toggle power to the Patient Alignment Light on and off.

b. Confirm that each time a button is pressed a unique number and a number showing the total of number of times a button was pressed are displayed in the Scan Time and Table Motion Display Windows, respectively.

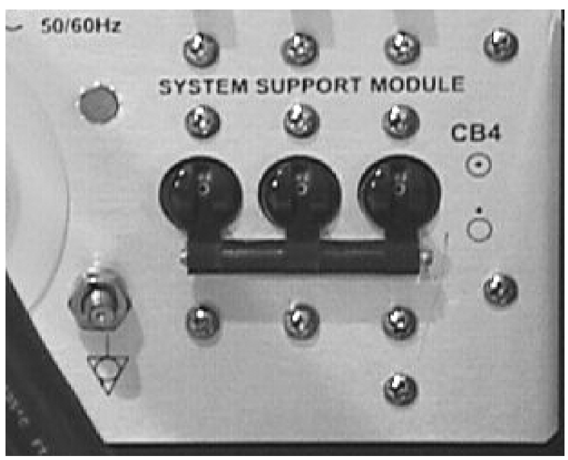
5. If one or both of the buttons don't respond then check the functionality of the SRI and the keypad(s).

6. If the test was successful then cancel out of the test and exit the diagnostics.

7. Locate the **CB4 SYSTEM SUPPORT MODULE** Circuit Breaker in the lower-right corner rear of the SSM and place it in the down (OFF) position. See Illustration 6-3.

CAUTION

Placing the CB4 System support module circuit breaker in the down (OFF) position will remove power from high voltage Dynamic Disable and Inductive Drive Disable Circuits but does not remove AC power from the NB AMP receptacle, SPECTRO AMP receptacle, Front Service Outlets, or Gradient Blower Fan.



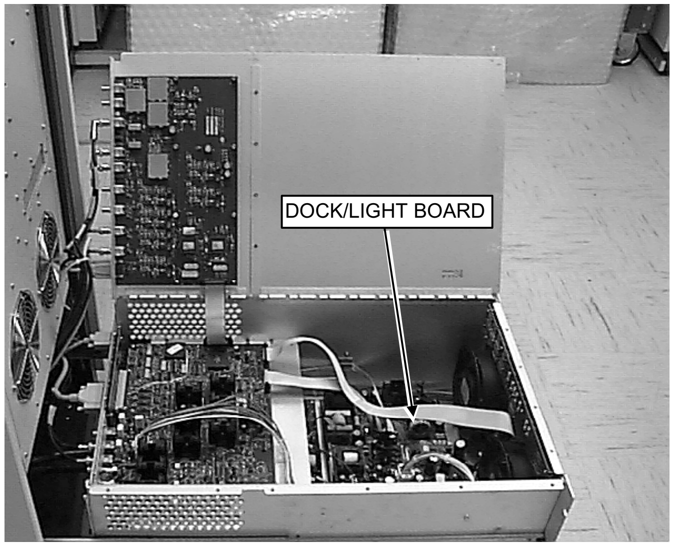
CB4 SYSTEM SUPPORT MODULE CIRCUIT BREAKER IN DOWN (OFF) POSITION
ILLUSTRATION 6-3

8. Refer to **APPENDIX A – REMOVING AND REPLACING SSM CHASSIS LID** and remove SSM lid.

9. Obtain Digital Multimeter (DMM) with clip-on test leads and configure to measure 5VDC.

6 – TROUBLESHOOTING PATIENT ALIGNMENT LIGHT FAILURE (CONTINUED)

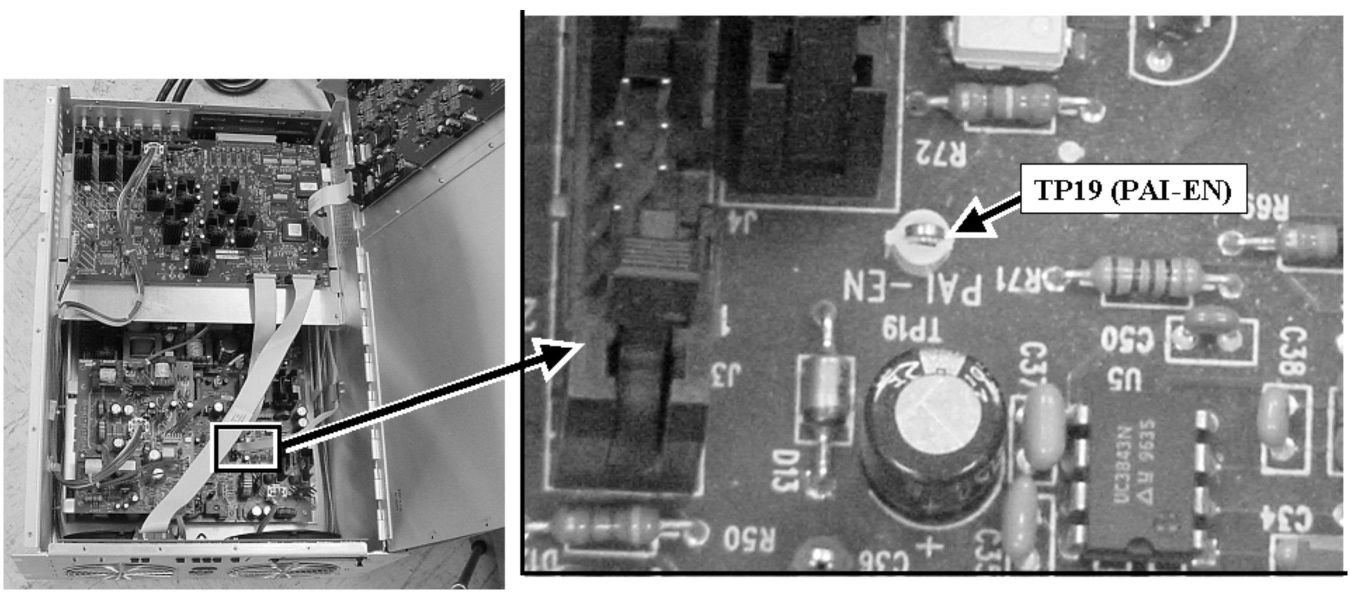
10. Refer to Illustration 6-4 and locate Dock/Light Board inside bottom of the SSM. This board is mounted on the bottom of the SSM chassis behind the fans.



LOCATION OF DOCK/LIGHT BOARD INSIDE SSM
ILLUSTRATION 6-4

11. Reseat the ribbon cable interconnecting connector J4 on the Dock/Light Board to J9 on the CPD Board.

12. Refer to Illustration 6-5 and locate **TP19 (PAL-EN)** on the Dock/Light Board. Connect one DMM test lead to TP19 and the other to chassis ground.



LOCATION OF TP19 ON THE DOCK / LIGHT BOARD
ILLUSTRATION 6-5

13. Locate the **CB4 SYSTEM SUPPORT MODULE** Circuit Breaker in the lower-right corner rear of the SSM and place it in the up (ON) position.

6 – TROUBLESHOOTING PATIENT ALIGNMENT LIGHT FAILURE (CONTINUED)

WARNING!

DO NOT LEAVE THE SSM LID OPEN IN HEAD OR SURFACE MODE WITH POWER ON. THE SSM WILL OVERHEAT AND COMPONENT DAMAGE WILL RESULT. IN HEAD OR SURFACE MODE THE LID MUST BE LEFT IN THE CLOSED POSITION WHEN POWER IS APPLIED TO INSURE ADEQUATE AIR FLOW THROUGH THE SSM.

14. Have someone push one of the Patient Alignment Light Buttons on either of the keypads on the front of the Magnet Enclosure in the Magnet Room.

15. Check the DMM display for any change in voltage level. If the optocoupler is active then the voltage shown on the DMM display should spike up to 5 volts, then down.

    a. **If the voltage is 0 (zero) VDC or close to this** then replace the CPD Board.

    b. **If the voltage is 5 VDC or close to this** then the signal has reached the optocoupler. Replace the Dock/Light Board.

16. Power off the SSM.

17. Carefully remove the DMM test leads from inside the SSM.

18. Perform actions in **Section A-2 Replacing SSM Chassis Lid**.

19. Perform one head and body scan, if possible.

20. If necessary, return to the troubleshooting flowchart.

7 – TROUBLESHOOTING SSM RELATED IMAGE QUALITY PROBLEMS

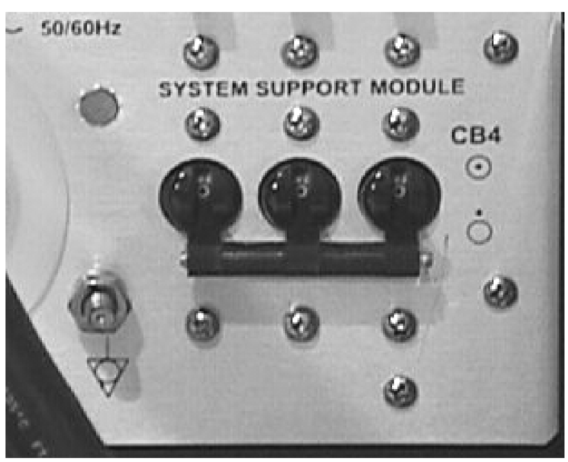
Most image quality (IQ) problems don't have anything to do with the SSM. The SSM is well filtered and is an unlikely source of screen room noise. It has good fault protection and sensing circuitry and, if not in manual bypass, it will fault on most failures that could cause an IQ problem. There are two instances when the SSM may cause an IQ problem but not register a fault. The first is if there is a poor electrical connection on one of the four dynamic disable bias lines from the Penetration Panel to the RF Coil. This will generally cause IQ problems with Head and Surface/PA scans. Methods for checking these connections are described in **Section 7-1 Poor Image Quality Due To Poor DD Bias Line Connections**. The second instance is if circuitry on the CPD Board fails to the extent that high voltage (1000 VDC) cannot reach the dynamic disable or direct drive bias lines but no faults are registered. This will generally start as a Body IQ problem but, as dynamic disable boxes fail over time due to the missing high voltage, it will quickly escalate to a Surface/Phased Array IQ problem. A process for checking this is described in **Section 7-2 Poor Image Quality Due to Loss of High Voltage Bias**.

7-1 Poor Image Quality Due To Poor DD Bias Line Connections

1. Locate the **CB4 SYSTEM SUPPORT MODULE** Circuit Breaker in the lower-right corner rear of the SSM and place it in the down (OFF) position. See Illustration 7-1.

CAUTION

Placing the CB4 System support module circuit breaker in the down (OFF) position will remove power from high voltage Dynamic Disable and Inductive Drive Disable Circuits but does not remove AC power from the NB AMP receptacle, SPECTRO AMP receptacle, Front Service Outlets, or Gradient Blower Fan.



CB4 SYSTEM SUPPORT MODULE CIRCUIT BREAKER IN DOWN (OFF) POSITION
ILLUSTRATION 7-1

2. Locate connectors J72, J73, J74, J75, and J76 on the Dynamic Disable and Inductive Drive Filter that is mounted on the Magnet Room side of the Penetration Panel.

7-1 Poor Image Quality Due To Poor DD Bias Line Connections (Continued)

CAUTION

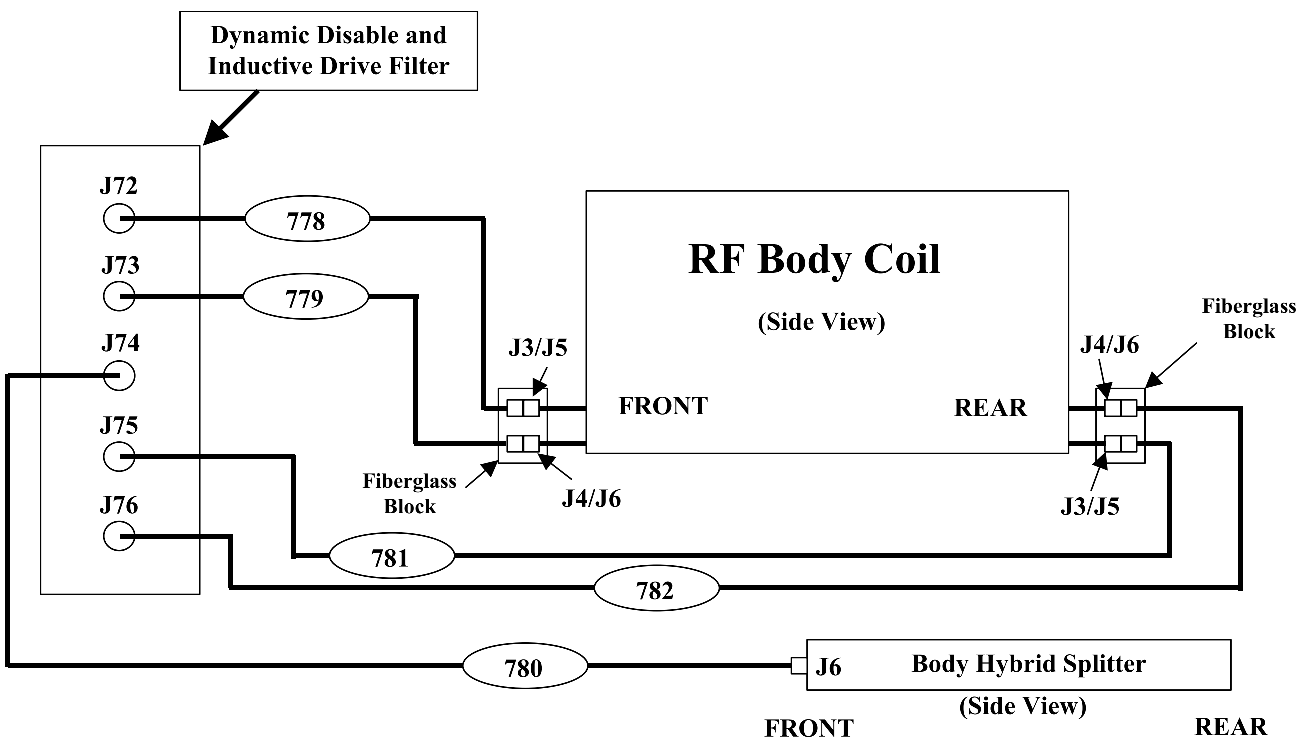
While MHV connectors may look somewhat like BNC connectors the construction of the two connectors is not the same. Forcing MHV and BNC connectors together can damage both connectors.

3. Refer to Illustration 7-2. Individually remove, inspect and reconnect the male MHV Dynamic Disable Cable connectors (Runs 778, 779, 780, 781, and 782) at the J72, J73, J74, J75, and J76 connectors on the Dynamic Disable and Inductive Drive Filter located on the Magnet Room side of the Penetration Panel. Make certain that each male MHV connector is locked onto its female MHV counterpart.

4. Confirm that the cables are routed to the proper locations as shown in Illustration 7-2.

CAUTION

Do not assume that cable labels are correct and that cables have the correct labels. Physically trace the path and verify the connection of the cables.



MAP OF DYNAMIC DISABLE AND INDUCTIVE DRIVE CABLE CONNECTIONS
ILLUSTRATION 7-2

5. Refer to Illustration 7-2 check the Dynamic Disable Cable Connections at the front and rear Body Coil J3/J5 and J4/J6 pigtail connections.

   a. Locate the two Body Coil pigtail cables coming from the front and rear of the Body Coil.

7-1 Poor Image Quality Due To Poor DD Bias Line Connections (Continued)

i. Disconnect each pigtail cable male MHV connector from the female adapter on the Fiberglass Block and inspect it for damage.

ii. Reconnect each cable when the inspection is done and make certain that the male MHV connector locks onto its female MHV counterpart.

b. Individually remove, inspect and reconnect the male MHV Dynamic Disable Cable connectors (Runs 778, 779, 781, and 782) at the J3/J5 and J4/J6 female MHV connectors on the Fiberglass Block. Make certain that each male MHV connector is locked onto its female MHV counterpart.

c. Refer to Illustration 7-2. Locate the Body Hybrid Splitter mounted in the Rear Pedestal behind the Magnet. Remove, inspect, and reconnect the Inductive Drive Bias Cable male MHV connector (Run 780) at J6 on the front of the Body Hybrid Splitter. Make certain that the male MHV connector is locked onto its female J6 MHV connector.

NOTE

It is generally not necessary to perform this type of check on the Equipment Room side of the Dynamic Disable and Inductive Drive Filter since poor connections on this side will, as long as the SSM is not bypassed, result in some type of system fault.

6. Locate the **CB4 SYSTEM SUPPORT MODULE** Circuit Breaker in the lower-right corner rear of the SSM and place it in the up (ON) position.

7. Reset the TPS from the Service Desktop Manager.

8. If any poor connections were found and corrected then press the **CAL** button on the front of the SSM once. If necessary, refer to the **CPD BOARD SETUP & CALIBRATION** procedure on the Service CD-ROM for more information.

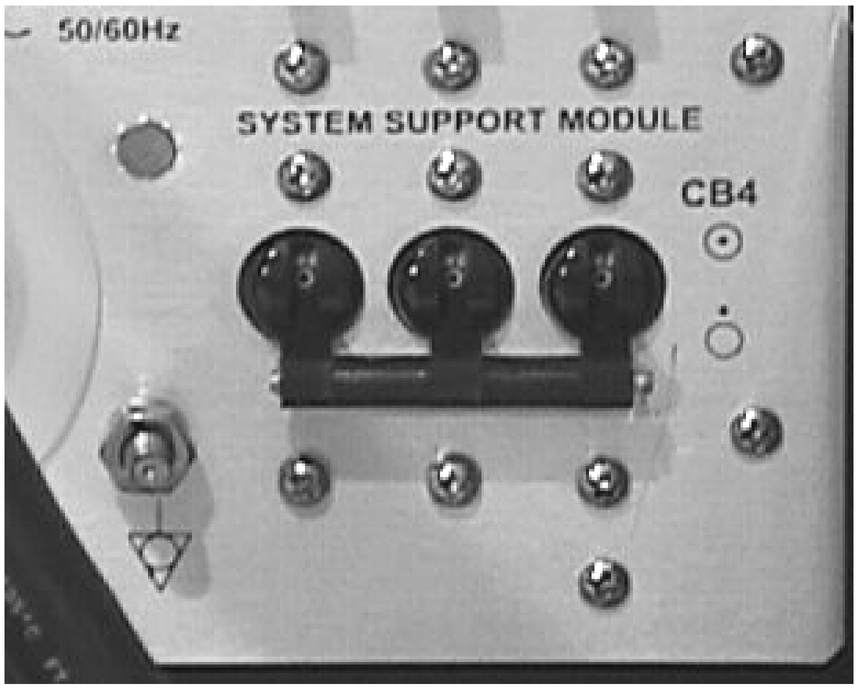
9. Perform one head and one body scan.

10. Return to the Troubleshooting Flowchart, if necessary.

7-2 Poor Image Quality Due to Loss of High Voltage Bias

This section assumes that there is a body or surface coil Image Quality (IQ) problem that the FE believes may be related to the loss of Dynamic Disable or Inductive Drive high voltage but the MR System is not reporting any Dynamic Disable or Inductive Drive faults. If the system is reporting faults then the FE should refer to the fault message and check the hardware reported as failing in the error message.

1. Locate the **CB4 SYSTEM SUPPORT MODULE** Circuit Breaker in the lower-right corner rear of the SSM and place it in the down (OFF) position. See Illustration 7-3.



CB4 SYSTEM SUPPORT MODULE CIRCUIT BREAKER IN DOWN (OFF) POSITION
ILLUSTRATION 7-3

CAUTION

Placing the CB4 System support module circuit breaker in the down (OFF) position will remove power from high voltage Dynamic Disable and Inductive Drive Disable Circuits but does not remove 120 VAC power from the Erbtec RF Amplifier, Spectroscopy RF Amplifier, Front Service Outlets, or Gradient Blower Fan.

2. Wait 5 minutes after placing CB4 in the down (OFF) position for the high voltage supply in the SSM to discharge before performing the next step.

3. Refer to **APPENDIX A – REMOVING AND REPLACING SSM CHASSIS LID** and remove SSM lid.

7-2 Poor Image Quality Due to Loss of High Voltage Bias (Continued)

4. Refer to Illustration 7-4 and remove the male MHV cable connectors from the **J42 BODY 1**, **J43 BODY 2**, and **J44 DD** connectors on the rear exterior, right side of the SSM.

DYNAMIC DISABLE AND INDUCTIVE DRIVE DISABLE MHV CONNECTORS
ILLUSTRATION 7-4

5. Remove the body RF transmission line from **J4 BODY OUTPUT** on the rear of the RFI Module inside the RF Cabinet and set it aside.

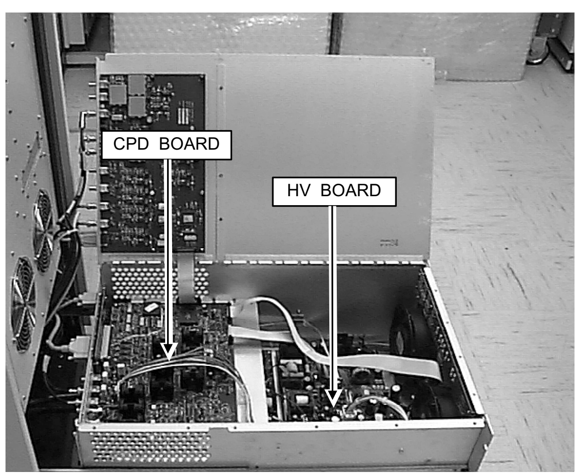
6. Remove the RF Input Cable (Run 229) from the RF Cabinet RF IN Interface Panel.

   a. **If this is a RF/PDU Cabinet:** Remove the RF Input Cable (Run 229) from **RF IN J3** on the rear RF/PDU Cabinet Interface.

   b. **If this is an SRF Cabinet:** Remove the RF Input Cable (Run 229) from the **RF IN J1** on the rear SRF Cabinet Interface.

7. Obtain a Digital Multimeter (DMM) with clip-on test leads that is capable of measuring 1000 VDC or more. Configure DMM to measure at least 1000VDC.

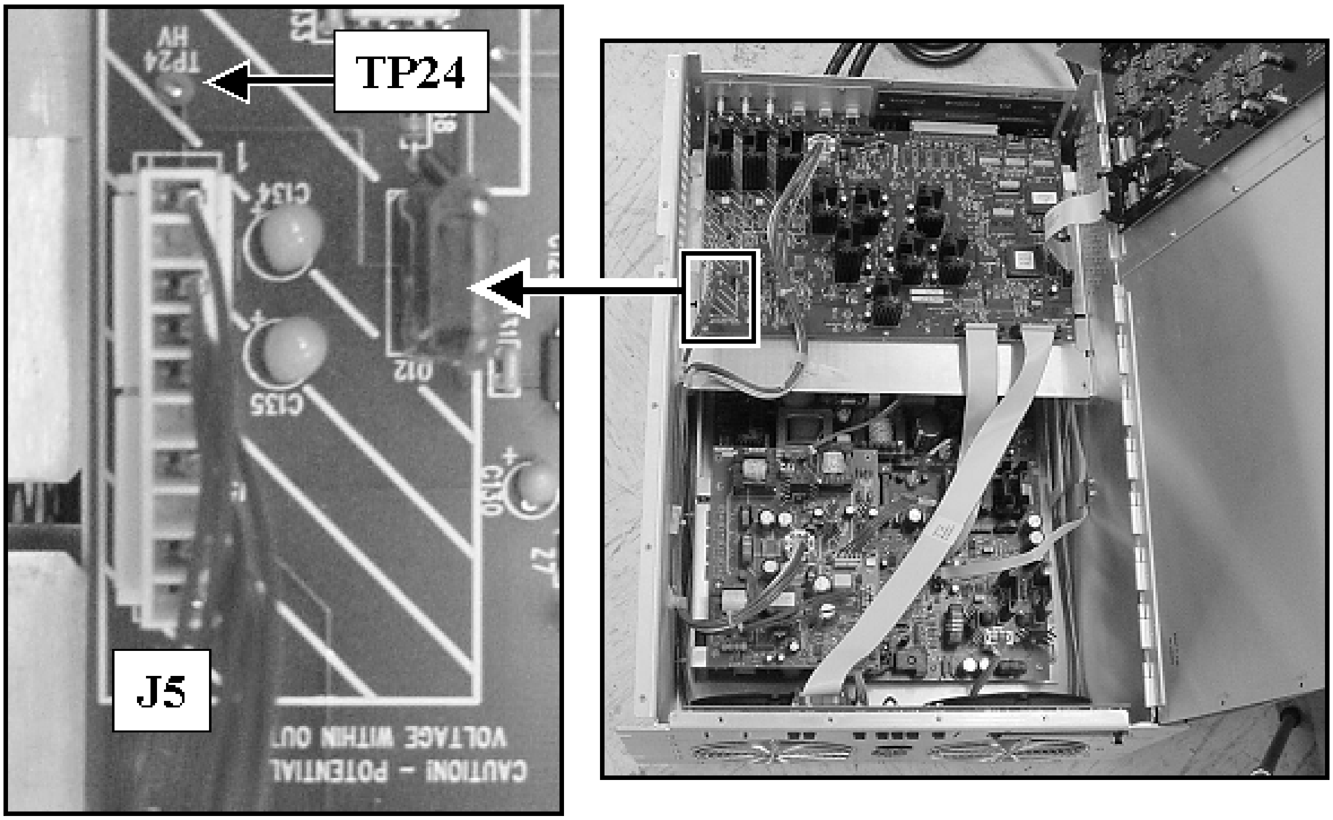
8. Refer to Illustration 7-5 and locate the CPD Board in the upper, rear-end of the SSM. The CPD Board is mounted on an aluminum plate above the Buck/H-Bridge Board.



LOCATION OF HV BOARD AND CPD BOARD INSIDE SSM
ILLUSTRATION 7-5

7-2 Poor Image Quality Due to Loss of High Voltage Bias (Continued)

9. Refer to Illustration 7-6 and attach one DMM test lead to **TP24 (HV)** on the CPD Board and the other to the SSM chassis.



LOCATION OF TP24 (HV) ON THE CPD BOARD
ILLUSTRATION 7-6

10. Close SSM lid and let it rest on test leads.

11. Place the **CB4 SYSTEM SUPPORT MODULE** Circuit Breaker in the lower-right corner rear of the SSM in the up (ON) position.

7-2 Poor Image Quality Due to Loss of High Voltage Bias (Continued)

12. Refer to Table 7-1 and setup for a body scan.

TABLE 7-1
SCAN PROTOCOL: BODY MODE

Note: This is the alternate proprietary procedure available for GE use and for sites with a valid Advanced Service Package Limited License. Any other protocol utilizing the body coil would also work for this check.

A. **[New Pt]**
 Id: **geservice<ENTER>**
 Name: **Bias Check**
 Weight (Lb.): **111<ENTER>**
 Set Patient Protocols to **Service**.
B. At front enclosure:
 Landmark in the Head area—remove any coils.
 press **LANDMARK**.
 press **MOVE TO SCAN**.
C. In the Patient Position Protocol field:
 type **o.45.1<ENTER>**(o=Other, 45.1 =series) to load the body protocol
 OR select **other** and select protocol **45** and select series **1**.
D. **[Save Series]**
E. **[Manual Prescan]** (System should pulse once and then fault with a TR fault. Since the body transmit cable is disconnected from J4 Body Output on the RFI, this fault is normal.)
F. **[Done]** (Select **Done** if for some reason system does not stop pulsing.)

13. Verify **BODY** LED is illuminated on front of RFI Module.

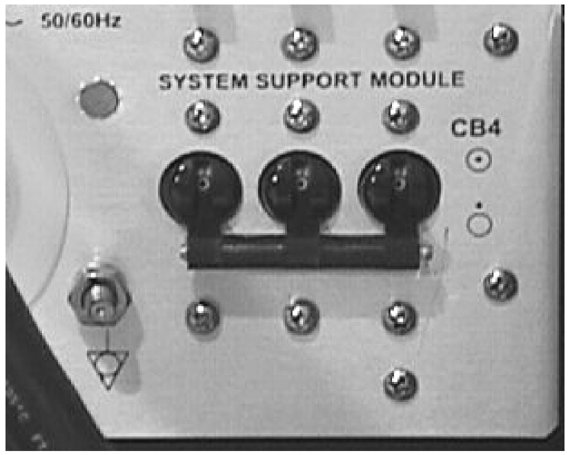
14. Look at the DMM display and observe the voltage reading.

    a. **If the voltage is between 950 VDC and 1050 VDC** then the HV hardware is working properly. Proceed to **Section 7-3 Restoration**.

    b. **If the voltage is low and not between 950 VDC and 1050 VDC** then the tests in the remainder of this section will be performed to determine the extent of the failure.

7-2 Poor Image Quality Due to Loss of High Voltage Bias (Continued)

15. Locate the **CB4 SYSTEM SUPPORT MODULE** Circuit Breaker in the lower-right corner rear of the SSM and place it in the down (OFF) position. See Illustration 7-7.

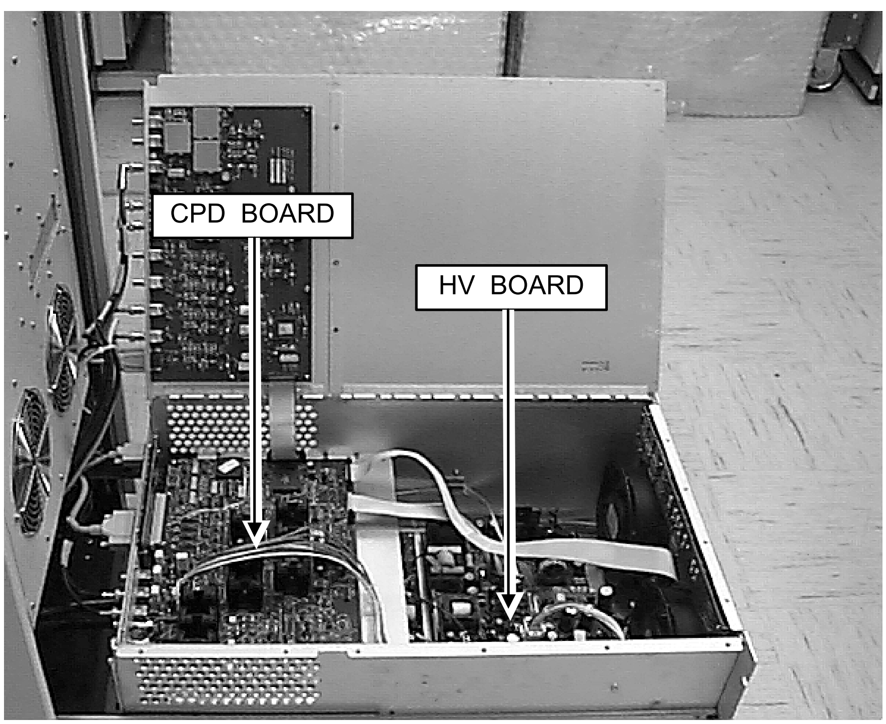


CB4 SYSTEM SUPPORT MODULE CIRCUIT BREAKER IN DOWN (OFF) POSITION
ILLUSTRATION 7-7

16. Wait 5 minutes after placing CB4 in the down (OFF) position for the high voltage supply in the SSM to discharge before performing the next step.

17. Open the SSM lid.

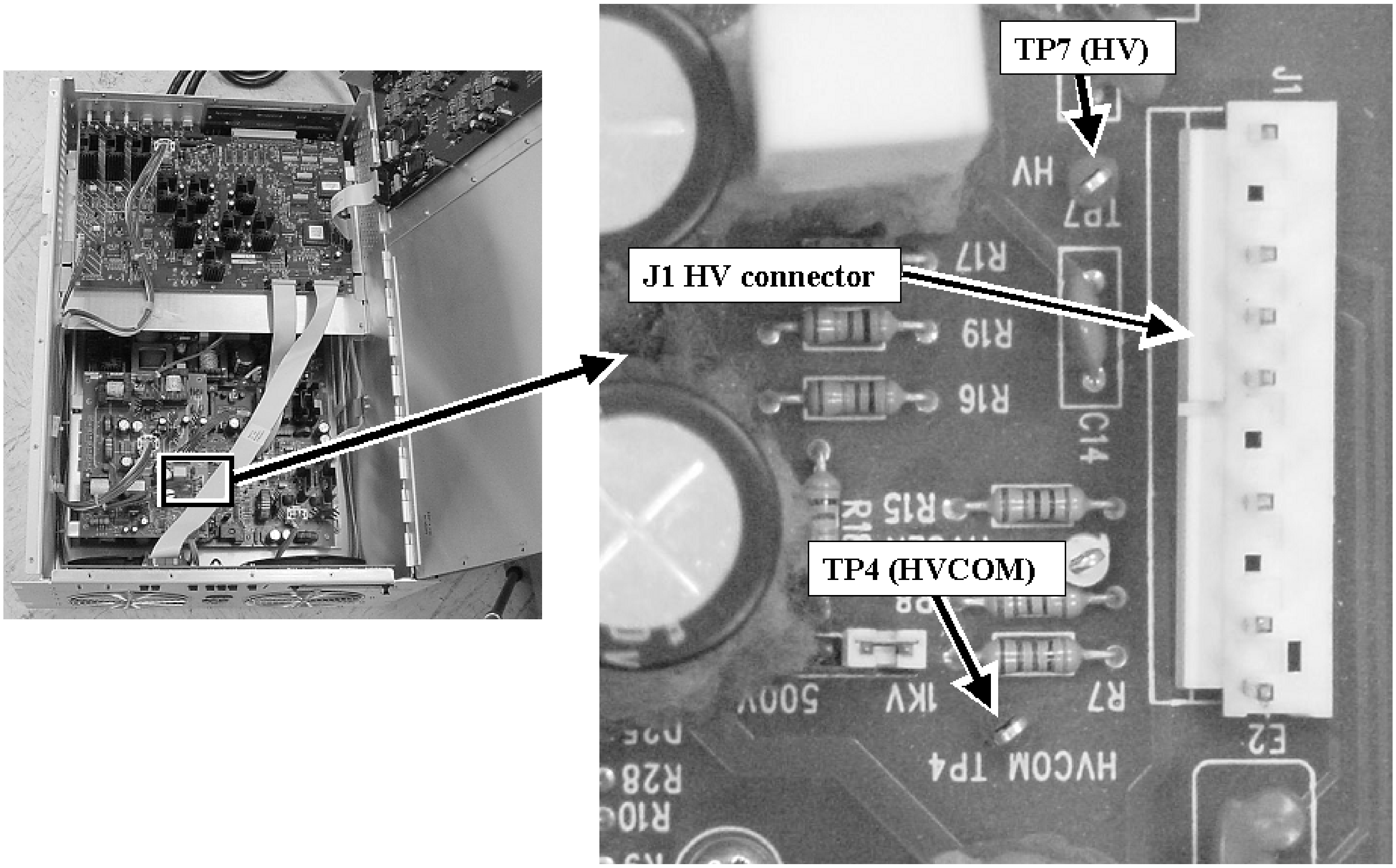
18. Refer to Illustration 7-8 and locate the HV Board in the upper, front-end of the SSM. The HV Board is mounted on stand-offs above the Dock/Light Board.



LOCATION OF HV BOARD AND CPD BOARD INSIDE SSM
ILLUSTRATION 7-8

7-2 Poor Image Quality Due to Loss of High Voltage Bias (Continued)

19. Refer to Illustration 7-9 and locate test points **TP7 (HV)** and **TP4 (HVCOM)** on the HV Board.



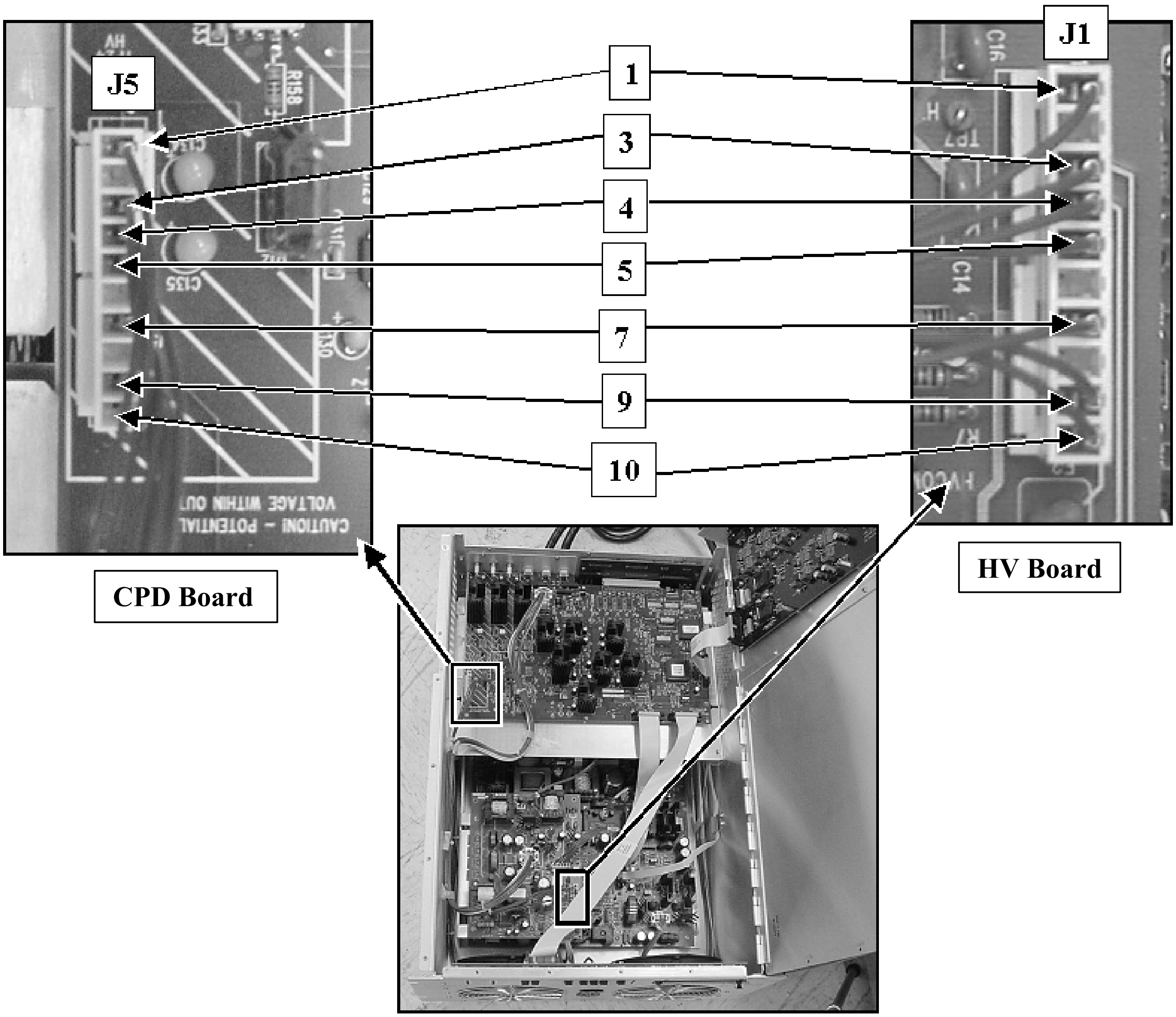
LOCATION OF J1, TP7, AND TP4 ON THE HV BOARD
ILLUSTRATION 7-9

20. Configure the DMM to measure 1000VDC and then carefully connect the positive DMM lead to **TP7 (HV)** and negative DMM lead to **TP4 (HVCOM)**. Avoid allowing the bare test leads to come into contact with fingers.

21. Verify that the DMM shows less than 10VDC between these two test points. If necessary, wait for the voltage to drop below 10VDC before proceeding.

7-2 Poor Image Quality Due to Loss of High Voltage Bias (Continued)

22. Refer to Illustration 7-10 and locate the red HV Power Cable (Erbtec part #550010) interconnecting J5 on the CPD Board to J1 on the HV Board.



HV POWER CABLE CONNECTIONS AND PINS
ILLUSTRATION 7-10

7-2 Poor Image Quality Due to Loss of High Voltage Bias (Continued)

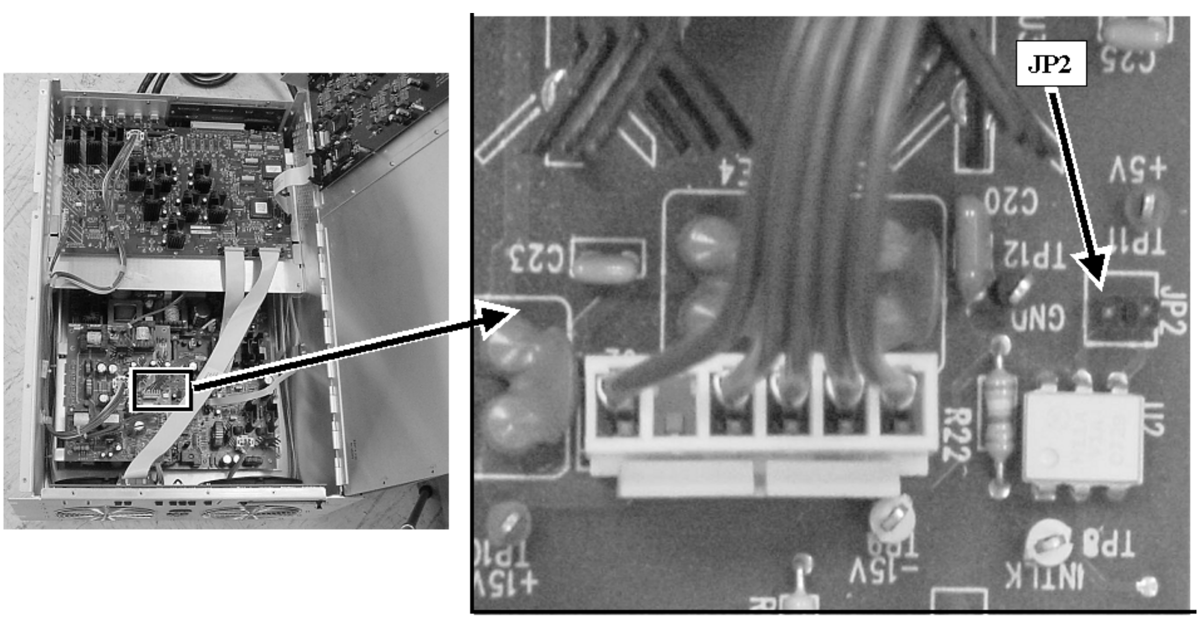
23. Refer to Table 7-2 and, with the HV Power Cable connected as shown in Illustration 7-10, use the DMM to perform a continuity check (measure the resistance) between each of the pins of CPD Board J5 and HV Board J1 and confirm that a good connection (near zero resistance) exists between the corresponding pins of each of the cable connectors.

TABLE 7-2
J5 CPD BOARD AND J1 HV BOARD CABLE CONTINUITY CHECKS

Measure Between
J1-1 and J5-1
J1-3 and J5-3
J1-4 and J5-4
J1-5 and J5-5
J1-7 and J5-7
J1-9 and J5-9
J1-10 and J5-10

24. Replace or repair the red HV Power Cable if a high resistance is measured between any of the corresponding connector pins referenced in Table 7-2.

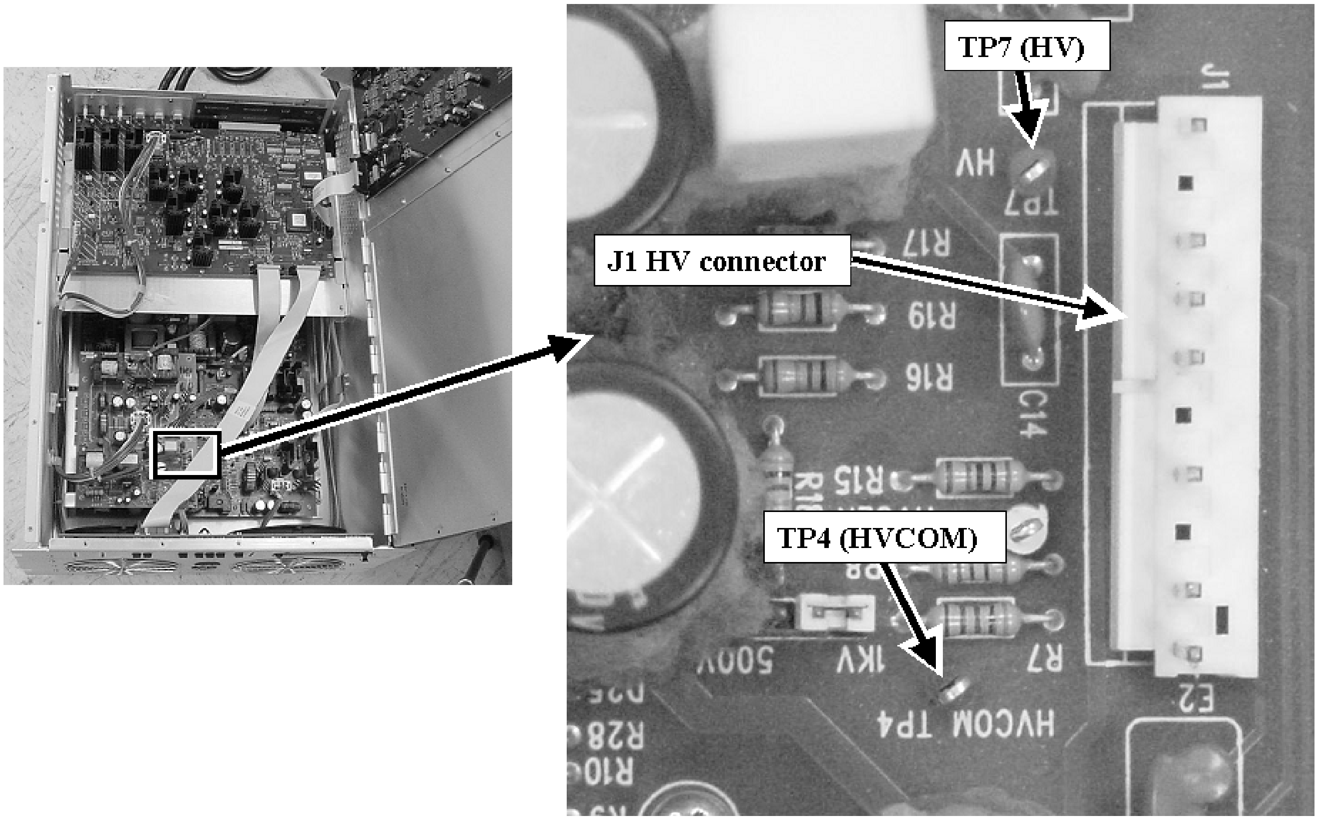
25. Refer to Illustration 7-11. Locate the JP2 jumper pins on the HV Board (High Voltage Board) and short these together with a spare jumper or piece of wire.



LOCATION OF JUMPER JP2 ON THE HV BOARD
ILLUSTRATION 7-11

7-2 Poor Image Quality Due to Loss of High Voltage Bias (Continued)

26. Refer to Illustration 7-12 and disconnect the red HV Power Cable at J1 on the HV Board and set it aside.



LOCATION OF J1, TP7, AND TP4 ON THE HV BOARD
ILLUSTRATION 7-12

27. Configure the DMM to measure 1000VDC and then connect the positive DMM lead to **TP7 (HV)** and negative DMM lead to **TP4 (HVCOM)**.

28. Close SSM lid and let it rest on test leads.

29. Place the **CB4 SYSTEM SUPPORT MODULE** Circuit Breaker in the lower-right corner rear of the SSM in the up (ON) position.

30. Verify that the reading on the DMM display is between 950 VDC and 1050 VDC.

    a. **If the voltage is low and is not between 950 VDC and 1050 VDC** then the HV Board has failed. The CPD Board may have shorted and caused it to fail. Replace both the CPD Board and the HV Board.

    b. **If the voltage is between 950 VDC and 1050 VDC** then the CPD Board failed but did not damage the HV Board. Replace the CPD Board.

31. Locate the **CB4 SYSTEM SUPPORT MODULE** Circuit Breaker in the lower-right corner rear of the SSM and place it in the down (OFF) position.

32. Watch the DMM display and wait for the high voltage to discharge down below 10 volts.

7-2 Poor Image Quality Due to Loss of High Voltage Bias (Continued)

33. Reconnect the HV Power Cable to J1 on the HV Board.

34. Remove the jumper or shorting wire from between the pins of JP2 on the HV Board.

35. Remove the DMM test leads from **TP7** and **TP4** on the HV Board and close the SSM lid cover.

36. Proceed to **Section 7-3 Restoration**.

7-3 Restoration

1. Verify that the male MHV connectors labeled J42, J43, and J44 are connected to the **J42 BODY 1**, **J43 BODY 2**, and **J44 DD** connectors on the rear of the SSM.

2. Reconnect the body RF transmission line cable to **J4 BODY OUTPUT** on the rear of the RFI Module.

3. Reconnect the RF Input Cable (Run 229) to the RF Cabinet RF IN Interface Panel.

   a. **If this is a RF/PDU Cabinet:** Reconnect the RF Input Cable (Run 229) to **RF IN J3** on the rear RF/PDU Cabinet Interface.

   b. **If this is an SRF Cabinet:** Reconnect the RF Input Cable (Run 229) to the **RF IN J1** on the rear SRF Cabinet Interface.

4. Confirm that the black fault switches on the front of the SSM are in the up (ENable) position.

5. Perform actions in **Section A-2 Replacing SSM Chassis Lid**.

6. Place the **CB4 SYSTEM SUPPORT MODULE** Circuit Breaker in the lower-right corner rear of the SSM in the up (ON) position.

7. Reset the TPS from the Service Desktop Manager Window.

8. If any Dynamic Disable Switch Boards or the Body Hybrid Splitter was replaced then press the **CAL** Button on the front of the SSM once. If necessary, refer to the **CPD BOARD SETUP & CALIBRATION** procedure on the Service CD-ROM for more information.

9. Perform one head and one body scan.

10. Replace the front cover on the RF Cabinet.

8 - TROUBLESHOOTING TR DRIVER FAULTS

8-1 General Theory Overview

Each time the MR system transmits or receives signal it switches a corresponding transmit or receive bias onto the body, head, and spectroscopy transmission lines. It also switches a similar type of bias onto each of the multicoil (phased-array) lines but the method and purpose for this is different enough that troubleshooting and theory for multicoil TR bias is not covered in this section but instead in **SECTION 10 – TROUBLESHOOTING MULTICOIL TR BIAS FAULTS**. This bias switching is done irrespective of whether RF transmit signal is applied to the transmission line or not. The level of the bias is the same for the body and spectroscopy lines but is different for the head transmission line. See Table 8-1. The purpose of this bias, referred to as TR bias, is to forward or reverse bias PIN diodes inside the Body Hybrid Splitter, Head TR Network, and, if the site has the MNS hardware option, Spectro TR Module. During system transmit this bias "switches" the PIN diodes so that RF transmit energy, if it is present on the line, is routed to the RF coil and not into the sensitive body receive circuit. Likewise, during system receive, this bias "switches" the PIN diodes so that the very small receive signal from the RF coil is routed into the body receive circuit.

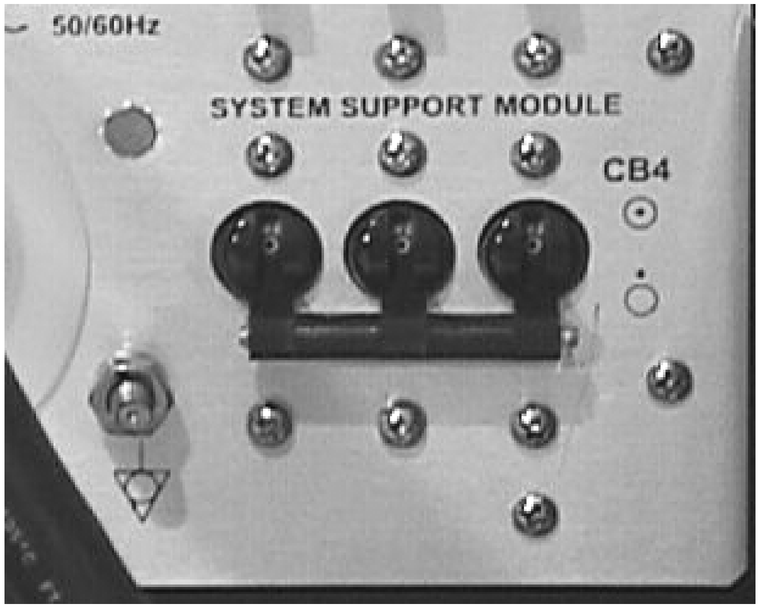
TABLE 8-1
TR DRIVER BIAS OVERVIEW

Transmission Line	Transmit Mode	Receive Mode
Body	+ 4 VDC	- 12.25 VDC
Head	+ 8 VDC	-12.25 VDC
Spectroscopy [1]	+ 4.3 VDC	-12.00 VDC

NOTE 1: REFERENCE THE SPECTROSCOPY MANUAL ON THE 8.X SERVICE CD-ROM FOR ALL QUESTIONS CONCERNING SPECTROSCOPY TR ADJUSTMENTS.

8-2 Troubleshooting Head, Body, and Spectroscopy TR Driver Faults

1. Locate the **CB4 SYSTEM SUPPORT MODULE** Circuit Breaker in the lower-right corner rear of the SSM and place it in the down (OFF) position. See Illustration 8-1.



CB4 SYSTEM SUPPORT MODULE CIRCUIT BREAKER IN DOWN (OFF) POSITION
ILLUSTRATION 8-1

CAUTION

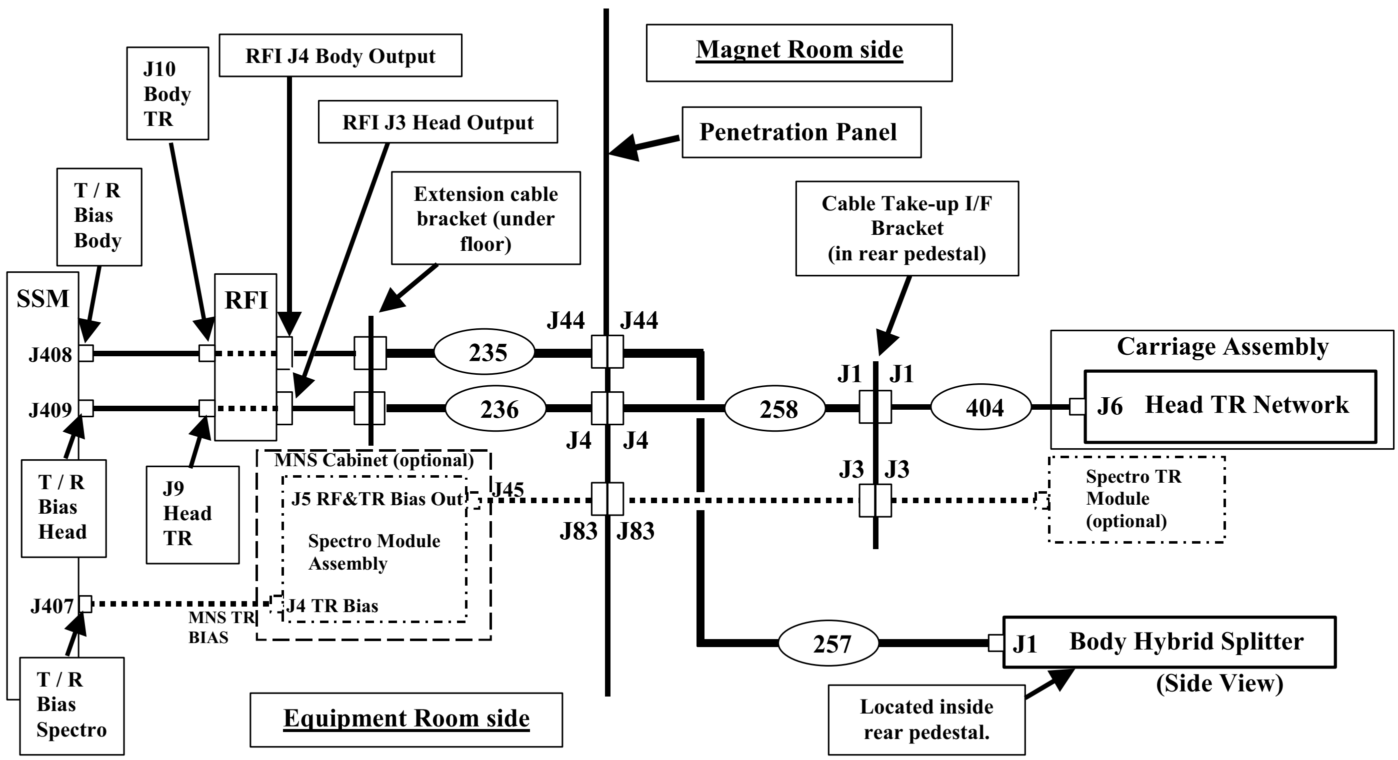
Placing the CB4 System support module circuit breaker in the down (OFF) position will remove power from high voltage Dynamic Disable and Inductive Drive Disable Circuits and the low-voltage TR driver circuits but does not remove AC power from the NB AMP receptacle, SPECTRO AMP receptacle, Front Service Outlets, or Gradient Blower Fan.

8-2 Troubleshooting Head, Body, and Spectroscopy TR Driver Faults (Continued)

2. Refer to Illustration 8-2 below. Check and make certain that all the cables are properly connected to the head or body circuit reporting the TR driver failure.

CAUTION

Do not assume that cable labels are correct and that cables have the correct labels. Physically trace the path and verify the connection of the cables.



PATH OF TR DRIVER BIAS ON BODY, HEAD, AND SPECTRO TRANSMISSION LINE CIRCUITS
ILLUSTRATION 8-2

3. Obtain a Digital MultiMeter (DMM) and place it in diode test mode.

4. Remove the coaxial cables from the rear of the SSM at **J408 T/R BIAS BODY**, **J409 T/R BIAS HEAD**, and, if there is a cable connected to it, at **J407 T/R BIAS SPECTRO**. Set these cables aside.

8-2 Troubleshooting Head, Body, and Spectroscopy TR Driver Faults (Continued)

5. Use the DMM to measure for a diode drop through each of the TR driver cables removed from the SSM and compare the results to those shown in Table 8-2.

TABLE 8-2
TR DIODE DROP MEASUREMENTS ON EACH CABLE

CABLE IDENTIFIER	DMM LEAD POSITION	TYPICAL READING	MEASURED READING
MR1-A7-J408	Black to center pin, Red to shell	Open circuit	____________
	Red to center pin, Black to shell	~ 0.3 – 0.6V	____________
MR1-A7-J409	Black to center pin, Red to shell	Open circuit	____________
	Red to center pin, Black to shell	~ 0.3 – 0.6V	____________
MR1-A7-J407 [1]	Black to center pin, Red to shell	Open circuit	____________
	Red to center pin, Black to shell	~ 0.3 – 0.6V	____________

NOTE 1: IF THE SITE DOES NOT HAVE THE MULTI-NUCLEAR SPECTROSCOPY (MNS) OPTION THEN THIS CABLE WILL NOT BE PRESENT.

6. Compare each of the measured values with the typical values provided in Table 8-2.

7. If all the measurement results look normal then the CPD Board inside the SSM appears to have failed. Replace the CPD Board.

8. If any of the measurement results look abnormal then hardware external to the SSM has failed and/or cable misconnection has occurred.

DANGER!!

THE DMM CONTAINS FERROUS COMPONENTS THAT, IF BROUGHT INTO CLOSE PROXIMITY TO THE MAGNET, WILL BE STRONGLY ATTRACTED TO IT. EXERCISE GREAT CARE IF USING THE DMM INSIDE THE MAGNET ROOM.

9. Refer to Illustration 8-2. Repeat the diode drop measurements taken earlier at different points in the TR driver circuit starting at the next point in the circuit in the Equipment Room and then working from there to the Penetration Panel and, if necessary, to the circuit in the Magnet Room. Repeat this process until the failed component in the circuit is found. Once the failed part has been located, either repair or replace it.

10. If the failure happened on a circuit other than the one that was reported in the system error log then the TR driver bias coaxial cables have been swapped at the SSM, RFI, or, if the customer has the MNS hardware option, at the MNS Cabinet. Refer to Illustration 8-2 and physically trace the cable routing and connections. Correct any mis-labeled cables.

8-2 Troubleshooting Head, Body, and Spectroscopy TR Driver Faults (Continued)

11. Once the problem has been found and resolved, reconnect all the system cables as shown in Illustration 8-2.

NOTE

Although in the course of troubleshooting it usually is not necessary to do so, it is possible to use an oscilloscope to view the TR driver bias waveforms from each of the SSM drivers. Instructions for doing this, adjusting the TR bias, and reference comparison waveforms can be found in **SECTION 6 – TR VOLTAGE ADJUSTMENT WITH PULSE TRAIN** in the **CPD BOARD SETUP & CALIBRATION (RC3SCC2.DOC)** procedure located in the SRF or RF/PDU Cabinet section on the 8.X Service CD-ROM.

12. Locate the **CB4 SYSTEM SUPPORT MODULE** Circuit Breaker in the lower-right corner rear of the SSM and place it in the up (ON) position.

CAUTION

If shorted or open diodes were located in the Body Hybrid Splitter, Head TR Network, or any of the body coil Dynamic Disable Boards then it is necessary to confirm that the RF amplifier output waveform is not the cause of the failure(s). A distorted or noisy RF waveform or one that is occurring outside of the TR unblank time period can cause PIN diode failures. Failure to check the RF amplifier output waveform after making a repair may result in frequent return visits to the site to repair more PIN diode failures.

13. If the failure was isolated to a damaged PIN diode(s) in the Head TR Network, Body Hybrid Splitter, or Dynamic Disable Board(s) then please check the appropriate RF output (Head or Body depending on what failed) from the RF amplifier to confirm that the waveform is symmetrical (not clipped), is not noisy, and is occurring within the TR unblank period. A dual channel oscilloscope will allow you to view both the RF output waveform and the appropriate TR bias together (see Section 6 in the **CPD Board Setup & Calibration** document for viewing TR bias).

    a. If the RF amplifier is an SRF with an RFI Module then reference this document to view the appropriate RF output power: **1.5T SRF Max Power RF Out Setup and Calibration**

    b. If the RF amplifier is an SRFD2 then reference this document to view the appropriate RF output power: **1.5T SRFD2 Max Power RF Out Setup and Calibration**

14. Please reference the appropriate RF troubleshooting documentation if an RF failure is found.

15. Confirm that system restoration from the appropriate procedure has been done.

16. Perform one head and body scan. If necessary, perform a spectroscopy scan.

17. Replace the SRF or RF/PDU Cabinet front cover.

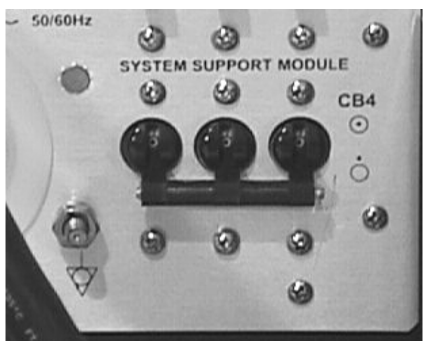
9 – TROUBLESHOOTING DYNAMIC DISABLE AND INDUCTIVE DRIVE DISABLE FAULTS

9-1 General Theory Overview

The role of the dynamic disable bias is to either tune (1000VDC) or detune (-15VDC) the body coil. During a body scan the dynamic disable bias remains constant at 1000VDC (coil tuned) while during a head scan the bias remains constant at –15VDC (coil detuned). During a surface or phased-array coil scan (assumed to be receive-only), the bias does not remain constant but instead switches from 1000VDC during body transmit to –15VDC during surface or phased-array coil receive. The System Support Module (SSM) supplies the dynamic disable bias and monitors the load circuit for open-circuit fault conditions during head or surface / phased-array coil scanning. The load is monitored for short-circuit fault conditions during body or surface / phased-array coil scanning. Provided that the RF/PDU or SRF Cabinet front cover has not been removed, system scanning will be inhibited and a fault message written to the error log when an open or short-circuit condition is detected by the SSM. Dynamic disable fault reporting can be inhibited by placing the **DD** Switch on the front of the SSM in the down (DISable) position. Fault reporting will remain inhibited until either the switch is put back into the up (ENable) position or the cabinet front cover is replaced.

9-2 Isolating Dynamic Disable and Inductive Drive Disable Faults

1. Locate the **CB4 SYSTEM SUPPORT MODULE** Circuit Breaker in the lower-right corner rear of the SSM and place it in the down (OFF) position. See Illustration 9-1.



CB4 SYSTEM SUPPORT MODULE CIRCUIT BREAKER IN DOWN (OFF) POSITION
ILLUSTRATION 9-1

CAUTION

Placing the CB4 System support module circuit breaker in the down (OFF) position will remove power from high voltage Dynamic Disable and Inductive Drive Disable Circuits and the low-voltage TR driver circuits but does not remove AC power from the NB AMP receptacle, SPECTRO AMP receptacle, Front Service Outlets, or Gradient Blower Fan.

2. Remove the male MHV cable connectors from the **J42 BODY 1**, **J43 BODY 2**, and **J44 DD** connectors on the rear exterior, right-side of the SSM.

CAUTION

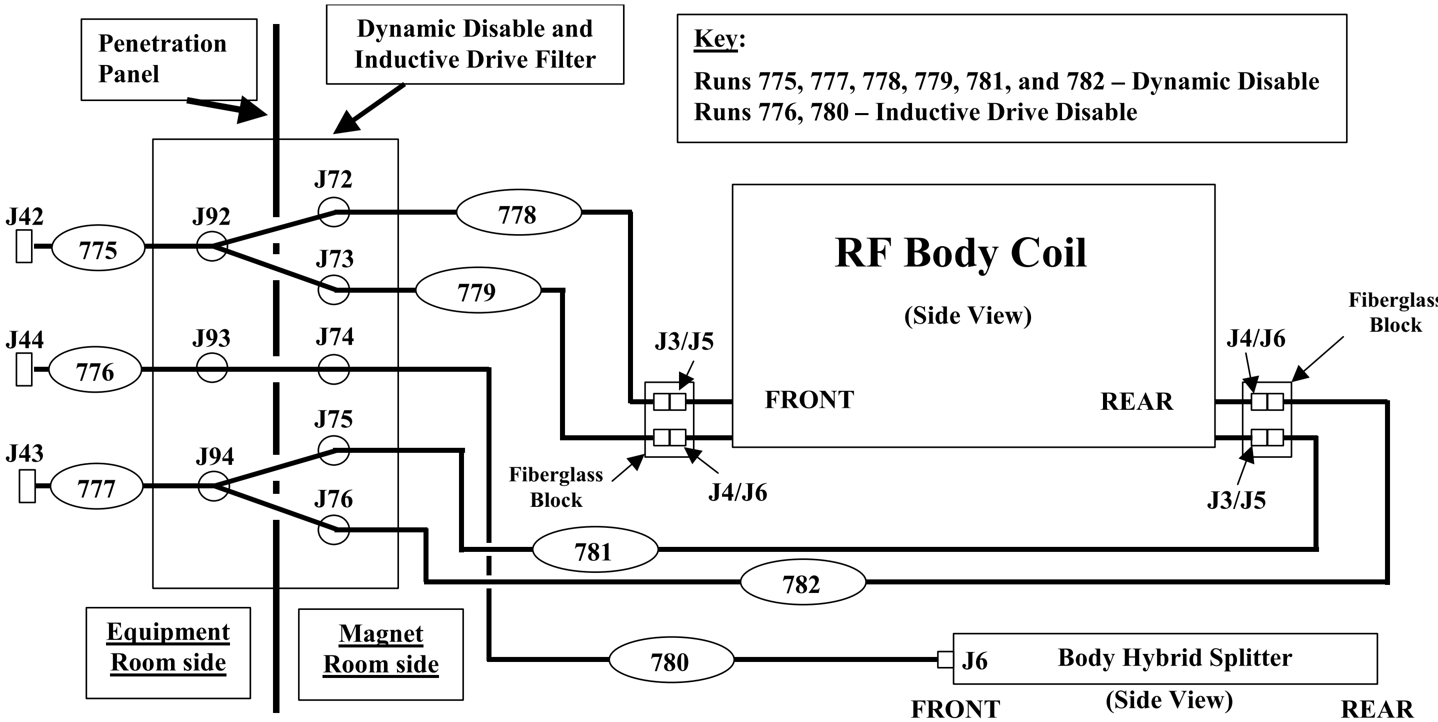
While MHV connectors may look somewhat like BNC connectors the construction of the two connectors is not the same. Forcing MHV and BNC connectors together can damage both connectors.

9-2 Isolating Dynamic Disable and Inductive Drive Disable Faults (Continued)

3. Refer to Illustration 9-2 below. Check and make certain that all the cables are properly connected to the dynamic disable or inductive drive disable circuit reporting the fault.

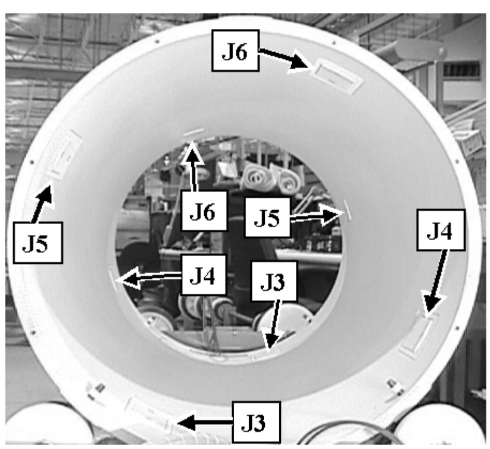
CAUTION

Do not assume that cable labels are correct and that cables have the correct labels. Physically trace the path and verify the connection of the cables.



CONNECTION MAP OF DYNAMIC DISABLE AND INDUCTIVE DRIVE CONNECTIONS
ILLUSTRATION 9-2

Note that while the Inductive Drive Disable Driver (**J44, Run 776**) only connects to **J6** on the Body Hybrid Splitter, each of the Dynamic Disable Drivers (**J42, Run 775** and **J43, Run 777**) control four Dynamic Disable Switch Boards (**J3/J5** and **J4/J6**, in each case) on each side of the RF coil. See Illustration 9-3 below.



REAR VIEW INTO RF COIL SHOWING BOARD POSITIONS (LOCATIONS SAME FOR BRM, TRM, CRM)
ILLUSTRATION 9-3

9-2 Isolating Dynamic Disable and Inductive Drive Disable Faults (Continued)

4. Obtain a Digital MultiMeter (DMM) and place it in diode test mode.

5. Use the DMM to measure for a diode drop through each of the Dynamic Disable and Inductive Drive Disable cables removed from the SSM and compare the results to those shown in Table 9-1.

TABLE 9-1
DYNAMIC DISABLE / INDUCTIVE DRIVE DISABLE DIODE DROP MEASUREMENTS ON EACH CABLE

RUN	CABLE IDENTIFIER	LEAD POSITION	TYPICAL	MEASURED
775	MR1-A7-J42	Red to center pin, Black to shell	Open circuit	
		Black to center pin, Red to shell	~ 0.3 – 0.6V	
777	MR1-A7-J43	Red to center pin, Black to shell	Open circuit	
		Black to center pin, Red to shell	~ 0.3 – 0.6V	
776	MR1-A7-J44	Red to center pin, Black to shell	Open circuit	
		Black to center pin, Red to shell	~ 0.3 – 0.6V	

6. Compare each of the measured values with the typical values provided in Table 9-1.

7. If all the measurement results look normal *and* Error Message 2225516 - “High voltage power supply low” **was not** reported at or near the same time that the Dynamic Disable or Inductive Drive Disable faults were reported in the system error log (avoid a misdiagnosis and please look closely) then the CPD Board inside the SSM appears to have failed. Replace the CPD Board.

8. If all the measurement results look normal *and* Error Message 2225516 - “High voltage power supply low” **was** reported at or near the same time that the Dynamic Disable or Inductive Drive Disable faults were reported in the system error log then either the CPD Board and HV Board or both boards have failed. Proceed to **Section 7-2 Poor Image Quality Due to Loss of High Voltage Bias** and isolate the fault to the failed board(s).

9. If any of the measurement results look abnormal then hardware external to the SSM has failed and/or cable misconnection has occurred.

DANGER!!

THE DMM CONTAINS FERROUS COMPONENTS THAT, IF BROUGHT INTO CLOSE PROXIMITY TO THE MAGNET, WILL BE STRONGLY ATTRACTED TO IT. EXERCISE GREAT CARE IF USING THE DMM INSIDE THE MAGNET ROOM.

9-2 Isolating Dynamic Disable and Inductive Drive Disable Faults (Continued)

10. Refer to Illustration 9-2. Repeat the diode drop measurements taken earlier at different points in the Dynamic Disable or Inductive Drive Disable circuit starting at the next point in the circuit in the Equipment Room and then working from there to the Penetration Panel and, if necessary, to the circuit in the Magnet Room. Repeat this process until the failed component in the circuit is found.

NOTE

Instructions on how to replace a failed Dynamic Disable Switch Board inside the RF body coil can be found in the Body Coil section of the 8.X Service Methods CD-ROM in **DYN DISABLE SWITCH (1.5T & 1.0T) REPLACEMENT (RB1REA3.DOC).**

11. If the failure happened on a circuit other than the one that was reported in the system error log then the Dynamic Disable and Inductive Drive Disable Driver bias coaxial cables have been swapped at the SSM, or at the Penetration Panel High-Voltage Filter (either side of filter), or at the magnet. Refer to Illustration 9-2 and physically trace the cable routing and connections. Correct any mis-labeled cables.

WARNING!

DO NOT ATTEMPT TO MEASURE THE DYNAMIC DISABLE OR INDUCTIVE DRIVE DISABLE BIAS WITH AN OSCILLOSCOPE. WHILE THIS CAN BE DONE USING A DIGITAL MULTIMETER RATED FOR HIGH VOLTAGE, 1000VDC INPUT DIRECTLY TO AN OSCILLOSCOPE WILL DAMAGE IT.

12. Once the problem has been found and resolved, reconnect all the system cables as shown in Illustration 9-2.

13. Locate the **CB4 SYSTEM SUPPORT MODULE** Circuit Breaker in the lower-right corner rear of the SSM and place it in the up (ON) position.

CAUTION

If shorted or open diodes were located in the Body Hybrid Splitter or any of the body coil Dynamic Disable Boards then it is necessary to confirm that the RF amplifier Body RF output waveform is not the cause of the failure(s). A distorted or noisy Body RF output waveform can cause PIN diode failures. Failure to check the RF amplifier output waveform after making a repair may result in frequent return visits to the site to repair more PIN diode failures.

14. If the failure was isolated to a damaged PIN diode(s) in the Body Hybrid Splitter or Dynamic Disable Board(s) then please check the Body RF output from the RF amplifier to confirm that the waveform is symmetrical (not clipped) and is not noisy.

    a. If the RF amplifier is an SRF with an RFI Module then reference this document to view the Body RF output power: **1.5T SRF Max Power RF Out Setup and Calibration.**

   b. If the RF amplifier is an SRFD2 then reference this document to view the Body RF output power: **1.5T SRFD2 Max Power RF Out Setup and Calibration.**

15. Please reference the appropriate RF troubleshooting documentation if an RF failure is found.

16. Confirm that system restoration from the appropriate procedure has been done.

17. Perform one head and body scan. If necessary, perform a spectroscopy scan.

18. Replace the SRF or RF/PDU Cabinet front cover.

10 – TROUBLESHOOTING MULTICOIL TR BIAS FAULTS

10-1 General Theory Overview

TR bias, supplied through a dedicated line from the SSM to the Multicoil Select Switch and the Preamp Protect Module, is provided to switch the PIN diode switching circuitry for the individual coils inside the multicoil (phased-array coil) during a multicoil scan. –5V is applied to the selected individual coils during the receive portion of the scan to reverse bias the multicoil PIN switch circuitry and thus tune the multicoil. +5V is applied during the transmit portion of the scan to forward bias the PIN switch circuitry and thus detune the multicoil. This switching also serves a dual purpose to connect or isolate coil outputs from various receive lines. Also, forward bias (+5V) is also applied during the receive portion of some scans to detune certain multicoils that are not being used so as to prevent them from collecting noise and spurious signal that could degrade the combined image. Biasing the coil in this way and then also switching the coil output from the receive line is sometimes referred to as switching the coil "hard" off. This is a very effective way to insure that the coil output is isolated from the receive line. A less effective way to minimize unwanted receive signal from a coil is to switch the coil "soft" off by leaving the coil reversed biased (tuned) but yet switched from the receive line.

10-2 Isolating Multicoil TR Driver Faults

1. Setup for a scan using the CTLTOP coil (even if site does not have this coil) and then **[Manual Prescan] [Done]**.

2. Refer to the CTLTOP Multicoil diagram on the 8.X Service CD-ROM under Coils, Multicoil Control Diagrams, "**1.5T Teledyne, USAI As Well As 1.0T CTL Coils**". This is the first coil on the list. This diagram can be used for 1.0T also.

Note that on the CTLTOP diagram, with the exception of the **J20 MC SIGNALS** connector supplying the multicoil driver signals on the SSM, all the pin numbers are furnished with the connectors.

DANGER!!

A DIGITAL MULTIMETER (DMM) CONTAINS FERROUS COMPONENTS THAT, IF BROUGHT INTO CLOSE PROXIMITY TO THE MAGNET, WILL BE STRONGLY ATTRACTED TO IT. EXERCISE GREAT CARE IF USING THE DMM INSIDE THE MAGNET ROOM.

10-2 Isolating Multicoil TR Driver Faults (Continued)

3. Use a Digital Multimeter (DMM) to measure for voltages between the designated Multicoil Driver pins and the common pins at **J20 MC SIGNALS** on the rear of the SSM. See Table 10-1 below for a listing of all the Multicoil Driver pin assignments and expected voltages for CTLTOP coil selection. If necessary, trace these through the chain all the way to the Preamp Protect Module in the Low Profile Carriage Assembly.

TABLE 10-1
J20 MC SIGNALS PIN ASSIGNMENTS AND EXPECTED VOLTAGES FOR CTLTOP COIL SELECTION

MULTICOIL DRIVER	PIN	Measured Voltage	COMMON PINS
1	8	-5 VDC	10, 11, 12, 13, 14, 15
2	6	-5 VDC	10, 11, 12, 13, 14, 15
3	3	-5 VDC	10, 11, 12, 13, 14, 15
4	1	+5 VDC	10, 11, 12, 13, 14, 15

4. If the problem is missing multicoil drive voltage from the SSM then replace the CPD Board.

5. When finished reconnect any disconnected cables and then, if necessary, return to the flowchart.

APPENDIX A – REMOVING AND REPLACING SSM CHASSIS LID

The process works the same regardless of whether the SSM is inside an RF/PDU or SRF Cabinet.

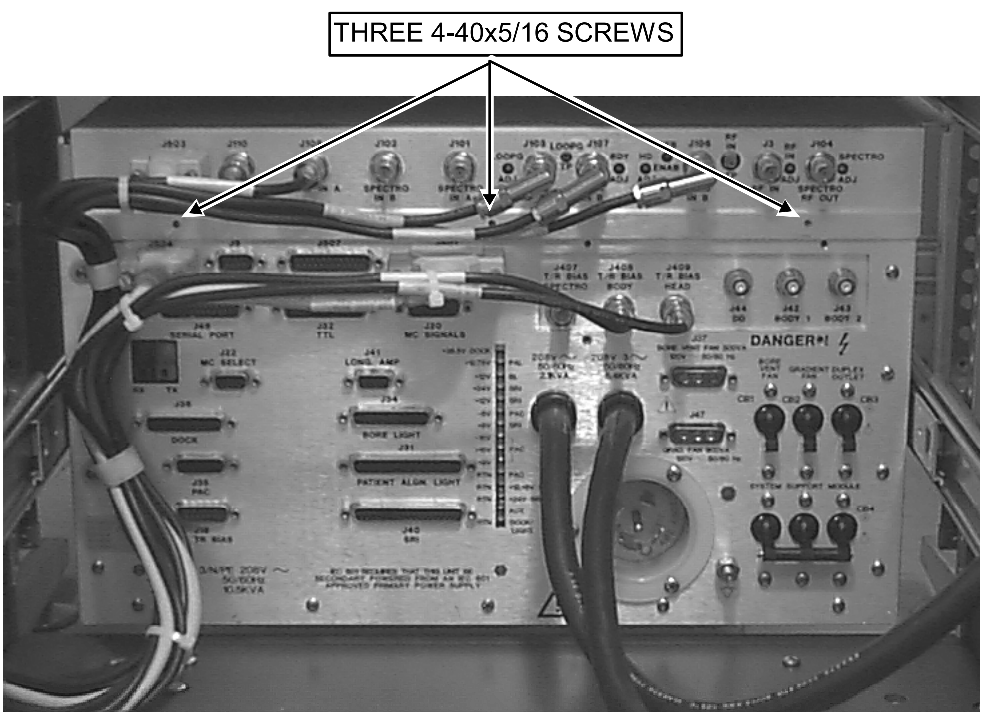
A-1 Removing SSM Chassis Lid

DANGER!!

POSSIBLE PERSONAL INJURY! AVOID SERIOUS INJURY OR DEATH BY ELECTROCUTION. MAKE CERTAIN THAT SYSTEM SUPPORT MODULE CB4 CIRCUIT BREAKER ON THE REAR OF THE SSM IS IN THE DOWN (OFF) POSITION. BE AWARE THAT THIS WILL REMOVE POWER FROM HIGH VOLTAGE DYNAMIC DISABLE AND INDUCTIVE DRIVE DISABLE CIRCUITS AND THE LOW-VOLTAGE TR DRIVER CIRCUITS BUT DOES NOT REMOVE AC POWER FROM THE NB AMP RECEPTACLE, SPECTRO AMP RECEPTACLE, FRONT SERVICE OUTLETS, OR GRADIENT BLOWER FAN.

1. Confirm that System Support Module CB4 circuit breaker on the rear of the SSM is in the down (OFF) position.

2. Remove the three 4-40x5/16 panhead screws that secure the APM lid to the rear panel (see Illustration A-1). A few cables may need to be disconnected first.

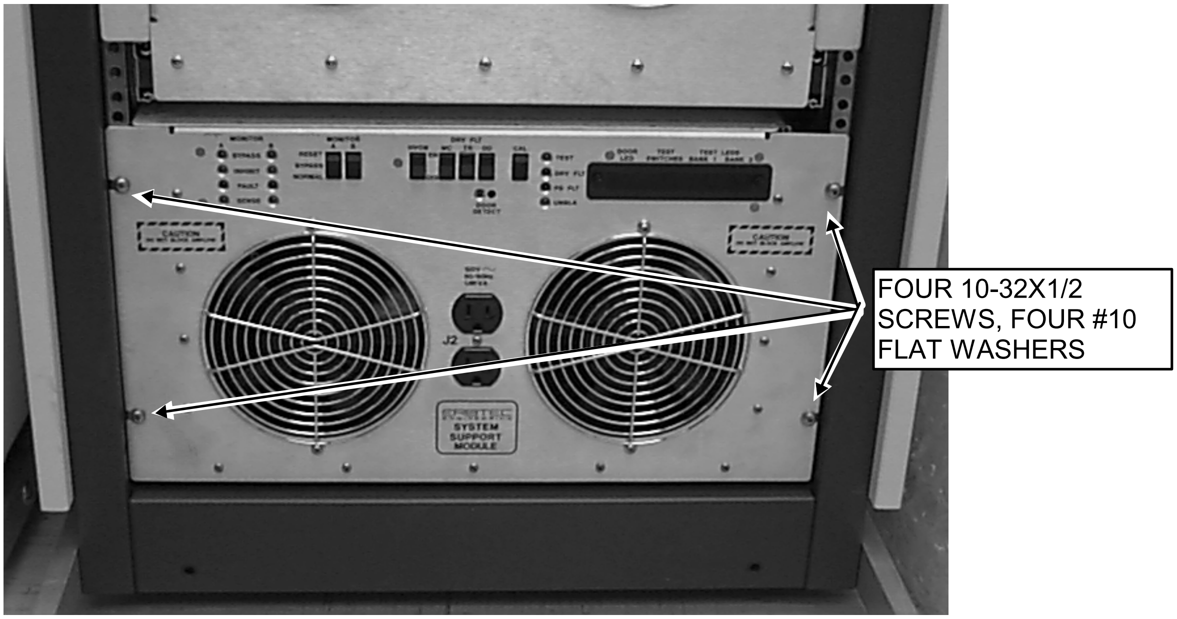


REMOVAL OF ANALOG PROCESSOR MODULE LID FROM REAR PANEL
ILLUSTRATION A-1

3. Remove the front door from the cabinet.

A-1 Removing SSM Chassis Lid (Continued)

4. Remove the four 10-32x1/2 screws and four #10 flat washers that secure the SSM chassis to the cabinet (see Illustration A-2).

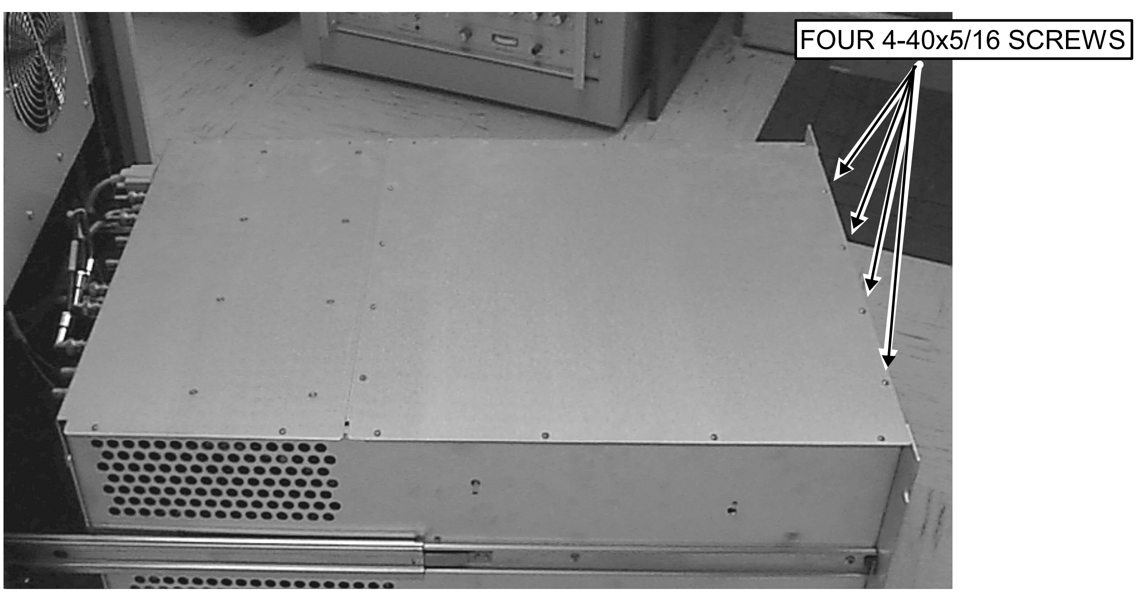


REMOVAL OF SYSTEM SUPPORT MODULE
ILLUSTRATION A-2

5. Remove all external cabling from the rear of the SSM.

6. Slide out the SSM from the cabinet.

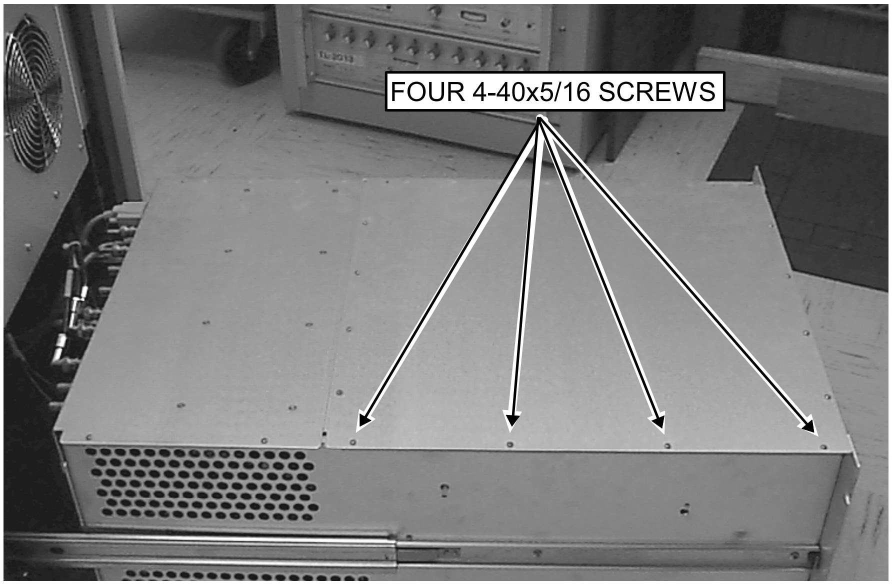
7. Remove the four 4-40x5/16 panhead screws that secure the front lid to the front panel (see Illustration A-3).



REMOVAL OF SYSTEM SUPPORT MODULE LID FROM FRONT PANEL
ILLUSTRATION A-3

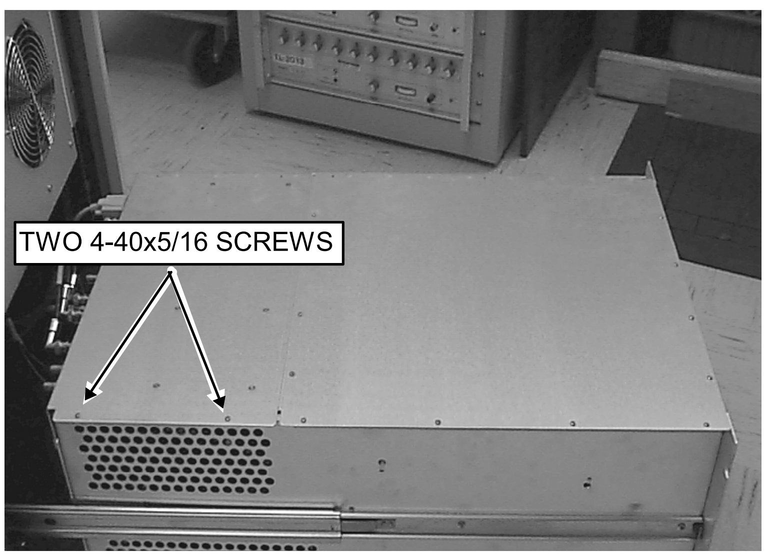
A-1 Removing SSM Chassis Lid (Continued)

8. Remove the four 4-40x5/16 panhead screws that secure the front lid to the chassis (see Illustration A-4).



REMOVAL OF SYSTEM SUPPORT MODULE LID FROM CHASSIS
ILLUSTRATION A-4

9. Remove the two 4-40x5/16 panhead screws that secure the APM lid to the chassis (see Illustration A-5).



REMOVAL OF ANALOG PROCESSOR MODULE LID FROM CHASSIS
ILLUSTRATION A-5

10. Grasp the edge of the module lid where the APM lid meets the front lid, and lift the lid to a vertical position. Use care to ensure that the APM cables do not catch on the bottom of the RFI and that there is sufficient slack.

11. Return to the previous procedure.

A-2 Replacing SSM Chassis Lid

1. Close the front lid. Install eight 4-40 x 5/16 panhead screws: four to the APM lid, and four to the front panel.

2. Attach six 4-40x5/16 panhead screws that secure both lids to chassis. Tighten all screws in steps 4, 5, and 6.

3. Slide the module back into the cabinet. (Press the slide locks to enable the module to slide completely into the cabinet.)

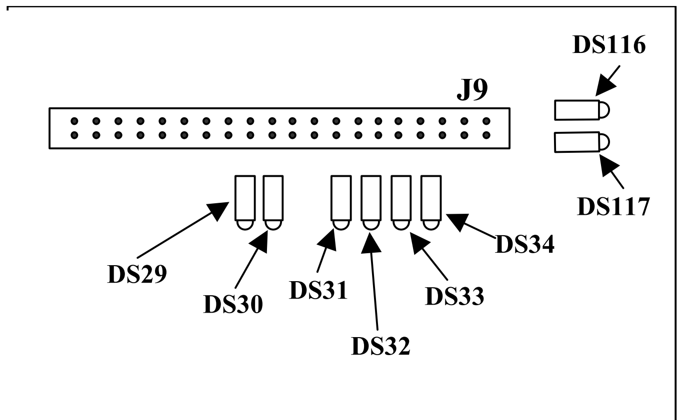
4. Secure SSM to cabinet with four 10-32x1/2 panhead screws and four #10 flat washers.

5. Install three 4-40x5/16 panhead screws that secure the APM lid to the rear panel.

6. Return to previous procedure.

APPENDIX B – IDENTIFICATION OF SRI LEDS

The Scan Room Interface (SRI) LEDS are located in the upper-right corner of the SRI internal main board. The LED identifiers are silk-screened on the SRI main board next to each LED. See Illustration B-1 below for the locations of the individual LEDs. See Table B-1 below for a listed name and function of each LED.



LOCATION OF LEDS ON SRI MAIN BOARD
ILLUSTRATION B-1

TABLE B-1
DESCRIPTION OF SRI LEDS

LED	DESCRIPTION
DS29	TEST (normally blinking)
DS30	MOTOR DISABLE (illuminated when longitudinal drive motor idle)
DS31	UPLIM (illuminated when table full up)
DS32	HOME (illuminated when table in “home” poisiton)
DS33	EOT (End Of Travel; normally dark)
DS34	DCKSTP (lit when dock motor rotating; LX SRI does not monitor)
DS116	ENC – B (flashes a binary code as encoder rotates; read with ENC – A)
DS117	ENC – A (flashes a binary code as encoder rotates; read with ENC – B)

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	June 3, 2002	Don Thome	Initial Release
1	June 29, 2004	Don Thome	Updated Sections 8 & 9 to check RF output waveform integrity.